

NOC Engineer

Interview Preparation Guide

108 Real-World Scenarios from L1 to L3

From your first NOC role to senior escalation

by Jiwan Bhattarai

CCIE #60261 | CEH

IT Education Nepal

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INTERVIEW PREPARATION GUIDE

108 real-world scenarios for Network Operations Center interviews, from your first NOC role to senior escalation

First edition

By
Jiwan Bhattarai
IT Education Nepal

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NOC Engineer Interview Preparation Guide: 108 Real-World Scenarios from L1 to L3

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Feedback and corrections. Write to contact@iteducationnepal.org. Include the scenario number, the platform and software version, the command, the expected result and the actual result. Vendor behaviour changes between releases, so a precise report helps the next reader. Corrections that add a primary standard reference are especially valued. A suggested new scenario should include a way to verify it.

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verify every command in your own environment and respect the safety label on each command block before using it on a production device. The author and publisher accept no liability for loss or disruption arising from the use of this material.

Mission and Motivation

Learn how to learn. Learn by doing.

Let us create 10,000 quality IT engineers, from Nepal to the world.

Jiwan Bhattarai

CCIE #60261 | CEH | Mentor | Education Activist

About IT Education Nepal

IT Education Nepal (ITEN), कम्प्युटर शिक्षा सबैका लागि (Computer education for all)

IT Education Nepal is a volunteer-driven, non-profit initiative providing practical computer education to Nepali speakers around the world. It was founded on the principle that technology literacy is a universal right, and it works to remove the geographic, language and financial barriers that keep many learners out of IT education.

Many learners finish formal study with strong theory and very little practice in the skills the job actually asks for. ITEN exists to close that gap.

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About the Author

Jiwan Bhattarai holds the Cisco Certified Internetwork Expert (CCIE #60261) and Certified Ethical Hacker (CEH) certifications. He has worked in the international IT industry and mentors students and early-career engineers on technical skill and career growth.

He founded IT Education Nepal with one goal: develop 10,000 skilled IT engineers from Nepal for the global market. His teaching joins technical practice with mindfulness, an approach he calls “where technology meets tranquility.”

LinkedIn: www.linkedin.com/in/jiwanbhattarai

Contents

Publication Details	3
Mission and Motivation.....	5
How to Use This Guide	7
Part I. Core Networking and Services.....	11
Chapter 1. Routing and Switching, Q1 to Q14.....	12
Chapter 2. DNS and DHCP, Q15 to Q22.....	23
Part II. Security, Remote Access and Wireless	31
Chapter 3. Firewall and Security, Q23 to Q34.....	32
Chapter 4. VPN and Remote Access, Q35 to Q41	44
Chapter 5. Wireless, Q42 to Q49.....	52
Part III. NOC Operations	60
Chapter 6. Monitoring, Alerting and NOC Process, Q50 to Q64.....	61
Part IV. Cloud, Voice and the Physical Layer.....	74
Chapter 7. Cloud and Load Balancing, Q65 to Q72	75
Chapter 8. VoIP and QoS, Q73 to Q78.....	84
Chapter 9. Physical Layer and Cabling, Q79 to Q84	90
Part V. Escalation and Professional Practice.....	96
Chapter 10. Escalation, ITIL and Communication, Q85 to Q91	97
Part VI. Hosts, Supporting Protocols and IPv6.....	104
Chapter 11. Server and OS Networking, Q92 to Q96	105
Chapter 12. Miscellaneous Protocols, Q97 to Q101	111
Chapter 13. IPv6, Q102 to Q108.....	117
Appendices	
Appendix A. Study Paths.....	125
Appendix B. Questions-Only Recall Edition.....	134
Appendix C. Self-Assessment Tracker.....	142
Appendix D. Command Reference	144
Appendix E. Ports, Protocols and Markings.....	152
Appendix F. Standards Index.....	158
Appendix G. Research Foundations	165
Appendix H. Glossary	169
Appendix I. Corrections in This Edition	Error! Bookmark not defined.

How to Use This Guide

Every scenario in this book is a question you might actually be asked, or a fault you might actually be handed at 2 a.m. Each one gives you a short answer you could say out loud in an interview, an explanation of what is really happening underneath, the commands to confirm it, and a link to the standard that governs the behaviour.

It is written to be read twice. The first time you will want the short answers. The second time, with a lab open and a specific gap to close, you will want the mechanism.

Who it is for

If you are new to networking, start with the L1 study path. Every chapter begins with a plain-language orientation, and the L1 questions assume no more than CCNA-level knowledge. You will not be thrown into BGP timers on page one.

If you already work in a NOC, the L2 study path is where the interview differentiators live. You probably know the symptoms already. What gets you the job is explaining the mechanism when the interviewer asks “why?” for the second time.

If you are interviewing for senior or escalation roles, the L3 study path is short on purpose. Senior interviews test judgement and protocol-level reasoning more than recall, so those scenarios go deeper on fewer things.

Where to start

I want to...	Go to
Follow a structured study route for my level	Appendix A, Study Paths
Go straight to a topic I am weak on	Contents
Test myself without seeing answers	Appendix B, Questions-Only Recall Edition
Score my readiness honestly	Appendix C, Self-Assessment Tracker
Look up a command, a port or a standard	Appendices D, E and F
Read the research behind the mechanisms	Appendix G, Research Foundations

How to study with this book

Reading this book front to back is the least useful way to use it. Three methods, in order of value:

1. Active recall, not recognition. Use the questions-only edition. Say your answer out loud, in full sentences, as if the interviewer is sitting opposite you. Then check.

Reading an answer and thinking “yes, I knew that” is recognition. Recognition feels like knowledge and disappears the moment someone is watching you. Recall is the thing that survives an interview.

2. The “why” test. For each scenario, try to explain the *What is actually going on* part without looking. If you can name the cause but cannot explain the mechanism, that is a gap.

Interviewers almost always follow up with “why?” once. Sometimes twice. The first “why” separates people who memorised a list from people who understand the protocol.

3. Break it in a lab. No book gives you the muscle memory of reading a real interface counter during a live incident. Build the scenarios you are weakest on and break them on purpose.

Free options include Containerlab (<https://containerlab.dev/>), GNS3 (<https://www.gns3.com/>), EVE-NG (<https://www.eve-ng.net/>) and Cisco Packet Tracer (<https://www.netacad.com/courses/packet-tracer>). For capture practice, the Wireshark sample captures (<https://wiki.wireshark.org/SampleCaptures>) are free and immediately useful.

This is the one gap no written guide closes, and it is the gap that separates candidates who sound prepared from candidates who are prepared.

A note on the behavioural questions

Chapter 10 covers escalation, handover and stakeholder communication. Do not memorise those answers. Interviewers hear textbook process answers constantly and discount them immediately.

Take the principle, then rebuild the answer around an incident you actually worked, using STAR: Situation, Task, Action, Result. Prepare three or four real stories in advance and you can adapt them to almost any behavioural question. A specific account of a real Friday night outage beats a correct description of change management every single time.

Tier definitions

Interview expectations differ sharply by tier. Answering an L1 question with L3 depth wastes everyone’s time. Answering an L3 question with an L1 answer ends the interview.

Tier	Role	What is being tested	Answer shape
L1	NOC Analyst, Tier 1	Do you gather correct data, follow the runbook, and escalate cleanly?	Symptom recognition, correct first commands, knowing when to hand off
L2	NOC Engineer, Tier 2	Can you isolate a fault to a layer and a device without help?	Hypothesis, ordered diagnostics, correct fix, rollback awareness
L3	Senior, Escalation	Can you reason about protocol behaviour and design the failure out?	Mechanism-level explanation, packet-level evidence, permanent remediation

The 108 scenarios divide into 25 at L1, 73 at L2 and 10 at L3. Appendix A explains why the distribution looks like that.

Environment note

Where an answer changes between environments, the scenario says so. The difference matters more than people expect:

- **Enterprise NOC.** You own the whole stack: access switches, wireless, firewalls, servers, users. Root cause is usually yours to find and yours to fix.
- **Service provider or ISP NOC.** You own transport and edge. Customer equipment is often out of scope, most escalations point toward transit or peering partners, and your instruments are BGP state, optical light levels and circuit IDs rather than endpoint troubleshooting.
- **MSP NOC.** Many small customers, hard contractual SLAs, constant context switching. Ticket hygiene and documentation are weighted as heavily as technical depth.

Command safety labels

Every command block carries a label. Read it before you paste anything into a production device.

Label	Meaning
read-only	Displays state. Safe on production.
active test	Generates traffic or probes. Low risk, but visible in logs and may trigger security alerts.
service-affecting	Can interrupt traffic. Needs a maintenance window.
change approval	Modifies configuration. Needs change control and a rollback plan.

Debug commands are marked service-affecting even when they look harmless, because debug output on a loaded control plane has taken production routers down before.

Anatomy of a scenario

Every scenario follows the same structure, so you always know where to look.

Part	What it gives you
Tier and Safety	The interview level the question targets, and the most disruptive command label in the scenario
Short answer	What you could say out loud in the first thirty seconds
What is actually going on	The mechanism, which is what the follow-up “why?” is testing
How to check	Commands and procedures, each block marked with its safety label
Watch out for	The trap, the overlooked cause, or the detail that lifts an answer
References	The primary standard or vendor document that governs the behaviour

Scenarios in Chapter 10 end with a STAR prompt instead of a command block, because they test judgement and experience rather than syntax.

Review status and tested platforms

Last technical review: 25 July 2026.

Commands were written against the platforms below. Syntax and defaults vary by release, so verify in your own environment before relying on anything here in production.

Platform	Tested against
Cisco IOS / IOS XE	15.x, 17.x
Cisco ASA	9.x
Cisco WLC / AireOS	8.x
Linux	Ubuntu 22.04, RHEL 9
Windows Server	2019, 2022
AWS CLI	v2

Where a command is vendor-specific but has not been verified on a named release, it is labelled a **command pattern** rather than presented as exact syntax.

Conventions

Placeholders appear in angle brackets, for example <if> for an interface name and <ip> for an address. Replace them, brackets included. Addresses in examples are illustrative, and none of them belongs to a real network you should probe.

Scenario numbers run from Q1 to Q108 through the whole book, and cross-references use those numbers, for example “see Q75”. Spelling follows British English. Protocol names, command syntax and the titles of standards keep their original spelling.

Part I. Core Networking and Services

Most NOC tickets begin as “I can’t reach something”, and most of those turn out to be routing, switching or name resolution. Part I covers the reasoning that settles those questions quickly: which layer is failing, which device owns the fault, and what evidence proves it.

Chapter 1. Routing and Switching

Scenarios Q1 to Q14. Tier mix: 5 L1, 7 L2, 2 L3.

Orientation

Almost everything in a NOC starts here. When a user says “I can’t reach the server,” you are being asked a routing and switching question whether or not anyone uses those words.

Two ideas carry most of the weight:

Switching happens inside a broadcast domain. Devices in the same VLAN talk to each other using MAC addresses. No router is involved, no routing table is consulted.

Routing happens between broadcast domains. The moment the destination is in a different subnet, the sending host gives up on finding it directly, sends the frame to its default gateway, and a Layer 3 device takes over.

That single distinction is the first fork in most troubleshooting trees. If something works inside a VLAN but not between VLANs, you have just eliminated cabling, the NIC, and most of Layer 2 in one step.

Scenarios in this chapter

ID	Scenario	Tier
Q1	Cross-VLAN reachability failure	L1
Q2	Repeated STP topology changes	L2
Q3	Broadcast storm after adding a switch	L2
Q4	50% packet loss with working traceroute	L2
Q5	OSPF adjacency failure	L2
Q6	VLAN not passing over a trunk	L1
Q7	Time-correlated connectivity loss	L1
Q8	Router control plane saturation	L2
Q9	Static route covering one prefix only	L1
Q10	CRC and input errors	L1
Q11	EtherChannel underperforming	L2
Q12	Preventing routing change outages	L2
Q13	BGP prefix not received or not installed	L3
Q14	First-hop redundancy failure	L3

Q1. A user can't reach a server in another VLAN, but can reach one in the same VLAN

Tier: L1. Safety: read-only.

Short answer. Working inside the VLAN and failing across VLANs points at Layer 3 or at policy. Check that a routed interface exists for both VLANs and is up, that the client's gateway and subnet mask are right, and that no ACL is blocking the flow in either direction.

Be careful how you phrase the reasoning. Same-VLAN success tells you the NIC, the cable and the local switching path are working for *that* flow. It does not prove the whole of Layer 2 is healthy, and it certainly does not prove that policy for a different protocol and port is open.

What is actually going on. Traffic inside a VLAN is switched on MAC address and never consults a routing table. Cross a subnet boundary and the host has to ARP for its default gateway and hand the frame over to a Layer 3 device. Several things can break that chain, and they produce identical symptoms:

- The SVI or router subinterface is missing, shut down, or in line-protocol down state
- The subnet mask makes the host think the remote subnet is local, so it ARPs for a host that will never answer
- An ACL permits the return path but not the forward path, or vice versa
- The server's own gateway is wrong, so replies never come back

How to check

Cisco IOS		[read-only]
show ip interface brief include Vlan		
show vlan brief		
show ip route <destination>		
show ip access-lists		
show run interface vlan <id>		
Client		[read-only]
ipconfig /all	Windows	
ip addr ; ip route	Linux	
arp -a	is the gateway MAC resolving?	

Watch out for. An SVI stays down until at least one access port in that VLAN is up and the VLAN exists in the VLAN database. A freshly configured VLAN with nothing plugged into it is the most common false alarm here, and it catches people every time.

References: IEEE 802.1Q · RFC 826, ARP

Q2. Two switches show a spanning-tree topology change every few minutes

Tier: L2. Safety: read-only.

Short answer. Something is flapping. Find the port that owns the topology change with show spanning-tree detail, then look at that port's flap history and error counters. Usual suspects are a failing link, a duplex mismatch, or an edge port without PortFast.

What is actually going on. When a non-edge port moves into forwarding, the switch floods a Topology Change Notification toward the root bridge. The root sets the topology change flag, and every switch in the domain shortens its MAC address table ageing time from 300 seconds to the forward delay, typically 15 seconds.

The consequence is out of proportion to the cause. One flapping access port produces network-wide unicast flooding, brief traffic loss and CPU load. Ports facing end devices should carry PortFast so that a user rebooting their PC does not generate a TCN at all.

How to check

```
[read-only]
show spanning-tree detail | include ieee|occurred|from
show spanning-tree interface <if> detail
show interface <if> | include flap|error|duplex|reset
show logging | include LINK-3|LINEPROTO
```

Watch out for. `show spanning-tree detail` names the last topology change and the port it arrived on. Follow that port toward the source, switch by switch. The flapping link is usually one or two hops away from where you started looking.

References: IEEE 802.1D · IEEE 802.1Q, RSTP

Q3. After adding a new switch, users report intermittent broadcast storms

Tier: L2. Safety: read-only.

Short answer. A forwarding loop. Either the new switch was cabled with two uplinks into the same domain without STP converging properly, a port was configured as access where it should trunk, or the switch arrived with STP disabled or running an incompatible mode.

What is actually going on. Ethernet frames carry no TTL. A broadcast that enters a loop is replicated forever, saturating links and CPU within seconds, while MAC tables thrash because the same source address keeps arriving on alternating ports.

STP exists to prevent this, but it can be defeated. The new switch may run a different STP flavour. A port may be configured as an edge port and then receive BPDUs. Somebody may have disabled STP on a VLAN to “fix” a slow-linking port. BPDU Guard shuts an edge port that receives a BPDU. Loop Guard stops a blocking port moving to forwarding when BPDUs go quiet.

How to check

```
[read-only]
show spanning-tree vlan <id>
show spanning-tree inconsistentports
show interface | include rate|drops          broadcast rate near 100%?
show mac address-table | include <flapping mac>
show logging | include MAC_MOVE|LOOP|BPDUGUARD
```

Watch out for. %SW_MATM-4-MACFLAP_NOTIF in the log, plus a MAC address alternating between two ports, is a loop until proven otherwise. Do not spend twenty minutes on other theories when that message is sitting in the log.

Reference: IEEE 802.1D

Q4. A route works via traceroute but pings drop 50% of the time

Tier: L2. Safety: active test.

Short answer. Two strong candidates. Either equal-cost multipath where one of the two paths is broken or filtered, or a genuinely lossy link. Run MTR from both ends and compare, then check interface error counters along the suspect direction.

What is actually going on. Loss sitting close to a clean 50% is a good hint at ECMP with one broken path. Traceroute keeps working because it only needs one probe in three to come back per hop, which hides the failure nicely.

Loss that varies randomly rather than sitting near 50% points somewhere else: CRC errors, an oversubscribed link dropping from the tail of a queue, or a failing optic. The distinction matters because one is fixed in routing and the other needs someone with a ladder.

How to check

Cisco IOS	[read-only]
show ip route <dest>	two next hops means ECMP
show ip cef exact-route <src> <dest>	
show interface <if> include errors CRC drops input rate	
From a host	[active test]
mtr -rwc 100 <dest>	run from both ends
hping3 -S -p 443 <dest>	TCP test, not ICMP

Watch out for. ICMP is often rate-limited or deprioritised on router control planes, so a router that drops your pings may be forwarding transit traffic perfectly. Confirm with a TCP-based test before you tell anyone there is an outage.

References: RFC 792, ICMP · RFC 6349, TCP Throughput Testing

Q5. OSPF neighbors won't form between two routers on the same subnet

Tier: L2. Safety: read-only.

Short answer. Work the checklist in order: same subnet and mask, matching area ID, matching hello and dead timers, matching authentication, matching MTU, matching stub flags, unique router IDs, interface not passive, and no ACL blocking 224.0.0.5 and 224.0.0.6.

What is actually going on. The adjacency state tells you the cause, which is why the mechanism matters more than the checklist.

Stuck at	Usual cause
Down / Init	Hellos are one-way. ACL, passive interface, or one side not sending.
2-Way	Normal between two DROthers on a broadcast segment. Not a fault.
ExStart / Exchange	Almost always MTU mismatch. Database Description packets carry MTU and are rejected when it differs.
Never forms at all	Mismatched hello or dead timers, area ID, or authentication.

How to check

show ip ospf neighbor	[read-only]
show ip ospf interface <if>	the STATE names the problem
show ip access-lists	area, timers, MTU, network type, auth
	permit 224.0.0.5 and 224.0.0.6
debug ip ospf adj	[service-affecting]
	use with care on a busy router

Watch out for. Interviewers ask this often and expect the state-to-cause mapping, not a recited parameter list. “Stuck in ExStart means MTU” is the answer that separates an L2 candidate from an L1 one.

Reference: RFC 2328, OSPF Version 2

Q6. A newly added VLAN isn’t passing traffic across a trunk link

Tier: L1. Safety: read-only.

Short answer. The VLAN is missing from the trunk’s allowed list, not created on one of the switches, pruned, or hit by a native VLAN mismatch. Check both ends of the trunk, not just the one you configured.

What is actually going on. A trunk only carries VLANs in its allowed list, and the VLAN has to exist and be active on both switches. Two details catch people out repeatedly:

switchport trunk allowed vlan 20 **replaces** the list. It does not add to it. One mistyped command can black-hole every other VLAN on that trunk, and it happens in production more often than anyone admits.

A native VLAN mismatch is worse than an outage. Untagged frames from one side land in a different VLAN on the other, merging two broadcast domains. That is a security problem and a loop risk rather than a clean failure. CDP or LLDP logs it.

How to check

show interface trunk	[read-only]
show vlan brief	both ends, compare allowed and active
show interface <if> switchport	
show logging include NATIVE_VLAN_MISMATCH	

Watch out for. Always use `switchport trunk allowed vlan add <id>` on a production trunk. The version without `add` is one of the most common causes of self-inflicted trunk outages.

Reference: IEEE 802.1Q

Q7. Users on one floor lose connectivity randomly, correlating with cleaning staff hours

Tier: L1. Safety: read-only.

Short answer. Physical disturbance. Match interface flap timestamps against the reported times, see which ports flap together, then inspect cabling and any shared power outlet feeding a floor switch or media converter.

What is actually going on. The time correlation does most of the diagnostic work. When several ports on one switch flap together at a repeating hour, the shared cause is usually power (an IDF switch plugged into a socket a cleaner can reach) or physical (a patch cord under a desk, a cable across a doorway).

One port flapping randomly is a cable or a NIC. Several ports flapping in unison is something upstream: power, the uplink, or the switch itself.

How to check

```
[read-only]
show interface <if> | include last input|flapped|reset
show logging | include LINK-3-UPDOWN
show interface counters errors
show power inline                PoE devices resetting?
show interface status err-disabled
```

Watch out for. Ask facilities before you ask engineering. A vacuum cleaner sharing a circuit with an unprotected IDF switch is a real and recurring root cause, and no amount of `show` output will ever reveal it.

Reference: ANSI/TIA Standards

Q8. Router CPU spikes to 99% during peak hours

Tier: L2. Safety: read-only.

Short answer. Find the consuming process with `show processes cpu sorted`, then work out whether the load is punted traffic or genuine control plane work. Confirm hardware or CEF switching has not fallen back to software.

What is actually going on. On a healthy router, transit traffic is switched in hardware or by CEF and barely touches the CPU.

High CPU with high **interrupt** time means packets are being punted to the CPU. Causes include an unsupported feature in the data path, an ACL with `log` on a busy rule, IP options, fragmentation, TTL expiry from a traceroute storm, or a TCAM that has overflowed and pushed forwarding into software.

High CPU in a **named process** is control plane work instead: OSPF or BGP recalculating after flaps, SNMP walking a large table, or NetFlow export.

How to check

```
[read-only]
show processes cpu sorted | exclude 0.00
show processes cpu history          5-second, 1-minute, 72-hour graphs
show ip cef                        is CEF enabled and populated?
show ip traffic
show controllers cpu-interface      punted packet counters
```

Watch out for. show processes cpu history is the fastest way to prove the spike is periodic. A spike that lands on the same clock minute every day is almost always a scheduled job: a backup, an SNMP discovery sweep, or a vulnerability scanner.

Reference: RFC 6192, Protecting the Router Control Plane

Q9. A static route works for one subnet but not its neighbour subnet

Tier: L1. Safety: read-only.

Short answer. A missing route or a wrong mask for the second prefix. Check the routing table for the exact destination address rather than reading the configuration and reasoning about it.

What is actually going on. Longest prefix match means a route for 10.1.1.0/24 tells you nothing at all about 10.1.2.0/24. The usual errors are a /24 typed where the design called for a /23, a summary route that does not actually cover the second subnet, or a next hop that has no route of its own so the recursive lookup fails.

A static route pointing at an interface on a multi-access segment also depends on proxy ARP, and fails quietly when proxy ARP is disabled.

How to check

```
[read-only]
show ip route <specific destination ip>
show ip route static
show ip route <prefix> longer-prefixes
ping <next-hop>                is the next hop itself reachable?
```

Watch out for. Always test with the exact destination address. show ip route 10.1.2.15 returns the route that will actually be used, which is a different question from what the config appears to say.

Reference: RFC 1812, Requirements for IPv4 Routers

Q10. Interface counters show high input errors and CRC errors

Tier: L1. Safety: read-only.

Short answer. Layer 1. CRC errors mean frames arrived corrupted: bad cable, bad or dirty transceiver, interference, or a duplex mismatch. Confirm speed and duplex match on both ends, then swap the cable and the optic in that order.

What is actually going on. The counter pattern tells you which fault you have:

Pattern	Likely cause
CRC, no collisions	Cable, connector, optic, or interference
Late collisions on one end, CRC errors and runts on the other	Duplex mismatch. The full-duplex side transmits whenever it likes. The half-duplex side counts late collisions; the full-duplex side counts CRC errors and runts from the frames the half-duplex side abandoned. Compare the counters on both ends. Link light stays green while throughput collapses.
Input errors without CRC	Runts, giants, or framing problems. Often an MTU or tagging mismatch.
Rising alignment errors	A cable that is failing rather than failed

How to check

	[read-only]
show interface <if>	CRC, runts, giants, late collisions
show interface <if> counters errors	
show interface <if> status	negotiated speed and duplex
show interface transceiver detail	optical Tx and Rx in dBm
	[read-only, resets counters]
clear counters <if>	then re-measure over a known window

Watch out for. Clear the counters before you draw conclusions. An interface up for 400 days may be showing errors accumulated during an incident someone fixed a year ago. Rate of increase matters. The total does not.

Reference: IEEE 802.3

Q11. Users behind an EtherChannel report slow throughput despite it showing “up”

Tier: L2. Safety: read-only.

Short answer. Confirm every member link is actually bundled and error-free, then look at the load-balancing hash. A bundle where one member carries everything gives you single-link throughput no matter how many members are up.

What is actually going on. A port channel does not stripe packets across members. It hashes selected header fields to pick one member per flow, which keeps packets in order but means a few large flows can all land on the same link.

The default hash on many platforms is source MAC only, which is close to useless when all traffic arrives via a router and therefore shares one source MAC. Separately, LACP keeps a bundle “up” while

individual members are suspended for mismatched speed, duplex or VLAN config, and one member with rising CRC errors drags the whole bundle down without taking it offline.

How to check

```

show etherchannel summary          [read-only]
                                   flags: P bundled, s suspended, I individual
show etherchannel load-balance
show interface port-channel<n> | include rate
show interface counters errors    compare members against each other
test etherchannel load-balance interface po1 ip <src> <dst>

```

Watch out for. Flag I (individual) means a member is forwarding outside the bundle. That is a loop risk, not just a performance problem. Move the hash to src-dst-ip or src-dst-port where the platform supports it.

Reference: IEEE 802.1AX, Link Aggregation

Q12. A default route was removed accidentally and internet access failed network-wide

Tier: L2. Safety: change approval.

Short answer. Three layers of defence. Change control with peer review for any routing change. Redundancy so no single route deletion is fatal, using a floating static or a dynamically learned default. And configuration versioning so rollback takes seconds instead of reconstruction from memory.

What is actually going on. The technical failure is trivial. The process failure is the real answer.

Mature environments add configuration change logging so you can see every command and who typed it, a pre-change snapshot, and a revert timer where the platform supports one. Junos has `commit confirmed`. IOS XE has `configure terminal revert timer`. Both undo the change automatically if you lose access or forget to confirm, which is exactly what happens when a change goes wrong.

How to harden

```

Cisco IOS                                [change approval]
  archive
    path <secure-transfer-target>
    write-memory
    log config
      logging enable
      notify syslog

  configure replace flash:pre-change.cfg force
  configure terminal revert timer 5      IOS XE

Junos                                     [change approval]
  commit confirmed 5
  rollback 1

```

Watch out for. Do not back up configurations over TFTP. It has no authentication and no encryption, and device configs contain SNMP strings, pre-shared keys and hashed credentials. Use SCP, SFTP, or HTTPS to an access-controlled store, and treat the backup repository as a secret store rather than a file share.

In a service provider environment the default route usually arrives via BGP, so the equivalent mistake is a mis-scoped route-map or prefix-list rather than a deleted static. The remediation is identical.

References: ITIL 4 Change Enablement · RFC 4271, BGP-4

Q13. A BGP peer is up, but a prefix you expect is missing

Tier: L3. Safety: read-only.

Short answer. Establish where the prefix stops. Is the neighbour sending it, are you receiving it, are you accepting it after inbound policy, and is it winning best-path selection? Those are four different failures and each has a different fix.

What is actually going on. “The peer is up” only means the TCP session and the BGP finite state machine reached Established. Everything interesting happens after that.

Walk it in order:

1. **Is it being advertised?** The neighbour’s outbound policy may filter it, or they may not have it in their table either.
2. **Is it arriving?** `show bgp neighbor <ip> received-routes` needs soft reconfiguration inbound or route refresh, otherwise you are looking at post-policy routes and drawing the wrong conclusion.
3. **Is inbound policy dropping it?** A prefix-list, route-map, AS-path filter, or an ORF can silently discard it.
4. **Did it lose best-path?** It may be in the BGP table but not the RIB, beaten by a shorter AS path, higher local preference, or a lower-origin route from elsewhere.
5. **Is the next hop reachable?** A prefix with an unresolvable next hop is valid but never installed. This one catches people constantly in iBGP designs without next-hop-self.
6. **Did you hit max-prefix?** Some implementations warn and keep going; others tear the session down. Check the log.

How to check

[read-only]	
<code>show bgp ipv4 unicast summary</code>	
<code>show bgp neighbor <ip> received-routes</code>	needs soft-reconfig or refresh
<code>show bgp neighbor <ip> routes</code>	post-policy
<code>show bgp neighbor <ip> advertised-routes</code>	what you send them
<code>show bgp ipv4 unicast <prefix></code>	why did it win or lose?
<code>show ip route <prefix></code>	
<code>show bgp neighbor <ip> include prefix max Notification</code>	

Watch out for. If you cannot see received-routes at all, you are probably missing soft-reconfiguration inbound or route refresh capability, and you are reasoning from incomplete data. Fix your visibility before you theorise about policy.

In an ISP NOC, this scenario often ends at the peer rather than at you, so know your route-server and looking-glass options in advance.

References: RFC 4271, BGP-4 · RFC 7454, BGP Operations and Security

Q14. A gateway failover didn't happen when the primary router failed

Tier: L3. Safety: read-only.

Short answer. First-hop redundancy failed to trigger or failed to complete. Check whether the standby actually saw the primary disappear, whether tracking was configured for the thing that actually broke, and whether the switching layer relearned the virtual MAC.

What is actually going on. HSRP, VRRP and GLBP protect against the *router* dying. They do not, by default, protect against the router's uplink dying, which is a far more common failure. If the primary keeps sending hellos while its WAN interface is down, the standby has no reason to take over and traffic disappears into a healthy-looking router.

Object tracking is what closes that gap. You track the uplink, or better, an IP SLA probe that tests real reachability, and decrement the priority when it fails. BFD gives you sub-second detection where the hardware supports it, which matters because default HSRP hold times are measured in seconds and voice notices.

The other half of the failure is Layer 2. After a failover, the new active router must gratuitously ARP so hosts and switches relearn the virtual MAC. If those frames are lost or filtered, hosts keep sending to the old MAC for the duration of their ARP cache.

How to check

```
show standby brief                                [read-only]
show vrrp brief                                  HSRP state, priority, preempt
show standby <if> | include Track|Priority|Preempt
show track                                       is anything actually tracked?
show ip sla statistics
show bfd neighbors detail
show logging | include HSRP|VRRP|TRACK

On the access switch                             [read-only]
show mac address-table address <virtual-mac>
```

Watch out for. Check preempt. Without it, a recovered primary never takes back the active role, which is fine until the day the standby fails and everyone discovers there was no redundancy left.

Also confirm the tracked object matches the real risk. Tracking the local interface line protocol misses a failure two hops upstream, and that is exactly the failure that takes your internet away.

References: RFC 5798, VRRPv3 · RFC 5880, Bidirectional Forwarding Detection

Chapter 2. DNS and DHCP

Scenarios Q15 to Q22. Tier mix: 4 L1, 4 L2.

Orientation

DNS and DHCP are the two services that make a network usable by humans rather than by people who memorise IP addresses. They break in ways that look like network faults but are not, which is why they generate so many misrouted tickets.

DHCP hands out addresses. A client shouts into the broadcast domain, a server answers, and the client leases an address for a set period. Because the request is a broadcast, it does not cross a router on its own. Something has to relay it.

DNS turns names into addresses. Your client asks a resolver, the resolver either knows the answer, has it cached, or goes and finds it by walking down from the root servers. Almost every DNS problem is really a question of *which* server answered and *when it cached that answer*.

One habit will save you hours: whenever a user reports something broken, test by IP address as well as by name. That single comparison tells you whether you are looking at a name resolution problem or a network problem.

Scenarios in this chapter

ID	Scenario	Tier
Q15	Name resolution failing, IP working	L1
Q16	External resolution intermittent	L2
Q17	DHCP failure on one client	L1
Q18	Scope exhaustion	L2
Q19	Single zone failing	L2
Q20	Link-local self-assignment	L1
Q21	Dynamic DNS registration race	L2
Q22	TTL and caching	L1

Q15. Users can reach IPs directly but not by hostname

Tier: L1. Safety: read-only.

Short answer. Reaching the same service on the same port by IP makes name resolution the leading hypothesis. Confirm the client's resolver settings, confirm that resolver is reachable on both UDP and TCP 53, then query the DNS server directly to separate a client problem from a server problem.

Say "leading hypothesis," not "proves." A successful connection by IP tells you that one flow, on one protocol and port, worked. It does not clear NAT, proxy policy, or application-layer filtering for

anything else, and it definitely does not clear the path your DNS queries take, which may be entirely different.

What is actually going on. Resolution has several independent failure points:

- The client's configured resolver, usually handed out by DHCP, so a scope option error breaks a whole subnet at once
- The resolver's cache, holding a stale or negative answer
- The DNS search suffix list, so a short name never becomes the FQDN you expect
- The local hosts file, which quietly overrides everything
- The upstream forwarder chain

Querying the authoritative server directly bypasses every cache and answers one question cleanly: does this record exist at all?

How to check

# Client configuration	[read-only]
ipconfig /all	# Windows
resolvectl status	# Linux, systemd-resolved
cat /etc/resolv.conf	# Linux, classic
# Split the problem in half	[active test]
nslookup host.example.com	# via the configured resolver
nslookup host.example.com 8.8.8.8	# bypassing it
dig @<internal-dns> host.example.com +norecurse	
dig host.example.com +trace	# walk down from the root
# Is port 53 reachable at all?	[active test]
nc -vz <dns-server> 53	# TCP

Watch out for. If `dig @<server>` works but the client fails, the problem is the client resolver, the search suffix, or a local cache. If `dig @<server>` also fails, the problem is the zone or the server. That single test splits the problem space in half, so make it your first move rather than your fifth.

References: RFC 1034 · RFC 1035, Domain Names

Q16. Internal DNS resolves fine but external domains fail intermittently

Tier: L2. Safety: active test.

Short answer. Internal zones are answered locally and never leave the building, so the fault sits somewhere in the recursion path. Several hypotheses fit, and you need evidence to choose between them rather than assuming one.

What is actually going on. Treat these as competing hypotheses, not a diagnosis:

Hypothesis	What it looks like	How to test
Forwarder failure or failover	Slow answers, then success; alternating results	Query each forwarder individually

Hypothesis	What it looks like	How to test
DNSSEC validation failure	SERVFAIL for signed zones only, consistent per domain	Compare with and without +cd
Packet loss on the path	Random, affects all external names equally	MTR to the forwarder
Rate limiting or query quota	Fails under load, recovers when quiet	Check resolver and firewall logs
Large response or EDNS problem	Small answers fine, large answers fail	Compare a small query with a large one
TCP 53 blocked	Fails only where fallback to TCP is needed	Query with +tcp

The last two are worth understanding together. DNS classically limits UDP responses to 512 bytes and falls back to TCP for anything larger. EDNS0 negotiates bigger UDP payloads, which some middleboxes then fragment or drop. Plenty of firewall policies permit UDP 53 and quietly omit TCP 53, and that produces exactly this “internal fine, external flaky” pattern.

How to check

# [active test]	
dig @<forwarder> example.com +short	
dig @<forwarder> example.com +tcp	# does TCP 53 work?
dig @<forwarder> example.com +bufsize=4096	# large EDNS0 response
dig @<forwarder> <signed-domain> +dnssec	# validation behaviour
dig @<forwarder> <signed-domain> +cd	# checking disabled
nc -vz <forwarder> 53	# TCP reachability
mtr -rwc 100 <forwarder>	# loss on the path
[read-only]	
Windows Server DNS	
Get-DnsServerForwarder	
Event Viewer > DNS Server log	

Watch out for. Comparing a query answered with +dnssec against the same query with +cd tells you whether validation is the cause, because +cd asks the resolver to skip validation. Do not use a domain that is deliberately broken as a general size test. Test size with +bufsize, and test validation with a real signed zone.

References: RFC 6891, EDNS(0) · RFC 7766, DNS over TCP · RFC 4035, DNSSEC Protocol

Q17. A new laptop can't get an IP address on the network

Tier: L1. Safety: read-only.

Short answer. Walk the DHCP path in order. Is the port up and in the right VLAN, is port security or 802.1X blocking it, does the gateway have an ip helper-address, and does the scope have free leases? If all four pass, capture at the switch.

What is actually going on. DHCP discovery is a broadcast, so it does not cross a router without a relay agent forwarding it as unicast to the server. On a brand new VLAN, the missing helper address is the classic cause.

On an existing VLAN where everyone else is fine, the cause is local to that port or that client. Port security may have err-disabled it after a MAC violation. 802.1X authentication may have failed and left the port unauthorised. The port may simply be in the wrong VLAN.

An exhausted scope affects every new client, not just one laptop, which is a useful way to tell the two apart quickly.

How to check

Switch	[read-only]
show interface <if> status	connected? err-disabled?
show port-security interface <if>	
show authentication sessions interface <if> details	
show interface <if> switchport	correct VLAN?
Router or SVI	[read-only]
show run interface vlan <id> include helper	
DHCP server	[read-only]
Get-DhcpServerv4ScopeStatistics -ComputerName <srv>	Windows
journalctl -u kea-dhcp4	Kea
dhcp-lease-list	ISC
Client	[active test]
ipconfig /release ; ipconfig /renew	
dhclient -v <interface>	

Watch out for. DHCP snooping enabled on the access switch without a trusted uplink port blocks every legitimate OFFER. It is a common self-inflicted cause after a security hardening project, and it looks exactly like a dead DHCP server.

References: RFC 2131, DHCP · RFC 3046, DHCP Relay Agent Information Option

Q18. A branch office DHCP scope is exhausted despite low headcount

Tier: L2. Safety: read-only.

Short answer. Addresses are being consumed faster than they are released. Look at the lease table before you look at anything else, because the pattern in it names the cause.

What is actually going on. A lease is held for its full duration whether or not the device is still there. In a site with guest phones, contractor laptops and roaming mobiles, an eight-day lease against a /24 can exhaust with only thirty staff.

The obvious explanation, a chatty client draining the pool, does not hold up. A device that repeatedly sends DISCOVER using the *same* client identifier normally gets offered the *same* address each time. Repeat requests alone do not consume the pool.

Scope exhaustion needs addresses to keep being allocated to apparently new clients, which means one of:

- **Randomised MAC addresses.** Modern phones and laptops present a randomised MAC address to each network by default. On most platforms that address stays stable for a given

SSID, but rotating modes, forgetting and rejoining the network, and periodic re-randomisation on some operating systems all make one device look like several. In guest and BYOD networks this is now a leading cause.

- **A changing client identifier.** Some clients send a DHCP client-ID that varies across reboots or firmware states, so the server treats each request as a new device.
- **Deliberate spoofing.** A starvation attack cycling through fabricated MAC addresses.
- **Stale leases plus genuine churn.** Not exhaustion by itself, but it shrinks the usable pool until normal churn overflows it.

How to check

```
Windows [read-only]
Get-DhcpServerv4Scope
Get-DhcpServerv4ScopeStatistics
Get-DhcpServerv4Lease -ScopeId <x> | Group-Object ClientId

ISC DHCP / Kea [read-only]
dhcp-lease-list --all
grep -c "DHCPACK" /var/log/syslog

Read the lease table for
- many leases, each with a distinct and unfamiliar MAC prefix
- lease start times clustered into one short window
- the same hostname appearing against several addresses
```

Watch out for. Before you extend the subnet, work out which of the causes above you have. Randomised MAC addresses call for scope sizing plus shorter leases. A starvation attack calls for DHCP snooping with rate limiting and port security. Those are very different responses to an identical-looking symptom.

Reference: RFC 2131, DHCP

Q19. A DNS server is up, but one specific domain always times out

Tier: L2. Safety: active test.

Short answer. The problem is scoped to that zone, not to the server. Either the domain's authoritative servers are unreachable from your resolver, a conditional forwarder for that zone points somewhere broken, or a delegation is stale. Walk the delegation and query the authority directly.

What is actually going on. A resolver answering every other query is healthy. A timeout on one zone means recursion for that zone is failing, and `dig +trace` shows exactly where by walking from the root through each delegation.

Common causes: a conditional forwarder still pointing at a domain controller that was decommissioned after a merger, a partner's firewall newly blocking your resolver's source address, a lame delegation where the parent names a server that is not authoritative, or a zone whose last remaining nameserver is offline.

How to check

```

# [active test]
dig example.com +trace          # where does the chain break?
dig NS example.com
dig @<authoritative-ns> example.com # query the authority directly
dig @<your-resolver> example.com +norecurse
nc -vz <authoritative-ns> 53

Windows Server                  [read-only]
  Get-DnsServerZone
  Get-DnsServerZone -Name example.com | Select MasterServers

```

Watch out for. `dig +trace` is one of the most useful DNS commands you can learn, and surprisingly few L1 candidates know it. Saying “I would run `dig +trace` and see which delegation stops responding” signals operational experience, because it describes what experienced engineers actually do.

References: RFC 1034 · RFC 8499, DNS Terminology

Q20. Devices are getting APIPA (169.254.x.x) addresses

Tier: L1. Safety: read-only.

Short answer. The client did not obtain a usable DHCP configuration, so it self-assigned a link-local address. Establish the blast radius first, then work the DHCP path: service, relay, and the network between them.

What is actually going on. A host that fails DHCP self-assigns from 169.254.0.0/16 under the link-local specification. That address works only on the local link and carries no gateway, which is why affected users can sometimes see each other and nothing else.

Do not over-read it. APIPA proves that no usable DHCP configuration was obtained. It does not prove the client’s stack is healthy. A host firewall blocking UDP 68, a broken or disabled DHCP client service, a NIC driver problem, a failed 802.1X supplicant, or the port sitting in the wrong VLAN all produce the same result.

How to check

```

Start with blast radius, because it points you at the layer
One host      -> port, VLAN, port security, 802.1X, host firewall, driver
One VLAN      -> ip helper-address, relay config, that subnet's scope
Everyone      -> DHCP service down, or scope database problem

Commands                                            [read-only]
show run interface vlan <id> | include helper
show ip dhcp binding                               if the switch is the DHCP server
Get-Service DHCPServer                             Windows
systemctl status isc-dhcp-server                   Linux

                                                    [active test]
tcpdump -i <if> -n port 67 or port 68
Do DISCOVERs leave the client?
Do they arrive at the server?
Does an OFFER come back and get dropped on the way?

```

Watch out for. Establishing the blast radius before touching anything cuts the search space immediately, and it is the structured habit an L1 role depends on. A capture at the client and at the server, compared, tells you which direction is failing.

Reference: RFC 3927, IPv4 Link-Local Addresses

Q21. A dual-homed server intermittently registers the wrong IP in DNS

Tier: L2. Safety: read-only.

Short answer. Both NICs are performing dynamic DNS registration and overwriting each other, so the A record alternates. Disable registration on the non-primary interface, or set interface metrics so one path is deterministically preferred.

What is actually going on. Under dynamic update, a host registers whatever address it believes is primary. With two NICs, each can register its own address and overwrite the other, producing a record that changes on every registration cycle. Clients cache whichever value they happened to receive, so roughly half of them break at any moment.

That is why the symptom is intermittent. The record is correct about half the time, which makes it maddening to reproduce on demand.

How to check

```
# Windows [read-only]
Get-DnsClient | Select InterfaceAlias, RegisterThisConnectionsAddress
Get-NetIPInterface | Select InterfaceAlias, InterfaceMetric

# Fix [change approval]
Set-DnsClient -InterfaceAlias "NIC2" -RegisterThisConnectionsAddress $false

On the DNS server [read-only]
Get-DnsServerResourceRecord -ZoneName <zone> -Name <host>
Check the timestamp. A record rewritten every few minutes confirms the race.
```

Watch out for. The same root cause produces intermittent Kerberos and service principal failures on multi-homed domain controllers and SQL servers, which is a much more expensive symptom than a wrong A record. Mention that downstream impact, because it is the part of the blast radius that costs most.

Reference: RFC 2136, Dynamic Updates in the DNS

Q22. After a DNS record change, some users still resolve the old IP

Tier: L1. Safety: read-only.

Short answer. Caching. The old answer is still held by client resolvers, intermediate forwarders and downstream ISP resolvers until the TTL expires. Flush what you control and wait out the TTL for everything else.

What is actually going on. Every cached answer carries the TTL that was in force *when it was cached*, not the TTL you set afterwards. Lowering the TTL after publishing a change does nothing for resolvers that already hold the old record.

That is why planned migrations lower the TTL to 300 seconds or less at least one full old-TTL period *before* cutover, then raise it again once things settle. Browsers and application runtimes add their own caching on top of the OS resolver, and some ignore TTL entirely.

How to check

```
# [read-only]
dig host.example.com           # watch the TTL count down
dig @<authoritative-ns> host.example.com # what the truth actually is
dig @8.8.8.8 host.example.com   # what a public resolver still holds

# Flush                               [read-only / low risk]
ipconfig /flushdns                # Windows client
resolvectl flush-caches          # Linux, systemd
sudo dscacheutil -flushcache     # macOS
Clear-DnsServerCache            # Windows DNS server
rndc flush                      # BIND
```

Watch out for. The mature answer includes the planning, not just the flush command. “We drop TTL to 300 seconds a day before the migration window, cut over, verify, then restore it” is a change management answer rather than a troubleshooting one.

References: RFC 1035 · RFC 8767, Serving Stale Data

Part II. Security, Remote Access and Wireless

Security controls, tunnels and radio links all fail in ways that look like something else. A firewall drop looks like an outage, a tunnel that is up can still pass nothing, and a strong Wi-Fi signal can still mean poor performance. Part II teaches you to find the stage where traffic actually stops.

Chapter 3. Firewall and Security

Scenarios Q23 to Q34. Tier mix: 9 L2, 3 L3.

Orientation

A firewall decides whether a packet is allowed through. That sounds simple until you realise a modern firewall makes that decision in several stages, and a packet can die at any of them.

Roughly, a packet passes through: routing lookup, NAT, session lookup, access policy, and then inspection engines like IPS, URL filtering and TLS decryption. Your job when something is blocked is to find out *which stage* dropped it, because “the firewall is blocking it” is not a diagnosis anyone can act on.

Two concepts worth understanding before you go further:

Stateful means the firewall remembers. Once a session is allowed, return traffic is permitted automatically because the firewall tracks the connection. This is why asymmetric routing is so painful behind firewalls: the return packet arrives at a device that has no record of the session and drops it.

Order matters. Access lists are evaluated top to bottom and stop at the first match. A permit rule at line 400 is irrelevant if a deny at line 12 matched first.

Most vendors give you a tool that answers the “which stage” question in one command. Learn yours. It will save you more time than anything else in this chapter.

Scenarios in this chapter

ID	Scenario	Tier
Q23	Policy evaluation order	L2
Q24	Calls dropping at a fixed interval	L2
Q25	Web filtering and decryption	L2
Q26	Scan or attack triage	L2
Q27	IPsec phase and proxy IDs	L2
Q28	NAT and publishing	L2
Q29	802.1X and posture	L2
Q30	Credential attack response	L2
Q31	Inspection capacity	L3
Q32	Change control failure	L2
Q33	Large transfers failing	L3
Q34	Interoperability	L3

Q23. Traffic is allowed by an ACL but still blocked

Tier: L2. Safety: read-only.

Short answer. Look for an earlier matching deny, since evaluation stops at the first match. Then check NAT and routing, and look for a second enforcement layer: zone policy, an object-group change, stateful session state, or the implicit deny at the end of a list applied in the other direction.

What is actually going on. “Allowed by an ACL” is a statement about one rule, not about the outcome. Packets can die in several other places:

- A higher rule in the same list matched first
- An ACL on the other interface, or the other direction, dropped it
- NAT rewrote the address so it no longer matches your permit
- Zone-based policy evaluated after the interface ACL denied it
- On a stateful device, a return packet with no matching session entry was dropped, which is what makes asymmetric routing so miserable behind a firewall pair

How to check

```
Cisco ASA [read-only]
packet-tracer input inside tcp 10.1.1.10 1234 10.2.2.20 443 detailed
show access-list <name> | include hitcnt
show conn address <ip>
show nat detail
show asp drop accelerated path drop reasons

Palo Alto [read-only]
test security-policy-match from trust to untrust source <ip> \
  destination <ip> protocol 6 destination-port 443
show session all filter source <ip>

Fortinet [service-affecting]
diagnose debug flow filter addr <ip>
diagnose debug flow trace start 20
```

Watch out for. packet-tracer on ASA and test security-policy-match on Palo Alto answer this question definitively by naming the exact phase and rule that dropped the packet. Knowing your platform’s version of that tool by name is a strong signal in an interview.

Reference: RFC 2979, Behavior of and Requirements for Internet Firewalls

Q24. After a firewall rule update, VoIP calls drop after about 30 seconds

Tier: L2. Safety: read-only.

Short answer. Start with the SIP dialogue, not with timers. Capture the signalling and find out which side sends BYE, or whether a transaction failed. Calls that fail at roughly 30 seconds most often point at a SIP transaction problem, commonly a missing ACK after the 200 OK, which is exactly the kind of thing a SIP ALG change causes.

What is actually going on. It is tempting to blame a UDP idle timeout, but think about whether that mechanism fits. Once a call is established, RTP flows continuously in both directions, and that traffic refreshes the session state. An idle timeout should not fire under a healthy bidirectional media stream.

What does fit a ~30 second failure:

- **A missing ACK to the 200 OK.** The caller answers, the server retransmits its 200 OK, gets no ACK, and eventually gives up and tears the call down. If your firewall's SIP ALG rewrites addresses in the signalling incorrectly, the ACK goes somewhere unreachable.
- **A broken or newly enabled SIP ALG.** Enabling it and disabling it both break calls, in different ways, which is why blanket advice about ALGs is useless.
- **One-way or absent media plus a media-inactivity timer.** If RTP only flows one way, the reverse pinhole can idle out and a media inactivity supervision timer can end the call. Here the UDP timeout genuinely does matter, but as a second-order effect of the media problem, not as the root cause.
- **Session or registration timers set unusually low.** Less likely at 30 seconds, but check.

How to check

Capture and read the SIP dialogue first	[read-only]
Who sends BYE, and with what Reason header?	
Is there a 200 OK with no matching ACK?	
Are the SDP addresses in the messages reachable from the other side?	
Cisco	[service-affecting]
debug ccsip messages	
show sip-ua connections	
show voip rtp connections	
Cisco ASA	[read-only]
show service-policy inspect sip	
show conn protocol udp	
show run timeout	
Media check	[active test]
tcpdump -i <if> -n udp portrange 16384-32768	
Is RTP flowing in both directions, or only one?	

Watch out for. Timing that lands on a precise, repeating value is a timer somewhere. Random drops are congestion or loss. The distinction is worth stating, but do not let it push you into guessing *which* timer before you have read the signalling.

References: RFC 3261, SIP · RFC 4787, NAT UDP Requirements

Q25. Users can't access a site that used to work, now blocked by URL filtering

Tier: L2. Safety: read-only.

Short answer. Read the logs to find which rule and which category produced the block. Check whether a category database update reclassified the site. Then rule out a TLS decryption failure, which looks like a block but is not one.

What is actually going on. Three different failures look identical to the user:

A **genuine policy block** logs a category and a rule ID. Straightforward.

A **reclassification** means the vendor's database moved the domain, often after it changed ownership or started hosting user content. The fix is a recategorisation request to the vendor, not a local rule.

A **decryption failure** is the sneaky one. If the site pins its certificate, requires a client certificate, or negotiates something the inspection engine cannot handle, the connection fails in a way that resembles a block but appears in the decryption log as an error rather than in the policy log as a hit.

How to check

In this order	[read-only]
1. Filter or threat log for that source IP at that time	
-> rule name, category, action	
2. Vendor category lookup for the URL	
3. Decryption log for the same session	
-> pinned certificate, unsupported cipher, expired CA	
4. Test with decryption bypassed for that destination	
Palo Alto	
show log url query equal "(addr.src eq <ip>)"	
show log decryption	
Fortinet	[service-affecting]
diagnose debug application urlfilter -1	

Watch out for. Certificate-pinned applications, including many banking and update clients, must go on a decryption exclusion list. They will never work through TLS inspection no matter what the URL policy says. Recognising that pattern saves hours.

Reference: RFC 8446, TLS 1.3

Q26. A perimeter firewall shows a spike in denied connections from one external IP

Tier: L2. Safety: read-only.

Short answer. Characterise before you react. Work out the pattern, check whether anything was *permitted* rather than denied, apply a temporary block or rate limit with an expiry, and escalate through the security incident process with the evidence attached.

What is actually going on. Internet-facing firewalls log denied traffic continuously. Background scanning is constant and mostly noise. What lifts it from noise to incident is any of:

- A permit alongside the denials, meaning something got through
- A scan walking your address space in order, meaning targeted reconnaissance
- A source that changes technique after being blocked, meaning a human is present
- A match against threat intelligence

The mistake to avoid is a permanent block from a single spike. Attacker infrastructure is disposable. Legitimate shared addresses, like a partner's NAT gateway or a cloud egress IP, are not.

How to check

	[read-only]
1. Denied vs permitted from that source in the same window	
2. Destination spread	many hosts = horizontal scan
3. Port spread	many ports on one host = vertical scan
4. Rate and timing	steady = automated, bursty = interactive
5. Threat intel and WHOIS on the source	
6. Correlate with IPS and authentication logs for the same window	
Then	[change approval]
- temporary block or rate limit, with an expiry date	
- security incident opened with the log excerpt attached	
- no permanent blacklist without confirming who owns the address	

Watch out for. The senior version of this answer leads with “check what was permitted.” Denied traffic is the firewall doing its job. Permitted traffic from a scanning source is the actual incident, and it is what junior analysts forget to look for.

Reference: NIST SP 800-61 Rev. 3, Incident Response

Q27. A site-to-site tunnel is up but no traffic passes

Tier: L2. Safety: read-only.

Short answer. Confirm both Phase 1 and Phase 2 are established, verify the proxy IDs match exactly on both peers, confirm routing actually sends traffic into the tunnel, and check for NAT rewriting addresses before they reach the encryption domain.

What is actually going on. “Tunnel is up” usually means Phase 1 completed, which proves the peers authenticated and nothing more.

Traffic needs a Phase 2 SA whose proxy IDs match on both sides. A /24 on one end against a /16 on the other fails to negotiate even though both configs look perfectly reasonable read on their own.

Beyond that, the tunnel is only used if routing directs traffic to it or the crypto ACL classifies it as interesting. A NAT rule that translates the source before encryption makes the packet miss the encryption domain entirely.

How to check

```

Cisco IOS / ASA [read-only]
  show crypto ikev2 sa          Phase 1
  show crypto ipsec sa          Phase 2
    -> compare pkts encaps against pkts decaps
  show crypto session detail

Palo Alto [read-only]
  show vpn ike-sa ; show vpn ipsec-sa ; show vpn flow

Checklist
  [ ] Phase 1 up          [ ] Phase 2 up
  [ ] Proxy IDs identical on both peers
  [ ] Route points into the tunnel
  [ ] No NAT applied before encryption, or a NAT-exempt rule exists
  [ ] Both peers agree on PFS group and lifetime

```

Watch out for. pkts encaps rising while pkts decaps stays at zero means your side is encrypting and the far side is not returning anything. That one counter comparison tells you which end owns the problem.

References: RFC 7296, IKEv2 · RFC 4301, Security Architecture for IP

Q28. An internal server suddenly can't be reached from outside after a change

Tier: L2. Safety: read-only.

Short answer. Three independent things must be true: the translation exists and points at the right internal address, policy permits the translated flow, and the server is actually listening. Test from outside and from inside separately.

What is actually going on. Publishing a server depends on objects that a change can break separately.

Object-group edits are a frequent culprit, because changing a group used by several rules silently changes all of them. A server that stopped listening, whether from a crashed service, a changed bind address, or a host firewall coming back on after a patch, produces an identical external symptom. A change to the outside interface address, or a new upstream device doing its own NAT, breaks publishing without anyone touching your config at all.

How to check

```

# From outside [active test]
nc -vz <public-ip> 443
curl -vk https://<public-ip>/

```

```

On the firewall [read-only]
show nat detail / show xlate
packet-tracer input outside tcp <src> 1234 <public-ip> 443 detailed
show access-list | include hitcnt

On the server [read-only]
netstat -tulpn | grep 443 Linux
Get-NetTCPConnection -State Listen Windows
systemctl status <service>

```

Watch out for. Run packet-tracer from the *outside* interface with the real public source address. Simulating from the wrong interface gives you a confident and entirely misleading result, and people act on it.

References: RFC 3022, Traditional IP NAT · RFC 5382, NAT Requirements for TCP

Q29. A user's laptop is quarantined by NAC without explanation

Tier: L2. Safety: read-only.

Short answer. Read the failure reason in the NAC console instead of guessing. It will tell you whether authentication failed or posture assessment failed, and those have completely different fixes.

What is actually going on. NAC combines two independent gates. Authentication answers “who are you” through 802.1X and EAP. Posture answers “are you compliant.” Fail either and you land in quarantine, and the user-visible symptom is the same.

The console separates them. An authentication failure shows an EAP or certificate error and a rejected RADIUS exchange. A posture failure shows a successful authentication followed by a compliance verdict.

In practice, the two most common real causes are expired machine certificates after a CA renewal, and antivirus definitions that aged out while a laptop was on leave.

How to check

```

Cisco ISE [read-only]
Operations > RADIUS > Live Logs filter by MAC or username
-> the Failure Reason field names the exact cause
Operations > Reports > Posture Assessment by Endpoint

Switch [read-only]
show authentication sessions interface <if> details
show dot1x interface <if> details

Client [read-only]
Windows: Event Viewer > Applications and Services Logs >
Microsoft > Windows > Wired-AutoConfig
certlm.msc is the machine certificate present and valid?

```

Watch out for. The Failure Reason field answers this question directly. Guessing at posture rules instead of reading the log suggests you have not used the platform.

References: IEEE 802.1X · RFC 3748, EAP

Q30. Repeated login failures from many source IPs against one admin account

Tier: L2. Safety: read-only.

Short answer. Treat it as credential stuffing or a distributed brute force. Establish first whether anything succeeded, because containment and investigation diverge completely at that point. Then lock or reset, verify lockout and MFA, block or rate-limit the sources, and escalate.

What is actually going on. Many sources against one account is a deliberate evasion of both per-source rate limiting and per-account lockout. Each source stays under the threshold while the aggregate rate stays high.

That pattern also implies the attacker has a validated username, which usually means it leaked or is guessable. Password spraying is the mirror image, one password against many accounts, and it evades lockout differently. Naming which pattern you are seeing matters.

How to check

	[read-only]
1. Any successful authentication from those sources?	
2. Was MFA satisfied on any success, or bypassed?	
3. Is the account privileged, and what does it reach?	
4. Source geography and ASN spread	
5. Are other accounts targeted, or only this one?	
Windows	[read-only]
Event ID 4625 failed, 4624 success, 4740 lockout	
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4625}	
Immediate actions	[change approval]
- reset credentials AND revoke active sessions and tokens	
- confirm lockout threshold and MFA enforcement	
- rate-limit or geo-restrict the admin surface	
- open a security incident; preserve logs before rotation	

Watch out for. Revoking sessions and tokens matters as much as resetting the password. An attacker holding a valid session cookie or refresh token is completely unaffected by a password change, and forgetting that step is a common real-world gap.

References: NIST SP 800-63 Revision 4, Digital Identity · MITRE ATT&CK T1110, Brute Force

Q31. After enabling deep packet inspection, throughput drops

Tier: L3. Safety: read-only.

Short answer. Inspection, and TLS decryption in particular, is expensive and often bypasses hardware acceleration. Measure CPU and session load, compare against the platform's *inspected* throughput rating rather than its headline figure, and apply inspection selectively.

What is actually going on. Vendor datasheets quote several throughput numbers and they differ by large multiples: raw firewall throughput, throughput with threat prevention, and throughput with TLS decryption.

Decryption requires terminating and re-establishing every session, multiplying session count, memory and CPU per connection. Many platforms accelerate plain forwarding in hardware and drop to software for inspected flows, so enabling inspection does not degrade performance gradually. It moves traffic onto a different and much slower path.

The engineering answer is selective policy, decryption exclusion lists, and capacity planning against the inspected figure.

How to check

Palo Alto	[read-only]
show system resources follow	
show session info	session count vs platform maximum
show running resource-monitor hour	
Fortinet	[read-only]
get system performance status	
diagnose sys session stat	
diagnose npu npu-feature	is hardware offload still engaged?
Method	[active test]
1. Baseline with iperf3 before the change	
2. Enable inspection on a limited policy only	
3. Re-measure and compare against the inspected datasheet figure	

Watch out for. Ask which datasheet number the platform was sized against. A firewall bought on a 20 Gbps raw figure and then asked to decrypt at 3 Gbps was under-specified at purchase, and no amount of tuning fixes that. Naming it as a sizing problem rather than a configuration problem is the L3 answer.

Reference: RFC 8404, Effects of Pervasive Encryption on Operators

Q32. A misconfigured ACL blocked a critical application at 2 AM

Tier: L2. Safety: change approval.

Short answer. The failure is process, not syntax. No peer review, no simulated test, no pre-approved rollback, no post-change verification. Fix it with mandatory review, policy simulation before commit, staged deployment, and an automatic revert timer.

What is actually going on. Firewall and ACL changes are unusually dangerous because their blast radius is invisible in the diff. Adding a deny above an existing permit changes behaviour for every flow that permit used to match, and nothing in the diff shows you which flows those were.

Mature controls attack that directly: hit-count review before deleting or modifying a rule, simulated policy tests run *before* commit, staged rollout to one site, and a monitored verification window before the change is declared complete.

How to harden

```

Pre-change                                     [change approval]
- peer review, with the reviewer named in the change ticket
- hit-count review on any rule being modified or removed
- simulate known-good flows: packet-tracer, test security-policy-match
- written rollback commands, tested, in the ticket

During
- Junos:      commit confirmed 10
- Palo Alto:  config snapshot plus scheduled revert
- IOS XE:     configure terminal revert timer 10

Post-change
- run the documented verification tests
- watch monitoring for one full interval before closing

```

Watch out for. The strongest version of this answer names the specific control that would have caught it, usually a hit-count review showing the rule was still in use, or a simulated test for the application's flow. Generic "we should have change control" is an L1 answer.

References: NIST SP 800-41 Rev. 1, Firewall Policy · ITIL 4 Change Enablement

Q33. Users can browse sites but file downloads consistently fail or hang

Tier: L3. Safety: active test.

Short answer. A path MTU discovery black hole is a strong hypothesis, especially if there is a tunnel in the path, but it is not the only thing that fits. Run the DF-bit test to gather evidence, and check the alternatives before you commit to it.

What is actually going on. The PMTUD story goes like this. TCP negotiates a maximum segment size at handshake, but handshake packets are small and succeed regardless of the real path MTU. When bulk transfer starts, full-size segments hit a link with a smaller MTU. With DF set, the intermediate router drops the packet and returns ICMP Type 3 Code 4 telling the sender the correct MTU. Firewalls that block all ICMP kill that message, so the sender never learns and retransmits the same oversized segment forever.

That fits the symptom well. So do several other things, and a good engineer rules them out rather than assuming:

Alternative	Tell
Proxy, DLP or antivirus file size limit	Fails at a consistent file size, logged on the proxy
Content filtering on file type	Fails by extension, not by size
Server-side timeout	Fails after a consistent duration, not a consistent size
Storage or backend fault	Server logs show the error, network is clean
Application-level limit	Fails only for one application

Use the DF-bit sweep as evidence for or against PMTUD, not as a formality on the way to a conclusion you already reached.

How to check

```
# Find the real path MTU by bisection                                # [active test]
ping -M do -s 1472 <dest>                                         # Linux: 1472 + 28 = 1500
ping -f -l 1472 <dest>                                           # Windows
tracert <dest>                                                    # reports MTU changes per hop

Cisco                                                            [active test]
  ping <dest> size 1500 df-bit

Fix, once PMTUD is confirmed                                     [change approval]
  interface tunnel0
  ip tcp adjust-mss 1360
Linux:
  iptables -t mangle -A FORWARD -p tcp --tcp-flags SYN,RST SYN \
  -j TCPMSS --clamp-mss-to-pmtu
Firewall:
  permit icmp type 3 code 4 (fragmentation needed)
```

Watch out for. MSS clamping fixes TCP only. UDP applications over the same path still need the MTU lowered or PMTUD working properly.

This is the classic symptom over GRE, IPsec, PPPoE and most overlays, because encapsulation reduces the effective MTU. If everyone affected is behind a tunnel, PMTUD moves to the front of your list.

References: RFC 1191, Path MTU Discovery · RFC 2923, TCP Problems with PMTUD · RFC 4821, Packetization Layer PMTUD

Q34. Remote sites behind different firewall vendors report inconsistent VPN compatibility

Tier: L3. Safety: read-only.

Short answer. Implementation differences in NAT traversal, IKE fragmentation and dead peer detection, made worse by firmware drift. Build a compatibility matrix across sites rather than troubleshooting them one at a time.

What is actually going on. IKEv2 and IPsec are standardised, but the surrounding behaviour is not uniformly implemented. NAT-T detection, IKE fragmentation support, DPD intervals, rekey behaviour and default proposal sets all vary by vendor and by release.

A site whose local equipment performs NAT changes the negotiation entirely, forcing UDP 4500 encapsulation that a stricter firewall may not permit. Where IKE fragmentation is unsupported, large certificate-based IKE_AUTH payloads fragment at the IP layer and get dropped by middleboxes, which produces a failure that affects only certificate authentication and only at some sites.

How to check

Per site, record and compare	[read-only]
- vendor and exact firmware version - is the site behind NAT? was NAT-T detected in IKE_SA_INIT? - are UDP 500 and UDP 4500 permitted both ways? - is IKE fragmentation supported and enabled? - negotiated proposal: DH group, encryption, integrity, PRF, lifetime - authentication: PSK or certificate	
Commands	
show crypto ikev2 sa detail	[read-only]
tcpdump -i <if> -n 'udp port 500 or udp port 4500'	[active test]
debug crypto ikev2 platform	[service-affecting]

Watch out for. Build the matrix. The pattern across sites names the cause far faster than any single packet capture, and the matrix stays useful as a living document, because the next firmware release will change it.

References: RFC 3947, NAT Traversal in IKE · RFC 3948, UDP Encapsulation of ESP · RFC 7383, IKEv2 Fragmentation

Chapter 4. VPN and Remote Access

Scenarios Q35 to Q41. Tier mix: 1 L1, 6 L2.

Orientation

A VPN builds an encrypted tunnel across an untrusted network. Two flavours dominate:

Site-to-site connects two offices permanently. Neither end knows or cares that a tunnel exists.

Remote access connects one user's device to the corporate network on demand.

Whichever you are troubleshooting, remember this: a tunnel coming up proves only that two devices authenticated to each other. It says nothing about whether traffic can flow. Routing, DNS, policy and NAT all still have to be right afterwards, and each one fails in its own way.

One of the most useful diagnostics in this chapter takes ten seconds. **Test by IP, then test by name.** If IP works and name does not, you have a DNS problem. If neither works, you have a routing or policy problem. That single comparison eliminates half the possibilities before you have opened a single config file.

Scenarios in this chapter

ID	Scenario	Tier
Q35	Connected but no access	L2
Q36	Repeating disconnect interval	L2
Q37	Failure triggered by volume	L2
Q38	Venue network restriction	L1
Q39	Certificate chain problem	L2
Q40	Tunnel policy and capacity	L2
Q41	Path performance	L2

Q35. A remote worker connects to VPN but can't access internal file shares

Tier: L2. Safety: read-only.

Short answer. Test by IP first, then by name. That tells you whether you are chasing routing and policy or chasing DNS. Then verify the tunnel routes include the file server subnet, the client received internal DNS and the right search suffix, and the firewall permits SMB from the VPN pool.

What is actually going on. The VPN establishing proves authentication and tunnel setup succeeded. Nothing more. Four separate things must also be true:

- A route for the file server subnet is pushed to the client
- Internal DNS is used for internal names, with the correct search suffix

- The firewall permits TCP 445 from the VPN address pool, which is often a separate zone with its own policy
- Kerberos can work, which needs the fully qualified name to match the service principal

That last point gives you a free diagnostic. Connecting by IP silently falls back to NTLM and can succeed where connecting by name fails, so a name-versus-IP difference here can mean DNS or Kerberos.

Interpreting the test:

Result	Points at
IP works, name fails	DNS servers, search suffix, or Kerberos SPN
Neither works	Routing or firewall policy
Both work, authentication fails	Share permissions, SPN, or account issue

How to check

On the client, while connected	[read-only]
route print / ip route	is the server subnet routed?
ipconfig /all	internal DNS? search suffix?
Get-DnsClientNrptPolicy	split DNS rules on Windows
Get-NetIPInterface sort InterfaceMetric	
resolvectl status	per-link DNS on Linux
	[active test]
nslookup fileserver.corp.local <internal-dns-ip>	
Test-NetConnection fileserver -Port 445	
net use \\10.1.5.20\share	by IP, to isolate DNS
On the firewall	[read-only]
packet-tracer input outside tcp <vpn-pool-ip> 1234 <fileserver> 445 detailed	

Watch out for. On Windows, the DNS Client service picks a resolver based on interface metric and connection-specific suffix. A physical adapter with a lower metric than the VPN adapter can win even when the VPN pushed perfectly correct servers. Querying the internal DNS server explicitly with `nslookup <name> <dns-ip>` bypasses that selection logic and tells you whether the record exists at all.

References: RFC 8598, Split DNS Configuration for IKEv2 · RFC 4120, Kerberos V5

Q36. A VPN client repeatedly disconnects at about the same interval

Tier: L2. Safety: read-only.

Short answer. A repeating interval means something with a timer, but draw the conclusion carefully. Under IKEv2 the peers do not negotiate a common SA lifetime, so a “lifetime mismatch” is not by itself the fault. Look instead for a rekey or reauthentication that is failing, an idle or session limit, or NAT state expiring between the client and the head end.

What is actually going on. This is a common interview trap, so get it right.

In **IKEv1**, lifetimes are negotiated and the shorter value wins. In **IKEv2**, RFC 7296 treats lifetime as a local matter. Each peer enforces its own policy and the side with the shorter lifetime simply initiates a rekey. Different lifetimes are normal and expected behaviour.

So when an IKEv2 tunnel dies at a repeating interval, the interesting question is why the rekey did not succeed:

- **The rekey exchange is being blocked.** A stateful device between the peers dropped the UDP 4500 mapping, so the exchange never completes.
- **Reauthentication is failing.** Some implementations require full reauthentication rather than a simple rekey, and that fails if a credential or certificate is no longer valid.
- **An idle or session timeout is enforced** by the head end group policy, independent of crypto.
- **NAT keepalive is too infrequent** for the NAT device's idle timeout, so the mapping is lost mid-session.
- **An implementation defect** in one peer's rekey handling, which is why firmware version matters.

How to check

```

[read-only]
show crypto ikev2 sa detail          life and remaining time, status
show crypto ipsec sa | include lifetime|sa timing
show run | include lifetime|keepalive|idle
show logging | include IKEV2|REKEY|RETRANSMIT|DELETE

Compare on both peers
IKE SA lifetime and reauth setting
IPsec SA lifetime
DPD interval and retry
NAT keepalive interval vs the NAT device idle timeout
group-policy vpn-idle-timeout and vpn-session-timeout

[active test]
tcpdump -n 'udp port 4500'          does the rekey exchange happen at all?
```

Watch out for. Get exact timestamps before theorising. A failure at a precise 3600 seconds points at a crypto lifetime and a failed rekey. A fuzzy “about an hour” points at an idle timeout or upstream instability. Those need different fixes and the difference is only visible in the numbers.

Reference: RFC 7296, IKEv2

Q37. A site-to-site VPN drops whenever large file transfers start

Tier: L2. Safety: read-only.

Short answer. Separate “the transfer stalls” from “the tunnel actually drops,” because they have different causes. Fragmentation and MTU problems usually stall or degrade the transfer while the SA stays up. If the SA genuinely goes down, look at volume-based lifetimes, crypto engine exhaustion, and anti-replay.

What is actually going on. Ask the first question properly: did the tunnel drop, or did the transfer fail while the tunnel stayed up? The answer changes everything.

If the transfer stalls and the SA stays up, you are almost certainly looking at MTU. IPsec adds roughly 50 to 70 bytes depending on mode, cipher, and whether NAT-T is in use. GRE adds 24 more. A 1500-byte payload becomes an oversized packet that must fragment or be dropped, and with ICMP filtered the sender never learns.

If the SA genuinely drops during bulk transfer, consider these instead:

- **Volume-based lifetime.** IPsec SAs can be limited by kilobytes transferred as well as by time. A large transfer hits the volume limit and triggers a rekey. If that rekey fails, the tunnel drops, and it will look like the transfer caused it because it did.
- **Crypto engine or CPU exhaustion.** Software crypto on an undersized platform saturates under sustained throughput.
- **Packet-per-second limits** on the platform or a middlebox.
- **Anti-replay window drops.** High rates plus QoS reordering can push packets outside the replay window, which shows in the SA counters.

How to check

```

[read-only]
show crypto ipsec sa | include lifetime|kilobytes|replay|frag
show crypto ipsec sa | include encaps|decaps
show crypto engine accelerator statistic
show processes cpu sorted
show logging | include REKEY|REPLAY|IPSEC

[active test]
ping <far-end> size 1500 df-bit      then bisect downward

Fix, on the tunnel interface, both ends      [change approval]
interface Tunnel0
  ip mtu 1400
  ip tcp adjust-mss 1360

Typical overheads
GRE                24 bytes
IPsec ESP tunnel mode  roughly 50 to 60 bytes
NAT-T (UDP 4500)    plus 8 bytes

```

Watch out for. MSS clamping helps TCP only. UDP traffic over the same tunnel still needs a lower MTU or working PMTUD. Clamping adjusts one TCP option; it is not a general MTU fix, and 1360 is not a magic number.

References: RFC 4459, MTU and Fragmentation Issues with Tunnels · RFC 4303, IP Encapsulating Security Payload

Q38. A user can VPN in from home but not from a hotel Wi-Fi

Tier: L1. Safety: active test.

Short answer. The venue network is filtering or intercepting something. Check for a captive portal first, then test whether the VPN's ports are usable, then fall back to a TLS-based VPN profile on TCP 443 or a phone hotspot.

What is actually going on. Guest and hospitality networks commonly permit very little outbound. IKE on UDP 500 and 4500 is a frequent casualty. Captive portals intercept everything until the user authenticates in a browser, and a VPN client that starts before the portal is satisfied fails in a way that looks like a server problem.

TCP 443 usually works, and that is why most vendors offer a TLS fallback. But do not claim that 443 always works, because it does not. Networks block or break VPN over 443 using authenticated proxies, TLS inspection, SNI-based policy, and application identification. Corporate guest networks and some national networks do this deliberately.

Double NAT and carrier-grade NAT are worth testing but are not automatically fatal. Standards-compliant NAT traversal is designed to cope with them. Treat them as hypotheses.

How to check

```
curl -I http://neverssl.com          # [active test]
nc -vz <vpn-head-end> 443           # a forced redirect means captive portal
traceroute <vpn-head-end>           # TCP handshake, this one is meaningful

A caution about testing UDP
nc -vz -u <host> 500 does NOT prove reachability.
UDP has no handshake, so nc reports success unless an ICMP port
unreachable comes back. Absence of an error is not evidence of a path.
To test UDP properly, send something the far end will answer, or
capture at both ends and compare.
```

Watch out for. This is a very common L1 ticket and the good answer is short and decisive. Confirm it is the venue network, move the user to the TLS profile or a hotspot, and move on. Spending an hour on someone else's hotel Wi-Fi is the wrong call.

References: RFC 8952, Captive Portal Architecture · RFC 3948, UDP Encapsulation of ESP

Q39. After a VPN concentrator upgrade, users get “certificate not trusted” errors

Tier: L2. Safety: read-only.

Short answer. The certificate now being presented does not chain to a CA the clients trust. Usually the appliance fell back to a self-signed certificate, or the intermediate CA certificate was not installed alongside the server certificate.

What is actually going on. A client validates a chain, not a single certificate. The server must present its own certificate *and* every intermediate up to a root the client already trusts.

Appliances often revert to a self-signed certificate after a firmware upgrade or a config restore. Administrators frequently import the server certificate and omit the intermediate, which works on

machines that happen to have cached that intermediate and fails everywhere else. That is where the maddening “works on my laptop” pattern comes from.

Two related causes look similar but produce different error text: a Subject Alternative Name list that omits the hostname users actually type, and a simple expiry.

How to check

```
# [active test]
openssl s_client -connect vpn.example.com:443 -showcerts

# Read the returned chain. Server certificate, then intermediates.
# The verify result names the fault directly:
#   "self signed certificate"
#   "unable to get local issuer certificate" = missing intermediate

openssl x509 -in server.crt -noout -text | grep -A1 "Subject Alternative Name"
openssl x509 -in server.crt -noout -dates

Fix [change approval]
- install the full chain, server plus intermediates
- confirm the SAN covers every hostname clients use
- push the issuing root to clients via GPO or MDM if it is a private CA
```

Watch out for. `openssl s_client -showcerts` answers this by printing exactly what the server sends. If only one certificate comes back and it is not self-signed, the intermediate is missing. Nothing more is needed to confirm it.

References: RFC 5280, X.509 Certificate Profile · RFC 6125, Service Identity Verification

Q40. Split-tunnel VPN users report both slow internet and slow internal resources

Tier: L2. Safety: read-only.

Short answer. Both symptoms sharing one cause is the clue. The split-tunnel policy is probably not being applied, so all traffic including web browsing is going through the concentrator and saturating it. Check the routes pushed to the client, then the concentrator’s load.

What is actually going on. Split tunnelling sends only corporate subnets through the VPN and lets everything else go direct. When that fails, whether from the wrong group policy being applied, an included subnet of 0.0.0.0/0, or a client profile overriding the server setting, every user’s video calls, streaming and software updates traverse the concentrator.

The device saturates, internal traffic queues behind internet traffic, and both degrade together.

In a deliberate full-tunnel design this is expected behaviour and the answer becomes capacity: the concentrator and its circuit must be sized for total user internet demand, not just corporate traffic.

How to check

```

On the client                                     [read-only]
route print | find "0.0.0.0"      is the default route via the tunnel?
ipconfig /all

tracert 8.8.8.8                                [active test]
                                         is hop 1 the concentrator?

On the concentrator                             [read-only]
show vpn-sessiondb detail anyconnect
show cpu usage ; show traffic
show run all group-policy <name> | include split
show run tunnel-group <name>

```

Watch out for. A traceroute to a public address settles this in seconds. If the first hop is the concentrator, split tunnelling is not in effect, whatever the configuration claims it is doing.

Reference: RFC 7296, IKEv2

Q41. A remote office has poor VPN performance despite a good local speed test

Tier: L2. Safety: active test.

Short answer. The speed test measured the path to a nearby test server, not the path to your head end. Suspect congestion or peering on that specific route, or asymmetric routing. Run MTR from both directions and compare against a control site.

What is actually going on. A fast result against a well-peered local server proves the access circuit is healthy. It proves nothing about the transit path to a head end several hundred kilometres away.

Poor performance on one path usually means congestion at a peering point, a saturated transit link, or a route taking an unexpected detour.

Latency also interacts badly with TCP over a tunnel. Throughput is bounded by window size divided by round trip time, so an extra 60 ms can halve throughput with zero packet loss. That is how a “good” 100 Mbps speed test coexists with 8 Mbps through the tunnel.

How to check

```

# [active test]
mtr -rwzc 200 <vpn-head-end>          # from the branch
mtr -rwzc 200 <branch-public-ip>      # from the head end, same window
iperf3 -c <internal-server> -t 30
iperf3 -c <internal-server> -t 30 -R  # reverse direction
ping -c 100 <head-end>                # jitter and loss baseline

How to read it
Loss at one hop that persists downstream -> real loss at that hop
Loss at one hop that clears afterwards   -> ICMP rate limit, ignore it
Asymmetric latency between the two MTRs -> asymmetric path
Throughput far below bandwidth-delay    -> window or latency limited

```

Watch out for. Loss shown at a single intermediate hop that disappears at later hops is a control-plane rate limit, not a fault. Misreading that is one of the most common false escalations to a carrier.

For structured throughput testing use RFC 6349, which is designed for live paths. Do not reach for RFC 2544 methods here. Those are laboratory benchmarks and RFC 6815 explains why running them on a production network is harmful.

References: RFC 6349, TCP Throughput Testing · RFC 6815, RFC 2544 on Production Networks Considered Harmful

Chapter 5. Wireless

Scenarios Q42 to Q49. Tier mix: 2 L1, 6 L2.

Orientation

Wireless breaks people's mental models because it behaves nothing like a switch port.

It is a shared, half-duplex medium. Every device on a radio takes turns. Twenty clients on one access point are sharing one conversation, not getting twenty dedicated connections. The "1.2 Gbps" on the box is an aggregate theoretical maximum, not what any one user gets.

Signal strength alone tells you very little. What matters is signal-to-noise ratio and the retry rate. A client with a strong signal in a noisy room performs worse than a client with a moderate signal in a quiet one.

The client decides where to connect, not you. The standard gives the access point no authority to move a client. You can influence the decision with power levels and standards like 802.11k and 802.11v, but you cannot command it.

Two numbers worth memorising, because they turn vague complaints into measurable facts:

- **RSSI better than -67 dBm** for voice, -70 dBm for data
- **SNR of 25 dB or better**, and **retries under 10 to 15 percent**

Scenarios in this chapter

ID	Scenario	Tier
Q42	Localised disconnects	L2
Q43	Sticky client roaming	L2
Q44	Expected vs faulty performance	L1
Q45	Segmentation failure	L2
Q46	802.1X and embedded devices	L2
Q47	AP join failure	L2
Q48	Non-Wi-Fi interference	L1
Q49	Carrier Wi-Fi calling	L2

Q42. Users experience random Wi-Fi disconnects only in one area of the building

Tier: L2. Safety: read-only.

Short answer. A location-bound symptom narrows things down, but do not jump straight to RF. Check the serving AP's own health first: uplink, switchport, PoE, and whether it is rebooting. Then, if the infrastructure is clean, classify the RF problem.

What is actually going on. It is tempting to say a fault affecting one area must be RF. That reasoning is wrong, and it sends people surveying with a spectrum analyser when the actual cause was a failing PoE injector.

Faults that are location-bound but not RF at all:

- The AP's uplink switchport is flapping or erroring
- PoE budget exhaustion, so the AP resets under load
- A cable run to that AP is damaged
- The AP itself is failing, or rebooting on a watchdog
- That AP has different configuration, a different tag or group, or a stale image

Once infrastructure is ruled out, the RF causes split three ways, and they need opposite remedies:

RF cause	Signature	Remedy
Coverage hole	Low RSSI, clients drop when moving	Add or reposition an AP
Co-channel interference	Good RSSI, high channel utilisation	Change channel plan, reduce power
Non-Wi-Fi interference	Good RSSI, high retries, high noise floor	Find and remove the emitter, or move to another band

How to check

Infrastructure first	[read-only]
show ap uptime <ap-name>	has it been rebooting?
show interface <ap-switchport> include error flap rate	
show power inline <ap-switchport>	
show logging include <ap-name>	
Then RF	[read-only]
show ap auto-rf <ap-name>	channel, power, noise, interference
show client detail <mac>	RSSI, SNR, retries, data rate
show ap dot11 5ghz summary	channel utilisation
Targets	
RSSI better than -67 dBm (voice) / -70 dBm (data)	
SNR 25 dB or better	
Retries under 10 to 15 percent	
Channel utilisation under 40 percent	

Watch out for. Good RSSI with high retries means noise, not coverage. Adding another AP in that situation makes things worse by adding another contender for airtime. That distinction is the difference between fixing the problem and spending money to amplify it.

Reference: IEEE 802.11

Q43. A new AP was added but clients still connect to a farther, weaker AP

Tier: L2. Safety: read-only.

Short answer. Sticky client behaviour. Roaming is the client's decision, and most clients hold their association until the signal becomes unusable. Size the cells properly with transmit power, then use 802.11k and 802.11v to influence the decision.

What is actually going on. The 802.11 standard gives the AP no authority to move a client. The client scans and decides, and vendor implementations vary enormously in how eager that decision is.

Oversized cells make it worse. If the old AP is still audible at -75 dBm, the client sees no reason to move, even though throughput at that signal level is a fraction of what the new AP would give it. Lowering transmit power shrinks the cell and forces the decision sooner.

The relevant amendments: **802.11k** gives the client a neighbour report so it scans efficiently. **802.11v** lets the infrastructure suggest a transition. **802.11r** speeds up the handoff once the client agrees to move.

How to check

	[read-only]
show client detail <mac>	associated AP, RSSI, data rate
show ap auto-rf <ap-name>	Tx power level, neighbour list
show wlan <id>	are 11k, 11v and 11r enabled?
Levers, in order of preference	
	[change approval]
1. Reduce Tx power so cells overlap about 15 to 20 percent, not more	
2. Enable 802.11k neighbour reports and 802.11v BSS transition	
3. Enable 802.11r fast transition for voice clients	
4. Disable the lowest legacy data rates (1, 2, 5.5, 11 Mbps)	
5. RSSI-based client steering, as a last resort	

Watch out for. Disabling low legacy data rates is the most under-used lever here. It shrinks the effective cell edge, forces earlier roaming, and reclaims airtime that slow clients would otherwise consume. No hardware required. Check for legacy devices that genuinely need those rates before you do it.

Reference: IEEE 802.11k, 802.11v, 802.11r amendments

Q44. Wireless throughput is much lower than wired for the same user

Tier: L1. Safety: read-only.

Short answer. Check the negotiated PHY rate and band, channel width, channel utilisation, how many clients share that radio, and the AP's uplink capacity. Then establish whether this is a fault at all, because a large gap between wired and wireless is normal.

What is actually going on. Several multipliers stack up:

- A client on 2.4 GHz with a 20 MHz channel and one spatial stream is capped far below a 5 GHz client on 80 MHz with two streams
- Airtime is shared, so twenty active clients each get a slice
- Legacy clients at low data rates consume disproportionate airtime for the same bytes, degrading everyone

- An AP with a 1 Gbps uplink serving radios capable of more in aggregate is itself the bottleneck

Real TCP throughput of 40 to 60 percent of the negotiated PHY rate is normal, not a fault.

How to check

```
[read-only]
show client detail <mac>
-> data rate, spatial streams, channel width, band, RSSI, SNR
show ap dot11 5ghz summary
-> client count and channel utilisation per radio
show interface <ap-uplink> | include rate
```

Watch out for. Ask what the user is comparing against. “Slower than wired” is expected physics and not a ticket. “Slower than it was last week, same laptop, same seat” is a real change and worth your time. Establishing which one you have should be your first question, not your last.

Reference: IEEE 802.11ac / 802.11ax

Q45. Guest Wi-Fi users can access internal VLANs unexpectedly

Tier: L2. Safety: read-only.

Short answer. Segmentation has failed somewhere along a chain, and VLAN mapping is only one of the links. Establish what address the guest client actually received and what it can reach, then work back through the chain. Treat it as a security incident from the start.

What is actually going on. Guest isolation depends on several controls in series, and any one of them failing produces the same result:

- **SSID to VLAN or policy mapping** on the controller or AP group is wrong
- **Switchport configuration** carrying AP traffic has a wrong native VLAN or allowed list, so untagged frames land in an internal VLAN
- **Firewall or ACL policy** between the guest segment and internal segments is missing, too broad, or was never written because everyone assumed the VLAN was enough
- **VRF or routing leak**, where the guest subnet is reachable through a route it should not have
- **Guest anchor tunnel failure**, so traffic is switched locally at the AP instead of being tunnelled to the anchor controller in the DMZ
- **Policy or role tag error**, where the client is assigned the wrong role after authentication

Client isolation within the SSID stops guest-to-guest traffic and does nothing at all about guest-to-internal, which is a distinction people get wrong under pressure.

How to check

Start from the client	[active test]
ipconfig /all	which subnet was actually issued?
ping <internal gateway>	
tracert <internal host>	which way does it go?
Then work back	[read-only]
show wlan <id>	interface, VLAN, policy mapping
show ap config general <ap-name>	AP group, local switching VLAN map
show interface <ap-switchport> trunk	native VLAN, allowed VLANs
show mobility anchor	is the anchor tunnel up?
show ip route <guest-subnet>	VRF leak?
Firewall: policy from guest zone to internal zones	

Watch out for. Raise this through the security incident process, not the routine change queue. Segmentation failure on a guest network is a reportable control failure under most compliance regimes, and treating it as business as usual is the wrong instinct.

References: IEEE 802.1Q · NIST SP 800-153, Securing WLANs

Q46. Certain IoT devices fail to connect to a WPA2-Enterprise SSID

Tier: L2. Safety: read-only.

Short answer. The devices probably do not implement 802.1X, or implement only an EAP method your RADIUS server is not offering. Give them a dedicated SSID and VLAN with a strong PSK or MAC authentication, then compensate for the weaker authentication with segmentation.

What is actually going on. WPA2-Enterprise needs a working supplicant, a trusted CA certificate, and a supported EAP method. Cheap or embedded devices often have none of those.

Common failure modes: no supplicant at all; support for PEAP-MSCHAPv2 only when the server requires EAP-TLS; certificate validation failing because the device has no trust store and no way to install one; and no real-time clock, so certificate validity checks fail on every boot until NTP syncs.

Forcing these devices onto enterprise authentication is not achievable. The correct answer is a design change.

How to check

ISE or RADIUS	[read-only]
Live Logs, filtered by device MAC	
"EAP method not supported"	
"certificate not trusted"	
"no supplicant response"	
Each names a different fix.	
Controller	[service-affecting]
debug client <mac>	
show client detail <mac>	does it reach EAP at all?

Design	[change approval]
Separate IoT SSID and VLAN	
WPA2-PSK with a unique key, or MAB with a device profile	
Deny IoT VLAN to internal VLANs by default	
Permit only the specific flows the devices actually need	

Watch out for. Say out loud that MAC authentication is weak because MAC addresses are trivially spoofed, and that the compensating control is segmentation rather than the authentication itself. Interviewers are listening for whether you know you are accepting a risk deliberately or by accident.

References: IEEE 802.1X · RFC 5216, EAP-TLS · NIST SP 800-213, IoT Device Cybersecurity

Q47. After a wireless controller firmware update, multiple APs go offline

Tier: L2. Safety: read-only.

Short answer. The APs cannot rejoin. Read the join statistics, because the controller records the specific reason. Usual causes are an AP model dropped from the new release, a CAPWAP control-plane problem, or certificate validation failing.

What is actually going on. APs download their image from the controller and rejoin after an upgrade, so a model that was dropped from the new release simply never comes back.

CAPWAP needs UDP 5246 for control and 5247 for data. A firewall rule or ACL that the old release tolerated can break the new one if discovery behaviour changed.

Certificate errors are common on older hardware whose manufacturer-installed certificates have expired, and on controllers whose clock is wrong, because certificate validation is time-sensitive. Check the clock before you blame the certificates.

How to check

Controller	[read-only]
show ap join stats summary all	
show ap join stats detailed <ap-mac>	names the failure reason
show ap image all	predownload status per model
show time	clock correct?
AP console	[service-affecting]
debug capwap client events	
debug pm pki enable	certificate validation detail
Network	[read-only]
Is UDP 5246 and 5247 permitted AP to controller?	
Is discovery working: DHCP option 43, DNS, or broadcast?	

Watch out for. Check the release notes for supported AP models *before* the upgrade window, and stage the upgrade on one AP first. Describing that as your standard practice answers the process half of this question, which is usually what is really being probed.

Reference: RFC 5415, CAPWAP Protocol Specification

Q48. Users near the kitchen experience 2.4 GHz Wi-Fi dropouts

Tier: L1. Safety: read-only.

Short answer. Interference from the microwave oven, which emits around 2.45 GHz and sits on top of the upper 2.4 GHz Wi-Fi channels. Move clients to 5 GHz, or shift the nearby AP to a channel further from the emitter.

What is actually going on. Microwave leakage raises the noise floor for the length of a cooking cycle. The symptom is intermittent and correlates with lunchtime rather than with network load, which is a useful thing to notice.

Wi-Fi cannot negotiate with a non-Wi-Fi emitter. It defers or suffers corruption, which shows up as retries and dropped associations. The 2.4 GHz band offers only three non-overlapping 20 MHz channels, so your avoidance options are limited. Moving clients to 5 GHz is the durable fix.

How to check

```
show ap auto-rf <ap-name> [read-only]
                             interference and noise per channel

Spectrum analysis (CleanAir, Ekahau, Chanalyzer)
  Microwave signatures are broad and vary over time.
  They look nothing like a Wi-Fi transmission.

Actions [change approval]
  1. Confirm the timing correlates with kitchen use
  2. Move that AP to channel 1, furthest from about 2.45 GHz
  3. Enable band steering; check 5 GHz coverage is adequate there
  4. Do not disable 2.4 GHz entirely if legacy or IoT clients need it
```

Watch out for. A packet capture cannot see a microwave oven at all, because the energy is not a decodable frame. Only spectrum analysis distinguishes non-Wi-Fi interference from co-channel Wi-Fi contention, and those need opposite remedies.

References: IEEE 802.11 · ITU-R Radio Regulations

Q49. Wi-Fi calling quality is choppy while data works fine

Tier: L2. Safety: read-only.

Short answer. First establish which kind of Wi-Fi calling you are dealing with, because the answer changes completely. Carrier Wi-Fi calling is encrypted to the mobile operator's gateway and your WLAN cannot see the voice inside it. Enterprise softphone traffic is a different problem entirely.

What is actually going on. This scenario is commonly answered wrongly.

Carrier Wi-Fi calling (the feature built into a mobile handset) establishes an IPsec tunnel from the phone to the operator's evolved packet data gateway. Everything inside is encrypted. Your WLAN sees ESP or UDP 4500 to an internet destination, not RTP, and it cannot read or trust any inner DSCP marking. Marking the outer packet is possible in principle but the phone chooses it, not you.

That means the levers available to you are:

- **WMM and airtime**, which apply to the encrypted flow just as they do to anything else
- **RF quality**, since retries add jitter regardless of what is inside the packet
- **Roaming delay**, which is audible on a call and invisible on data
- **The internet path** to the operator gateway, including latency and loss
- **Reachability of the operator gateway**, since some firewalls block IKE and ESP outbound
- **Whether your guest or corporate policy permits IPsec outbound at all**

Enterprise voice over WLAN, meaning a softphone or a Wi-Fi handset on your own call platform, is the case where end-to-end DSCP EF and a trusted QoS boundary genuinely apply. See Q75 for that one.

How to check

```
Identify the traffic first [active test]
  Capture at the AP or switchport for the client.
  ESP or UDP 4500 to an operator address = carrier Wi-Fi calling.
  RTP on 16384-32768 to your call platform = enterprise voice.

Then, for carrier Wi-Fi calling [read-only]
  show client detail <mac>      retries, data rate, RSSI, SNR
  show ap auto-rf <ap-name>     channel utilisation
  show wlan <id> | include WMM|Fast Transition
  Firewall: is UDP 500 and 4500 permitted outbound from that SSID?
  Path quality to the operator gateway: latency, jitter, loss
```

Watch out for. If someone tells you to “trust DSCP EF end to end” for carrier Wi-Fi calling, they are describing a different problem. You cannot mark what you cannot see. Many candidates give the enterprise VoIP answer to both questions, so make the distinction explicitly.

References: RFC 7296, IKEv2 · RFC 8325, Mapping DiffServ to 802.11 · IEEE 802.11e / WMM

Part III. NOC Operations

Technical skill gets you shortlisted. Working an incident without making it worse gets you hired. Part III covers alerting, prioritisation, escalation, handover and documentation: the operational judgement that interviewers test at every tier.

Chapter 6. Monitoring, Alerting and NOC Process

Scenarios Q50 to Q64. Tier mix: 6 L1, 9 L2.

Orientation

This is the chapter people skip, and it is the chapter that gets them hired.

Technical skill gets you shortlisted. What separates two technically similar candidates is whether they can work an incident without making it worse: prioritising sensibly, escalating at the right moment, handing over cleanly, and writing something afterwards that helps the next person.

A few terms you will hear constantly:

- **P1 to P4.** Incident priority, usually derived from business impact and urgency together. A P1 is a major outage. A P4 is a single user with a minor problem.
- **SLA.** The contractual time you have to respond and resolve. Missing one has commercial consequences, which is why escalation timing matters.
- **RCA.** Root cause analysis, written after the fact.
- **CAB.** Change Advisory Board, the group that approves changes.
- **Runbook.** The documented procedure for a known scenario.

If you have never worked in a NOC, read this chapter twice. The scenarios describe how the job actually feels on a night shift, and interviewers can tell within one answer whether you have thought about it.

Scenarios in this chapter

ID	Scenario	Tier
Q50	Alert storm	L1
Q51	Alerting pipeline failure	L2
Q52	Alert noise vs real fault	L2
Q53	Major incident handling	L1
Q54	Monitoring path vs user path	L1
Q55	Escalation judgement	L1
Q56	Suppression	L1
Q57	Knowledge management	L1
Q58	Capacity management	L2
Q59	Escalation automation	L2
Q60	Reconciling tool disagreement	L2
Q61	Verification	L2

ID	Scenario	Tier
Q62	Change visibility	L2
Q63	Handover	L2
Q64	Telemetry and automation	L2

Q50. Monitoring floods with hundreds of alerts during a single outage

Tier: L1. Safety: read-only.

Short answer. Find the root-cause alert, which is usually the highest device in the dependency tree that went down, and treat the rest as symptoms. Raise one parent incident, link the children to it, and work the parent.

What is actually going on. Most monitoring systems poll each device independently, so a single core switch failure alerts on everything behind it. Without dependency mapping the platform cannot tell cause from consequence.

The technique is to sort by timestamp and map the alerting devices onto the topology. The first alert, on the device closest to the core, is usually the cause. Everything behind it that alerted afterwards is a symptom.

How to work it

1. Sort by first-occurrence timestamp, not by severity
2. Map the alerting devices onto the topology
3. Identify the highest common upstream device
4. Open one parent incident; link children to it
5. Communicate once, from the parent
6. Afterwards: configure dependency suppression so this is automatic

Watch out for. The recovery pattern confirms your diagnosis. Fix the root cause and the downstream alerts clear on their own within a poll cycle or two. If they do not, you had more than one fault, and that is worth knowing before you close anything.

Reference: ITIL 4 Incident Management

Q51. A critical alert didn't page anyone during last night's outage

Tier: L2. Safety: read-only.

Short answer. Walk the alerting chain end to end. Each stage fails differently and only the audit trail tells you which one broke.

What is actually going on. The pipeline has more independent failure points than people expect, and every one produces the same outcome:

- The check did not exist for that condition
- The threshold was set too high, so it never fired
- A suppression or maintenance window was still active from an earlier change

- Severity mapping placed it below paging level
- The routing rule matched a team or schedule with a gap
- The on-call schedule had nobody assigned at that moment
- The notification was generated and delivery failed: expired integration key, SMS gateway problem, phone in do-not-disturb

How to work it

```
Confirm each stage in order [read-only]
[ ] Check existed and was enabled for that device or service
[ ] Condition breached the configured threshold
[ ] No suppression or maintenance window was active
[ ] Severity mapping placed it at paging level
[ ] Routing rule matched and selected an escalation policy
[ ] Someone was on call at that exact moment
[ ] Notification was generated (platform audit log)
[ ] Notification was delivered and acknowledged
```

Then

- fix the specific stage that failed
- add a synthetic end-to-end alert test to the monthly routine

Watch out for. Schedule gaps at shift handover and around public holidays are the most common cause and the easiest to miss, because the schedule looks fine when you check it during working hours. Check it as it was at the time of the outage, not as it is now.

Reference: Google SRE Workbook, Alerting on SLOs

Q52. A flapping “interface down/up” alert every 5 minutes on a core switch

Tier: L2. Safety: read-only.

Short answer. Investigate the physical cause first. Only after confirming the flap is genuinely not affecting service should you dampen the alert, and on a core device dampening should never be your first action.

What is actually going on. A repeating five-minute interval suggests a timer somewhere: a device rebooting on a watchdog, a PoE device power-cycling, a failing optic crossing a thermal threshold, or a negotiation loop. Cable faults usually flap irregularly.

The whole point of this question is whether you can tell “suppress the noise” apart from “fix the fault.” On core infrastructure, suppressing an alert is how a genuine failure gets missed three weeks later.

How to check

```
[read-only]
show interface <if> | include flap|last input|reset|error
show logging | include <if>
show interface transceiver detail      Rx power drifting toward threshold?
show power inline <if>                 PoE device cycling?
show interface <if> counters errors
```

Only after confirming it is not service-affecting [change approval]
 interface <if>
 carrier-delay msec 2000
 In the monitoring tool: raise the flap-count threshold.
 Do not disable the check.

Watch out for. show interface transceiver detail catching an Rx power reading drifting toward the low-warning threshold predicts an optic failure before it becomes an outage. Bringing that up unprompted signals real operational maturity.

Reference: RFC 2863, Interfaces Group MIB

Q53. Ticket volume for the same issue is coming in from multiple sites

Tier: L1. Safety: read-only.

Short answer. Declare a major incident, make one ticket the parent, link the rest, publish a single status update, and investigate the common upstream element that all affected sites share.

What is actually going on. Multiple sites reporting at once implies a shared dependency: a common transit provider, a central DNS or authentication service, a data centre, a cloud region, or a change pushed everywhere simultaneously.

Working the tickets individually duplicates effort, produces contradictory updates to different customers, and delays finding the shared cause.

How to work it

1. Declare major incident; assign an incident manager
2. Parent ticket created; inbound tickets linked as children
3. Build the affected AND unaffected site list
 What do affected sites share that unaffected sites do not?
4. Check the change calendar for the previous 24 hours
5. Check upstream provider status pages
6. One channel, one cadence, one message
7. Resolve the parent; children close against it

Watch out for. Building the *unaffected* list matters as much as the affected one. It is what turns “everything is broken” into a bounded fault domain, and it is the step junior engineers consistently skip.

Reference: ITIL 4 Incident Management

Q54. Monitoring shows 100% packet loss to a device users say is working fine

Tier: L1. Safety: active test.

Short answer. The monitoring server’s path differs from the user path, or ICMP is filtered somewhere on the monitoring path. Verify from a second vantage point and with a non-ICMP check before declaring an outage.

What is actually going on. Monitoring usually polls from a management network with its own routing and its own firewall policy. A change to a management ACL, a route withdrawal, or a newly enabled control-plane policer breaks monitoring reachability while user traffic on a different path is unaffected.

ICMP is also often deprioritised on device control planes, so a busy router may drop your pings while forwarding transit traffic perfectly.

The general principle is worth stating plainly: a monitoring check measures the path from the monitoring system. It does not measure the health of the service.

How to check

- [active test]

 1. Ping from a different source: another site, a jump host
 2. Test a non-ICMP service: SSH 22, SNMP 161, HTTPS 443
 3. Does SNMP polling still work while ICMP fails?
That combination proves the device is up and ICMP is filtered.
 4. Check the management path for a recent routing or ACL change
 5. Confirm with a user-side transaction test

Watch out for. The strategic answer is to monitor the service rather than the device. Synthetic transactions from the user's perspective, meaning "can a client resolve, connect and authenticate," do not produce this class of false positive at all.

References: RFC 792, ICMP · RFC 6192, Protecting the Router Control Plane

Q55. How do you decide whether to escalate or keep troubleshooting?

Tier: L1. Safety: read-only.

Short answer. Escalate when the runbook is exhausted, when the fault needs access or authority you do not have, or when continuing would consume a meaningful share of the SLA. Escalate proactively rather than at the breach point, and escalate with a complete handover rather than a bare reassignment.

What is actually going on. Escalation is a judgement about time and scope, not an admission of failure. Interviewers use this question to find out whether a candidate hoards tickets out of pride.

Most NOCs use a time box tied to priority: a fixed number of minutes on a P1 before escalation is mandatory, longer for lower priorities. Scope is the second trigger. If the fault is in a system you cannot access, more time does not help. Impact is the third: a P1 escalates faster than a P3 for identical technical difficulty.

How to work it

Escalate when any of these is true

- runbook exhausted with no working hypothesis
- the fix needs access or change authority you do not hold
- elapsed time has used roughly half the SLA window
- impact is expanding rather than contained
- a vendor or carrier must be engaged

Escalate WITH

- symptom, exact start time, blast radius
- what has been tested, and the result of each test
- what has been RULED OUT, and the evidence for it
- your current working hypothesis
- any vendor case numbers already open

Watch out for. “What has been ruled out, and how” is the part junior engineers omit and the part the receiving engineer needs most. Say it explicitly; it is what separates a useful handover from a reassignment.

Reference: ITIL 4 Incident Management

Q56. A maintenance window causes unexpected alerts to page on-call staff

Tier: L1. Safety: change approval.

Short answer. Schedule the suppression window before the change begins, confirm every affected device *and its dependents* are tagged into it, and verify the suppression is actually active at the start rather than assuming.

What is actually going on. Suppression usually fails for one of three reasons:

- The window covers the device being changed but not the devices downstream of it, so the dependents alert instead
- The window was scheduled in the wrong timezone, which is endemic in multi-region operations
- The change overran and the window expired mid-work, so alerts fired during the tail end when everyone had stopped watching

Each has a specific control: dependency-aware suppression, timezone-explicit scheduling, and padding.

How to work it

```

Before                                     [change approval]
[ ] Suppression scheduled for the device AND its dependents
[ ] Timezone stated explicitly, UTC preferred for multi-region
[ ] Window padded 30 minutes either side of the planned work
[ ] On-call told the window exists, in case anything leaks

At the start
[ ] Verify suppression is ACTIVE. Do not assume.

After
[ ] Confirm monitoring returns to normal and alerts resume
[ ] Close the window explicitly if the work finishes early

```

Watch out for. Padding the *end* is the control that saves you most often, because changes overrun far more than they finish early. An unpadded 60-minute window that expires at minute 61 pages the whole team during the riskiest part of the work.

Reference: ITIL 4 Change Enablement

Q57. How do you document a resolved incident for future reference?

Tier: L1. Safety: read-only.

Short answer. Record the symptom in the user's own words, the diagnostic path including what you ruled out, the root cause distinguished from the trigger, the exact resolution, and a preventive action with a named owner. Tag it so it is findable by symptom.

What is actually going on. Incident documentation fails at retrieval, not at capture. The next engineer searches by symptom, because the cause is precisely what they do not yet know. A record titled by root cause is invisible to them.

Separating trigger from root cause matters too. "The link failed" is a trigger. "The redundant path did not take over because the floating static had the wrong administrative distance" is a root cause, and only the second one prevents a repeat.

Structure

Title	symptom-first, in user language
Impact	who, how many, how long
Timeline	detection, acknowledgement, mitigation, resolution
Diagnosis	what was tested, what was ruled out, and how
Trigger	the immediate event
Root cause	the underlying condition that allowed it
Resolution	exact commands and configuration applied
Preventive	action, named owner, due date
Tags	symptom keywords, device, vendor, service, site

Watch out for. Title it as the user's complaint, "branch office users lost internet at 09:15," not as the engineering finding, "BGP session flap on WAN2." Search hits on symptoms are what actually save the next person time.

References: ITIL 4 Problem Management · Google SRE, Postmortem Culture

Q58. Bandwidth graphs show a steady climb toward saturation on a WAN link

Tier: L2. Safety: read-only.

Short answer. Treat it as capacity planning, not an incident. Identify top talkers and applications with flow data, decide whether the growth is legitimate business demand or something that should not be on the link, then either upgrade with lead time or shape what does not belong.

What is actually going on. Circuit upgrades take weeks to provision. The same trend spotted at 70 percent utilisation is manageable; spotted at 98 percent it is an outage already in progress.

Flow data decides which of two very different responses applies. Legitimate growth means procurement plus a QoS policy protecting interactive and voice traffic during peaks. Illegitimate growth, meaning unscheduled backups, cloud sync, a runaway replication job, or streaming, means shaping or scheduling and costs nothing.

How to check

```
Flow analysis [read-only]
show flow monitor <name> cache aggregate ipv4 source address
nfdump -R /flows -s srcip/bytes -n 20

Answer these before recommending anything
- Is the peak sustained or bursty?
- What is the 95th percentile, not the maximum?
- Is growth linear, or a step change? A step means a new service.
- Does it correlate with a business change or a scheduled job?
```

Watch out for. Most carrier contracts bill on the 95th percentile, so quote that rather than the peak when you build the business case. Quoting the maximum overstates the problem and costs you credibility with the people who approve the spend.

References: RFC 7011, IPFIX · RFC 3176, sFlow

Q59. A P1 ticket sits untouched for 20 minutes because the engineer is on break

Tier: L2. Safety: change approval.

Short answer. Automatic escalation on unacknowledged alerts. If a P1 is not acknowledged within a short defined window, it escalates to the secondary, then the team lead, without anyone having to notice. Acknowledgement must be a separate tracked action from resolution.

What is actually going on. Assignment is not acknowledgement. A ticket routed to a named person sits with that person whether or not they are available, and nothing detects the silence.

The design detail that matters is separating acknowledgement, meaning a human has seen this and is acting, from resolution, meaning the problem is fixed. A system tracking only resolution cannot tell “nobody is looking” apart from “someone is working on it.”

How to work it

```
Escalation ladder for P1
T+0    page primary on-call
T+5    unacknowledged -> page secondary
T+10   unacknowledged -> page team lead
T+15   unacknowledged -> page duty manager

Supporting controls
- acknowledgement is a distinct, tracked action
- break coverage is explicit; no single point of on-call in a shift
- a monitored group channel receives P1s, not one individual
- measure time-to-acknowledge as a KPI alongside time-to-resolve
```

Watch out for. Treat this as a process design failure, not a personal one. Answering “the engineer should have taken their phone” is the wrong instinct and reads badly.

Reference: Google SRE Book, Being On-Call

Q60. Multiple monitoring tools report the same outage at different severities

Tier: L2. Safety: read-only.

Short answer. Set priority from your organisation’s impact and urgency matrix, using validated blast radius and service criticality. Tool severity is evidence feeding that decision, not the decision itself.

What is actually going on. Different tools measure different things and were configured by different people at different times. A network platform may see an interface down as a warning while an application monitor sees the dependent service as critical. Both are correct from their own vantage point.

It is tempting to say “always take the highest severity,” and that instinct is understandable, but it is wrong as a *rule*. It lets a noisy or misconfigured tool promote a P3 to a P1 and consume resources that a genuine P1 needs. Tool severity reflects what that tool was configured to think, which is not the same as business impact.

The practical middle ground: while you are still validating, err toward treating it as the more serious case, because under-responding costs service and over-responding costs effort. But set the *recorded priority* from the matrix once you have established actual impact, and do it quickly.

How to work it

```
During the incident
- determine real user and business impact first
- apply the impact/urgency matrix to set priority
- use tool severities as inputs, not as the verdict
- while validating, act on the more cautious assumption
- do not spend incident time arguing about tool disagreement

After the incident
- document which tool reported what, and why
- agree one severity taxonomy across tools
- designate an authoritative source per service
- reduce overlap where duplication adds no signal
```

Watch out for. Persistent disagreement usually means one tool is monitoring the wrong thing. That is worth chasing afterwards, because it will eventually produce a false negative rather than just a false positive, and false negatives are the ones that hurt.

Reference: ITIL 4 Monitoring and Event Management

Q61. You’ve fixed an issue but the dashboard still shows red

Tier: L2. Safety: active test.

Short answer. Verify the service from a user's perspective before you trust or dismiss the dashboard. If it is genuinely restored, force a poll or clear the state. If the dashboard is right and your fix was too narrow, keep working.

What is actually going on. Two possibilities that lead in opposite directions.

The monitoring state may be stale. Polling intervals of five minutes are common and some platforms hold a state until acknowledged.

Or the dashboard is right and you fixed one component while the end-to-end service is still broken. That is the dangerous case, and an engineer who assumes stale data and closes the ticket has just declared a live outage resolved.

Testing the service the way a user experiences it resolves the ambiguity in under a minute.

How to work it

1. Test end to end as a user would, not just the component you changed
resolve the name, connect, authenticate, complete a transaction
2. If genuinely restored: force a poll, clear or acknowledge the state,
confirm it goes green and stays green for a full interval
3. If still failing: the dashboard is right. Keep troubleshooting.
4. Watch one complete monitoring interval before closing

Watch out for. Never close on the assumption that monitoring is lagging. Verifying the user experience takes a minute and is the difference between a resolved incident and a reopened one with an angry customer attached to it.

Reference: Google SRE Workbook, Monitoring

Q62. Another team's change caused an outage and the NOC wasn't told

Tier: L2. Safety: change approval.

Short answer. Mandatory CAB notification to the NOC for any network-impacting change, a change calendar visible to every team, freeze windows for high-risk periods, and a standing rule that checking the change calendar is the first step of every incident.

What is actually going on. This is an organisational boundary failure, not a technical one. The NOC detects the impact and is structurally the last to learn the cause, which inverts the correct information flow.

The fix works in both directions. Changes must be visible to the NOC before they happen, and the NOC must have change visibility built into its incident workflow. Adding a required "recent changes checked" field to the incident template is a small control with a disproportionate effect, because it turns a good habit into a process step.

How to work it

Preventive

- CAB approval for any network-impacting change
- NOC on the notification list for every approved change
- shared change calendar readable by all teams, no exceptions
- freeze windows for month-end, peak trading, major events

Detective

- incident template includes "changes in last 24h checked: Y/N"
- automated configuration-diff alerting on network devices
- change timestamps correlated automatically against alert timestamps

Cultural

- blameless post-incident reviews, so teams disclose changes readily

Watch out for. Automated configuration-diff alerting is the strongest control here, because it catches undeclared changes regardless of whether anyone followed the process. The changes that cause outages are disproportionately the ones that skipped the process, and a calendar will never show those.

Reference: ITIL 4 Change Enablement

Q63. Your shift is ending but an incident isn't resolved

Tier: L2. Safety: read-only.

Short answer. Give a structured verbal handover backed by a written record: current status, impact, everything tried and its result, everything ruled out and the evidence, current hypothesis, planned next steps, and any open vendor case numbers.

What is actually going on. Handover failures cause repeated work, and repeated work on a live incident extends the outage.

The most commonly omitted item is negative evidence. Without it, the incoming engineer re-tests the same hypotheses, sometimes for an hour, before arriving where their colleague already was.

A verbal handover on its own is not enough. The incoming engineer will need the detail again at 3 a.m. when the person who told them has gone home.

Structure

Written in the ticket, then walked through verbally

Current status	what is broken now, and the blast radius
Timeline	start, detection, key events
Tried	each action and its result
Ruled out	each hypothesis and the evidence that killed it
Hypothesis	what you currently believe is happening
Next steps	the specific action you would take next
Dependencies	vendor case numbers, contacts, callback times
Stakeholders	who has been told what, next update due when

Confirm the incoming engineer can restate it before you leave.

Watch out for. The teach-back step, where the incoming engineer summarises the situation in their own words, catches misunderstandings that a one-way briefing hides completely. It costs two minutes and is one of the most effective handover controls available.

Reference: ITIL 4 Service Desk and Incident Management

Q64. Configuration drift is causing repeat incidents across similar devices

Tier: L2. Safety: read-only.

Short answer. You cannot fix drift by hand at scale. Establish a source of truth for intended configuration, detect deviation automatically, and remediate through templates rather than through individual logs. Telemetry and config compliance tooling exist precisely for this.

What is actually going on. Drift happens because devices are configured individually over years by different people under different pressures. Eventually no two “identical” branch routers are identical, and an incident that was fixed on one recurs on the others.

Three capabilities close the gap, and they build on each other:

- **Source of truth.** A version-controlled intended configuration, per device role, from which actual configuration is derived rather than the other way round.
- **Continuous compliance checking.** Automated comparison of running config against intent, alerting on deviation. This is where a config-diff alert catches undeclared changes as a side benefit.
- **Streaming telemetry rather than polling.** SNMP polling every five minutes samples state. Model-driven telemetry pushes state changes as they happen, which turns “we noticed at the next poll” into “we knew immediately.”

The interfaces worth knowing by name: NETCONF and RESTCONF for configuration, YANG as the data model, gNMI for telemetry, and whichever automation tool your organisation uses to apply templates.

How to work it

Detect	[read-only]
- scheduled config collection into version control - automated diff against the intended template - alert on deviation, with the diff attached	
Prevent	[change approval]
- role-based templates as the only sanctioned change path - peer-reviewed changes to the template, not to the device - pre-deployment validation in a lab or a virtual instance	
Modernise	
- streaming telemetry (gNMI) alongside or instead of SNMP polling - NETCONF/RESTCONF with YANG models for structured change - dry-run and idempotent automation, so re-running is safe	

Watch out for. The credible version of this answer is modest. Say that you would start with read-only collection and diffing before automating any change, because an automation tool applying a wrong template to two hundred devices is a far larger outage than the drift it was meant to fix.

References: RFC 6241, NETCONF · RFC 8040, RESTCONF · RFC 7950, YANG 1.1

Part IV. Cloud, Voice and the Physical Layer

Part IV spans the top and the bottom of the stack. Cloud platforms use familiar words for unfamiliar behaviour, voice punishes inconsistency rather than slowness, and the physical layer still causes a large share of faults that are blamed on everything above it.

Chapter 7. Cloud and Load Balancing

Scenarios Q65 to Q72. Tier mix: 6 L2, 2 L3.

Orientation

Cloud networking uses familiar words for unfamiliar things, and that is where most confusion comes from.

A security group is not a firewall rule set in the traditional sense. It is stateful and attached to an instance. Allow traffic outbound and the return is permitted automatically.

A network ACL is stateless. Allow outbound and you must *also* allow the inbound return traffic on ephemeral ports, or the reply is silently dropped. This single asymmetry causes more “it should work” cloud tickets than anything else.

A public IP address does not mean internet access. The subnet also needs a route to an internet gateway. Without it, the address exists and does nothing.

A load balancer only sends traffic to targets it believes are healthy. Almost every load balancer problem is really a health check problem.

Vendor note. The examples in this chapter use AWS, because it is the most common in NOC interviews and the CLI output is easy to reason about. Azure and GCP have equivalent concepts under different names, and the *principles* transfer. The *behaviour* sometimes does not, so verify against your own provider’s documentation rather than assuming.

Scenarios in this chapter

ID	Scenario	Tier
Q65	Health check depth	L2
Q66	Session persistence	L2
Q67	Routing and stateless ACLs	L2
Q68	BGP session instability	L3
Q69	Hybrid routing	L2
Q70	Target registration and health	L2
Q71	Path and physics	L2
Q72	Source address translation	L3

Q65. A load balancer keeps sending traffic to a backend server that’s down

Tier: L2. Safety: read-only.

Short answer. The health check is not testing what your users test. It is probing the wrong port or path, checking too shallowly to notice the failure, or polling too infrequently with too tolerant a threshold.

What is actually going on. A TCP health check confirms only that a listener accepts connections. A hung application process will happily keep accepting connections while returning errors to every request.

An HTTP check against / may hit a static page served by the web tier while the application and database behind it are broken.

The interval and unhealthy threshold multiply into your detection window. A 30-second interval with a threshold of three means 90 seconds of traffic going into a dead server before anything happens.

Deep health checks that exercise the real dependency chain are the right answer, but tune them carefully. A check that queries the database on every probe can become a load problem in its own right.

How to check

```
                                # [read-only]
aws elbv2 describe-target-health --target-group-arn <arn>
aws elbv2 describe-target-group-attributes --target-group-arn <arn>

F5                                [read-only]
  show ltm monitor http <name>
  show ltm pool <name> members

Design questions
- does the check path exercise the application, or just the web server?
- interval x unhealthy threshold = detection window. Acceptable?
- does the check follow redirects, or accept a 3xx as healthy by mistake?
- is there a dedicated health endpoint that verifies real dependencies?
```

Watch out for. The right pattern is a dedicated health endpoint that checks the service's actual dependencies and returns a non-200 when any of them is unavailable. That pattern fixes the problem; tuning the timer only shortens it.

Reference: RFC 9110, HTTP Semantics

Q66. Users experience session drops with a round-robin load balancer

Tier: L2. Safety: read-only.

Short answer. No session persistence. Round robin sends consecutive requests from the same user to different backends, and a backend holding session state locally does not recognise a session created elsewhere.

What is actually going on. Round robin distributes evenly and knows nothing about sessions. An application keeping session state in local memory sees an authenticated user arrive as a stranger on every second request.

Cookie-based persistence is the usual fix and works well behind NAT. Source-IP persistence fails badly when many users share one NAT address, because it concentrates an entire office onto a single backend.

Persistence also complicates maintenance, since draining a node means waiting for sessions to expire. The architecturally better answer is to externalise session state to a shared store so any backend can serve any request.

How to check

```
# [read-only]
aws elbv2 describe-target-group-attributes --target-group-arn <arn>
# stickiness.enabled, stickiness.type, stickiness.lb_cookie.duration_seconds

F5 [read-only]
show ltm persistence persist-records
show ltm pool <name> members is one member carrying everything?

Options, in order of architectural preference
1. Externalise session state (Redis, Memcached, database)
2. Cookie-based persistence works behind NAT
3. Source-IP persistence breaks behind NAT
```

Watch out for. Saying that source-IP persistence is a poor choice when clients sit behind a corporate NAT distinguishes someone who has operated a load balancer from someone who has read about one.

Reference: RFC 6265, HTTP State Management

Q67. Cloud instances can't reach the internet despite having a public IP

Tier: L2. Safety: read-only.

Short answer. A public IP is not connectivity. The subnet needs a route to an internet gateway, the security group must permit the traffic, and the network ACL must permit it in *both* directions because NACLs are stateless.

What is actually going on. Cloud networking separates concerns that on-premises networking bundles together.

An instance can hold a public address while sitting in a subnet whose route table has no `0.0.0.0/0` entry pointing at an internet gateway. That is precisely what makes a subnet private.

The stateful versus stateless distinction is the one interviewers probe. Security groups are stateful, so an outbound allow implies the return is permitted. Network ACLs are stateless, so an outbound allow without a matching inbound rule for the ephemeral port range silently drops every response.

How to check

```

# [read-only]
aws ec2 describe-route-tables \
  --filters Name=association.subnet-id,Values=<subnet>
aws ec2 describe-internet-gateways \
  --filters Name=attachment.vpc-id,Values=<vpc>
aws ec2 describe-security-groups --group-ids <sg>
aws ec2 describe-network-acls \
  --filters Name=association.subnet-id,Values=<subnet>

# Let the platform answer it definitively
aws ec2 create-network-insights-path ...
aws ec2 start-network-insights-analysis ...

Checklist
[ ] Route table has 0.0.0.0/0 pointing at an internet gateway
[ ] Internet gateway attached to the VPC
[ ] Security group permits the outbound traffic
[ ] NACL permits outbound AND inbound ephemeral ports (1024-65535)
[ ] The instance actually has a public or Elastic IP associated

```

Watch out for. Reachability Analyzer answers this deterministically and names the exact blocking component. It is faster and more reliable than checking five configurations by hand.

References: RFC 1918, Private Address Space · AWS VPC Documentation

Q68. A hybrid cloud connection shows intermittent BGP flaps

Tier: L3. Safety: read-only.

Short answer. Read the actual reset reason before theorising. The log and the notification code tell you which class of failure you have, and the classes need different responses.

What is actually going on. Two popular explanations are wrong.

BGP peers **negotiate** the hold time and use the smaller of the two values, so a configured “mismatch” is not by itself a cause of intermittent flapping. And an MD5 authentication mismatch normally prevents the session from establishing at all rather than producing periodic drops.

What actually causes recurring flaps:

Cause	Evidence
Sustained loss or congestion on the circuit	Interface errors, optical levels, provider status
Control-plane policing or CPU starvation	High CPU, CoPP drops, keepalives generated late
Physical layer degradation	Rising CRC, Rx power drifting toward the threshold
LAG member failure under a bundled circuit	One member down or erroring, bundle still “up”
BFD timers too aggressive for the path	BFD down while BGP alone would have stayed up
Provider maintenance or churn	Their status page, their case notes
Max-prefix limit exceeded	A specific notification, session torn down deliberately

How to check

```
[read-only]
show bgp neighbor <ip> | include state|Last reset|hold time|keepalive
show logging | include BGP-5-ADJCHANGE|BGP-3-NOTIFICATION
show interface <if> | include errors|CRC|input rate
show interface transceiver detail          Rx dBm within spec?
show bfd neighbors detail
show processes cpu sorted                  is the control plane starved?
show policy-map control-plane              CoPP drops?
```

Watch out for. Open the provider ticket with the circuit ID, exact flap timestamps, and your optical readings attached. Carriers deprioritise vague reports, and a ticket carrying light levels and timestamps moves through their process substantially faster than one that says “our BGP keeps flapping.”

References: RFC 4271, BGP-4 · RFC 5880, BFD

Q69. An app in the cloud is reachable internally but not from on-prem over VPN

Tier: L2. Safety: read-only.

Short answer. Either route propagation is not delivering the prefixes, the subnet is associated with the wrong route table, the security group does not permit the on-premises source range, or the two address spaces overlap.

What is actually going on. Working internally tells you the application is running and that the rules covering *internal* sources are correct. It does not tell you the security group permits your on-premises CIDR, and those are different rules.

Cloud route tables do not automatically learn from a VPN gateway. Propagation must be enabled explicitly per route table, and a subnet associated with the wrong route table has no path back on-premises even though the tunnel itself is fine.

Overlapping CIDR ranges produce a particularly confusing symptom, because each side resolves the destination locally and traffic never enters the tunnel at all.

How to check

```
# [read-only]
aws ec2 describe-route-tables --route-table-ids <rtb>
aws ec2 describe-vpn-connections --vpn-connection-ids <vpn>
aws ec2 describe-security-groups --group-ids <sg>
```

```

On-premises [read-only]
  show ip route <cloud-cidr>
  show crypto ipsec sa | include encaps|decaps

Checklist
[ ] Route propagation enabled on the correct route table
[ ] The subnet is ASSOCIATED with that route table
[ ] Security group permits the on-premises source CIDR specifically
[ ] NACL permits it in both directions
[ ] No CIDR overlap between on-prem and the VPC
[ ] Return path exists: check both encaps and decaps counters

```

Watch out for. Check the subnet-to-route-table *association*, not just the route table contents. A correct route table that the subnet is not associated with is a very common cause of this symptom.

References: RFC 1918 · AWS VPC Route Tables

Q70. Auto-scaling added instances but the load balancer isn't sending them traffic

Tier: L2. Safety: read-only.

Short answer. Read the target health state and its reason code. The state narrows it down, but each state has several possible reasons and the reason code is what actually names the fault.

What is actually going on. Target health states are commonly over-simplified. In reality:

State	What it can mean
unused	The target group is not used by a listener, or the Availability Zone is not enabled for the load balancer, or the target is in an invalid state, or its IP is unusable. Not simply “not registered.”
initial	Registration is still in progress, or initial health checks have not completed yet.
unhealthy	Registered, checked, and failing. Read the reason code for which check failed.
draining	Deregistration in progress, finishing existing connections.

The relationship between Auto Scaling and load balancer health is also worth stating correctly. The **Auto Scaling health check grace period** prevents Auto Scaling from replacing an instance prematurely while it boots. It does not pause load balancer health checks, which run in parallel and independently decide whether the target receives traffic.

So a grace period that is too short causes premature *replacement*, producing a loop of instances that boot, get killed and get replaced. A health check that is too aggressive for the application's startup causes the target to sit unhealthy and never take traffic. Related, but different failures.

How to check

```

# [read-only]
aws elbv2 describe-target-health --target-group-arn <arn>
#   read both State and Reason for each target

aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names <asg>
#   confirm TargetGroupARNs is populated

aws autoscaling describe-auto-scaling-activities --auto-scaling-group-name <asg>
#   are instances being terminated and replaced in a loop?

Also check
- health check grace period vs actual application startup time
- health check interval and threshold vs startup time
- security group on instances permits the load balancer's check port
- the new Availability Zone's subnet is enabled on the load balancer

```

Watch out for. A grace period shorter than application startup causes an endless scale-up-and-terminate loop that also runs up cost quietly. It is a common real-world failure and a good detail to raise unprompted.

Reference: AWS Target Group Health Checks

Q71. Cross-region cloud traffic has unexpectedly high latency

Tier: L2. Safety: active test.

Short answer. Establish the physical floor for that distance first, then determine whether the measured latency is close to it. If it is, this is geography and not a fault. If it is well above it, look at the path and at whether traffic is leaving the provider's network.

What is actually going on. Start with physics, because it bounds everything else.

Light in fibre travels at roughly **200 km per millisecond**. So:

```

one-way distance / 200 = one-way propagation in ms
round-trip distance / 200 = round-trip propagation in ms

```

A round trip covering 15,000 km cannot be faster than about **75 ms** of pure propagation. Real fibre routes are longer than the straight-line distance, and equipment and queuing add more, so 90 to 110 ms would be entirely normal for that path. What tells you something is wrong is measured latency several times the floor, or high jitter.

On path behaviour, assert only what you can verify. AWS states that traffic between EC2 instances in different Regions stays on the AWS global network, including when public IP addresses are used. Do not claim that cross-region traffic “defaults to the public internet,” because for that provider it is not true. Other providers and other service types behave differently, so verify rather than generalise.

What genuinely causes worse-than-expected cross-region latency: an indirect route, a hop through an on-premises or third-party transit point in a hybrid design, congestion, or an application making many sequential round trips so latency multiplies.

How to check

```
# [active test]
mtr -rwzc 100 <remote-region-endpoint> # does the path leave the provider?
ping -c 100 <remote-endpoint> # jitter as well as latency
```

Assess in this order

1. Theoretical floor for the distance (round trip km / 200)
2. Measured latency: near the floor, or several times it?
3. Jitter: high jitter means congestion; steady high means distance
4. Path: any hops outside the provider's network?
5. Application behaviour: how many sequential round trips per transaction?

Options if it is genuinely path-related

VPC or Transit Gateway peering, provider backbone services,
global accelerator products, or moving the workload closer to the user

Watch out for. Establishing the physical lower bound before escalating prevents an embarrassing carrier ticket. It also stops you raising a P2 against the laws of physics, which does happen.

References: RFC 6349, TCP Throughput Testing · AWS EC2 Instance IP Addressing

Q72. A security group update blocked traffic that used to work through the NAT gateway

Tier: L3. Safety: read-only.

Short answer. Traffic leaving through a NAT gateway arrives at the destination carrying the gateway's public address, not the instance's private one. If the destination's allow list references private addresses or security group IDs, it no longer matches. Confirm it from flow log *records*, not from the flow log configuration.

What is actually going on. Security groups can reference other security groups as sources, which is the correct pattern for traffic staying inside the VPC.

The moment traffic exits through a NAT gateway, the source becomes the gateway's public IP. A security-group reference cannot match that. This bites hardest when a service is reached over the internet rather than through a private endpoint.

The architecturally better fix is often a private endpoint, which keeps traffic on the provider's network. Do not overstate source preservation, though. Whether the client's original address is preserved depends on the endpoint type, the target type, the protocol, and whether client IP preservation is enabled. It is not a universal guarantee.

How to check

```
# [read-only]
aws ec2 describe-nat-gateways --nat-gateway-ids <nat>
#   note the associated Elastic IP

aws ec2 describe-security-groups --group-ids <sg>

# Flow logs: describe-flow-logs lists CONFIGURATIONS, not records.
# To read the actual REJECT entries, query the destination:
aws logs start-query \
  --log-group-name <flow-log-group> \
  --start-time <epoch> --end-time <epoch> \
  --query-string 'fields srcAddr, dstAddr, dstPort, action
                  | filter action = "REJECT"
                  | sort @timestamp desc | limit 50'

# Or, if flow logs go to S3, query them with Athena.

Reading the result
  If srcAddr shows the NAT gateway's Elastic IP, the destination's
  allow list is the problem.

Fixes, in order of preference
1. Private endpoint, so traffic never leaves the private network
2. Allow-list the NAT gateway's Elastic IP at the destination
3. Document the NAT Elastic IP as a managed dependency, so the next
   change does not forget it
```

Watch out for. The command that lists flow log configurations and the query that returns flow log records are different things, and confusing them wastes real time during an incident. Know which one gives you evidence.

References: RFC 3022, Traditional IP NAT · AWS VPC Documentation

Chapter 8. VoIP and QoS

Scenarios Q73 to Q78. Tier mix: 6 L2.

Orientation

Voice traffic is unusual because it cares about *consistency* far more than *speed*. A call needs small packets delivered on time, every time. A file transfer just needs the bytes to arrive eventually.

Three numbers define acceptable voice quality, and knowing them turns a vague complaint into a measurable fact:

Metric	Target	What happens beyond it
One-way latency	under 150 ms	People start talking over each other
Jitter	under 30 ms	Choppy, robotic audio
Packet loss	under 1 percent	Dropouts and clipped words

Two protocols do the work. **SIP** sets up, modifies and tears down the call. **RTP** carries the actual audio. They travel separately, often through different ports and sometimes different paths, which is why a call can connect perfectly and then have no sound.

QoS is how you protect voice from everything else. It works by marking packets, usually DSCP EF for voice, and then treating marked packets preferentially at every congestion point. The catch is that it only works if the marking survives every hop. One device that does not trust DSCP and rewrites it to zero disables the entire scheme downstream.

Scenarios in this chapter

ID	Scenario	Tier
Q73	Time-correlated congestion	L2
Q74	RTP direction and NAT	L2
Q75	Voice over WLAN	L2
Q76	Session timers	L2
Q77	QoS after a change	L2
Q78	Media server capacity	L2

Q73. Calls are fine most of the day but become choppy around 5 PM

Tier: L2. Safety: read-only.

Short answer. Bandwidth contention at end of day from backups, updates and cloud sync. Confirm the correlation on the interface graphs, verify voice is marked correctly and that the marking survives every hop, and check the priority queue is doing its job on the WAN edge.

What is actually going on. Time-correlated degradation is congestion until proven otherwise.

QoS protects voice only if three things all hold: packets are marked at the source, every device on the path preserves that marking, and the egress interface has a priority queue with enough bandwidth reserved. A break at any single hop, typically a switch that does not trust DSCP and zeroes it, silently disables the whole scheme from that point onward.

How to check

```

[read-only]
show policy-map interface <wan-if>
-> priority queue drops? total drops rising at the same time daily?
show interface <wan-if> | include rate|drops
show mls qos interface <if>          is the trust boundary correct?

Per-hop marking check [active test]
Capture at each hop and confirm DSCP 46 (EF) survives:
client -> switch -> WAN router -> provider handoff

```

Watch out for. Quote the three voice targets by number. Numbers define what “good” means, and they let you prove or disprove the complaint from monitoring data rather than from user perception.

References: RFC 3550, RTP · RFC 4594, DiffServ Service Classes · ITU-T G.114

Q74. One-way audio is reported on calls between two sites

Tier: L2. Safety: read-only.

Short answer. RTP is flowing in one direction only. Read the SDP in the signalling to find out where each endpoint expects to receive media, then capture at both ends to see which stream is missing.

What is actually going on. SIP signalling and RTP media travel separately. The SDP body inside the SIP messages advertises the address and port where each endpoint expects to receive audio.

When an endpoint sits behind NAT, that advertised address is a private one the far side cannot reach, so media flows one way and not the other. SIP ALG exists to rewrite those SDP addresses, but implementations are inconsistent and often rewrite them wrongly. That is why disabling ALG sometimes fixes this and sometimes causes it, and why blanket advice about ALGs is unhelpful.

How to check

```

[read-only / active test]
1. Capture SIP signalling; read the SDP c= and m= lines from both ends
2. Is each advertised address routable from the other side?
3. Capture RTP at both ends:
   tcpdump -i <if> -n udp portrange 16384-32768
   Which direction is missing?
4. Check SIP ALG state on every firewall in the path
5. Confirm the RTP port range is permitted in both directions

Cisco [service-affecting]
debug ccsip messages
show voip rtp connections

```

Watch out for. Get the direction right. Media flows *from* the side that is not being heard. If site A cannot hear site B, the missing stream is B to A, so investigate B's outbound path and A's inbound policy. Many candidates fumble this and immediately look in the wrong place.

References: RFC 3261, SIP · RFC 4566, SDP · RFC 3550, RTP

Q75. VoIP quality complaints only come from wireless users, not wired ones

Tier: L2. Safety: read-only.

Short answer. The wireless segment is the only difference, so look there: WMM prioritisation, airtime contention, retransmissions, and roaming delay. Verify WMM is enabled and that DSCP survives the wired-to-wireless translation.

What is actually going on. Wired switches give every port full duplex and dedicated bandwidth. Wireless is a shared, half-duplex medium where every station contends for airtime.

Without WMM mapping voice into the AC_VO access category, a voice frame waits behind bulk transfers. Retransmissions add jitter directly, because a retried frame arrives late rather than not at all, and jitter is what destroys call quality.

Roaming adds a second failure mode. A full reauthentication between access points takes long enough to be audible on a call while being completely invisible on a data session. That is what 802.11r fast transition solves.

This scenario covers **enterprise voice on your own call platform**. Carrier Wi-Fi calling built into a mobile handset is a different problem, because the media is encrypted to the operator's gateway and your WLAN cannot see or mark it. See Q49.

How to check

```
[read-only]
show wlan <id> | include WMM|QoS|Fast Transition
show client detail <mac>
  -> retry percentage, data rate, RSSI, SNR, access category
show ap auto-rf <ap-name>          channel utilisation

Thresholds for voice over wireless
RSSI                better than -67 dBm
SNR                  25 dB or better
Retries              under 10 percent
Channel utilisation  under 40 percent
Roam time            under 50 ms, which needs 802.11r
```

Watch out for. Voice needs a stricter RF design than data: more cell overlap, lower transmit power, and coverage validated at -67 dBm rather than -70 dBm. A network that is perfectly adequate for laptops can be genuinely unfit for handsets, and saying so shows design awareness rather than just troubleshooting skill.

References: IEEE 802.11e / WMM · RFC 8325, Mapping DiffServ to 802.11

Q76. Softphone calls drop at a precise repeating interval

Tier: L2. Safety: read-only.

Short answer. Precision means a timer. Capture the signalling, find which side sends BYE and why, then compare the interval against the configured session timer, registration refresh, and NAT idle timeout until one of them matches.

What is actually going on. SIP session timers require a periodic re-INVITE or UPDATE to confirm the session is alive. If the refresh does not arrive, or its response is lost, both ends tear the call down at expiry.

The other candidate is NAT binding expiry. The firewall drops the UDP mapping after an idle period, the refresh cannot traverse it, and the call ends.

Both produce a precisely repeating interval, which is why the method is to compare configured timer values against the measured one until you find the match rather than guessing.

How to check

```
Capture the SIP dialogue [read-only]
  Session-Expires header value and refresher (uac or uas)
  Is the re-INVITE or UPDATE sent, and answered?
  Who sends the BYE? That side is enforcing the timer.

Compare the measured interval against
  SIP Session-Expires
  SIP registration refresh interval
  Firewall UDP idle timeout
  NAT keepalive interval on the client

Cisco [read-only]
  show sip-ua timers
```

Watch out for. Identifying which side sends the BYE tells you whose configuration to change. Without it you are adjusting timers on both ends and hoping, which is not troubleshooting.

References: RFC 4028, Session Timers in SIP · RFC 3261 · RFC 4787

Q77. After a WAN bandwidth upgrade, voice quality got worse

Tier: L2. Safety: read-only.

Short answer. Something about the change broke the QoS treatment. Look for the policy being detached, misapplied to a new interface, or referencing a stale bandwidth, and check whether the provider handoff now treats your markings differently.

What is actually going on. The popular version of this answer does not hold up. A shaper left at the old rate would preserve the *previous* bottleneck, which is not obviously worse than before. To explain quality getting genuinely *worse*, you need something that changed for the worse:

- **The service policy is no longer attached.** A new or replaced WAN interface does not inherit the old configuration, so voice is now best-effort where it was previously prioritised.

- **The trust boundary was reset.** A replaced interface or line card can default to untrusted, zeroing DSCP on ingress.
- **The provider handoff changed.** A new circuit often means a new handoff with a different marking agreement. Your EF may now be remarked or ignored in their network.
- **The parent shape rate is wrong in the other direction.** Shaping to more than the circuit can actually deliver moves the queue into the provider's network, where your QoS policy has no effect at all.
- **Queue or serialisation parameters changed** with the new interface type or speed.

A stale shaper set to the old, lower rate is still worth finding and fixing, but as a capacity problem rather than as the explanation for worse audio.

How to check

```

[read-only]
show policy-map interface <wan-if>      is a policy attached at all?
show run interface <wan-if> | include bandwidth|service-policy
show interface <wan-if> | include BW|rate
show mls qos interface <wan-if>          trust state after the change
show policy-map interface | include drops

Fix
interface <wan-if>                        [change approval]
  bandwidth <new-kbps>                    reference for percentage-based classes
  service-policy output PARENT-SHAPER
policy-map PARENT-SHAPER
  class class-default
    shape average <rate at or just below the contracted rate>
  service-policy CHILD-QOS

```

Watch out for. Add “verify and retune QoS, and confirm the service policy is still attached” as a standard line item in every circuit change record. A checklist item prevents the problem; a troubleshooting victory only fixes it once.

References: RFC 2474, DiffServ Field · RFC 4594

Q78. Conference calls degrade only above about 10 participants

Tier: L2. Safety: read-only.

Short answer. A resource ceiling on the media server or bridge: CPU, memory, transcoding capacity, or the aggregate bandwidth of the mixed streams. A consistent participant threshold points at capacity, not at the network.

What is actually going on. A mixing bridge decodes every incoming stream, mixes, and re-encodes for each participant. Cost scales sharply with participant count and even more sharply when participants use different codecs and require transcoding.

Bandwidth scales too. Twenty participants at 100 kbps each is 2 Mbps inbound, plus the mixed output streams.

A consistent threshold is the signature of a resource limit. Identify which resource by watching CPU, memory, transcoding session count and interface utilisation as participants join.

How to check

```
While a call ramps up, watch [read-only]
media server CPU and memory
active transcoding sessions vs licensed or hardware capacity
interface utilisation on the media server uplink
loss and jitter on the RTP streams

Cisco [read-only]
show call active voice brief
show voice dsp group all          DSP resource exhaustion

How to read it
Consistent threshold      -> capacity limit
Random degradation        -> network path problem
Only certain participants -> codec mismatch forcing transcoding
```

Watch out for. Check whether degradation hits everyone or only some participants. Selective degradation usually means a codec mismatch driving transcoding for a subset, and that is fixed by standardising the codec rather than by buying hardware.

References: RFC 3550, RTP · RFC 7667, RTP Topologies

Chapter 9. Physical Layer and Cabling

Scenarios Q79 to Q84. Tier mix: 4 L1, 2 L2.

Orientation

Layer 1 is where a surprising share of “network” problems actually live, and it is the layer most often skipped because it involves walking somewhere with a torch.

A few things worth knowing before you start:

A link light means almost nothing. It confirms two devices detected each other electrically. It does not confirm the cable meets its specification, that all four pairs work, or that data is arriving intact.

Copper has a distance limit of 100 metres, and that includes the patch cords at both ends, not just the run inside the wall.

Fibre is about light budget. Every connector, splice and kilometre of glass costs you a fraction of a decibel. Exceed the budget and the link fails or errors, and the most common cause by a wide margin is a dirty connector.

Counters tell the story. CRC errors mean corruption. Late collisions mean duplex mismatch. Learn to read them and you will diagnose most Layer 1 faults from the CLI before touching anything.

Scenarios in this chapter

ID	Scenario	Tier
Q79	Termination quality	L1
Q80	Optical loss	L1
Q81	Segment isolation	L1
Q82	Pair count and category	L1
Q83	Optic compatibility	L2
Q84	Hardware damage	L2

Q79. A newly terminated cable passes a link light but has a high error rate

Tier: L1. Safety: read-only.

Short answer. Poor termination. Either the pairs do not follow the T568A or T568B assignment, or they were untwisted too far at the connector, causing crosstalk. Re-terminate and test with a certifier, not a continuity tester.

What is actually going on. A link light requires far less than a compliant cable. Fast Ethernet uses two pairs, so a badly wired cable can link at 100 Mbps while failing entirely at gigabit.

The specific fault is usually a **split pair**, where the wires sit in the correct pin positions but are paired incorrectly. Continuity testers pass this happily while crosstalk is severe. Untwisting more than about 13 mm at the termination destroys the noise cancellation the twist provides.

Only a certifier measuring NEXT, return loss and insertion loss will find either problem.

How to check

Switch side	[read-only]
show interface <if> counters errors	CRC, alignment, symbol errors
show interface <if> status	did it negotiate gigabit or not?
Cable side	
Certifier test (Fluke DSX or equivalent), not a continuity tester	
Check wiremap, NEXT, PS-NEXT, return loss, insertion loss, length	
Common findings	
split pair	passes continuity, fails NEXT
untwist beyond 13 mm	fails NEXT at the connector
mixed T568A / T568B	crossover where a straight cable was intended
run beyond 100 m	fails insertion loss

Watch out for. A cable negotiating 100 Mbps rather than 1 Gbps strongly suggests a two-pair fault. That single observation from `show interface status` narrows the diagnosis before anyone walks to the patch panel.

References: ANSI/TIA Standards · IEEE 802.3ab, 1000BASE-T

Q80. A fibre link shows light-loss readings well beyond spec

Tier: L1. Safety: read-only.

Short answer. Contamination, damage, or too much loss along the path. Clean the connectors first, because that resolves the majority of cases, then measure and locate faults with an OTDR if it is still out of spec.

What is actually going on. Contamination is the leading cause of optical failure in the field. A single dust particle on a core narrower than a human hair scatters enough light to break the budget, and inspecting a connector without a scope tells you nothing at all.

Beyond contamination, the budget adds up:

Element	Typical loss
Mated connector pair	0.3 to 0.75 dB
Fusion splice	around 0.1 dB
Multi-mode fibre at 850 nm	around 3.0 dB per km
Single-mode fibre at 1310 nm	around 0.4 dB per km

Mixing fibre types, whether single-mode with multi-mode or 50 micron with 62.5 micron, produces large mismatch losses that no amount of cleaning fixes.

How to check

```

[read-only]
show interface transceiver detail
  -> Tx power, Rx power, and the low and high warning thresholds

Field procedure
1. Inspect with a fibre scope BEFORE cleaning, and after
2. Clean with a proper cleaner. Not compressed air. Not a sleeve.
3. Re-measure Rx power
4. Still out of spec? OTDR to locate the event by distance

```

Watch out for. Inspect before cleaning as well as after. Photographing the contamination gives you evidence for the incident record and, in a contracted environment, for the conversation with whoever installed it.

References: ISO/IEC 11801 · IEC 61300-3-35, Connector Cleanliness

Q81. A patch panel port works but the same run at the wall jack doesn't

Tier: L1. Safety: read-only.

Short answer. The fault is between the patch panel and the wall jack, so test each segment individually to find which one fails.

What is actually going on. A structured cabling channel is several segments in series, and a working test at the panel only validates the equipment side.

The horizontal run inside the wall is the segment most likely to have been damaged, by drilling, rodents, tight bends, or being pulled during someone else's work, and it is the segment nobody can see. The jack termination is the next candidate, particularly if it was terminated at a different time or by a different installer than the panel.

How to check

```

Test each segment separately
1. Patch panel port to switch      known good?
2. Panel to jack, permanent link test  certifier at both ends
3. Jack to device, with a known-good patch cord

Certifier readings that name the fault
Length shorter than expected      break in the run at that distance
Wiremap fault at one end only     that termination
High NEXT on the permanent link   run damage or bad termination
Open on specific pins             partial break or bad punch-down

```

Watch out for. Record the certifier's reported distance to the fault. It turns a wall-opening exercise into a targeted repair at a known point, which is the difference between a two-hour job and a two-day one.

Reference: ANSI/TIA Standards

Q82. A copper run works at 100 Mbps but fails at gigabit

Tier: L1. Safety: read-only.

Short answer. Gigabit uses all four pairs; Fast Ethernet uses two. A fault on the pairs that only gigabit needs, a run over 100 metres, or degraded crosstalk performance will produce exactly this. Certify the run and check the total channel length.

What is actually going on. 100BASE-TX transmits on two pairs and ignores the other two entirely, so any fault on those pairs is invisible until you try to run gigabit. 1000BASE-T uses all four pairs bidirectionally with echo cancellation, which is far less tolerant of crosstalk and return loss.

A word of care on cable category, because this is commonly stated wrongly. IEEE 802.3ab specified 1000BASE-T to operate over four-pair **Category 5** balanced copper cabling. Cat5e is the safer certification baseline and is what you should install today, because it adds specified limits for return loss and far-end crosstalk that make gigabit reliable in practice. But saying “Cat5 was never specified for gigabit” is not accurate, and an interviewer who knows the standard will notice.

The practical point stands: an old Cat5 run that was marginal at 100 Mbps often fails certification for gigabit, and the fix is to re-terminate or replace it.

How to check

```
show interface <if> status          [read-only]
show interface <if> capabilities    what did it negotiate?
show interface <if> counters errors

Certify and confirm
[ ] All four pairs continuous and correctly mapped
[ ] Cable category printed on the jacket
[ ] Total channel length under 100 m INCLUDING patch cords
[ ] NEXT and return loss within the relevant category limits
[ ] No mid-span joints or unrated couplers in the run
```

Watch out for. Cheap unrated couplers used to extend a run are a frequent hidden cause. They pass continuity, degrade crosstalk badly, and are usually tucked above a ceiling tile where nobody thinks to look.

References: IEEE 802.3ab interpretation, 1000BASE-T over Category 5 · ANSI/TIA Standards

Q83. An SFP transceiver shows link down despite being properly seated

Tier: L2. Safety: read-only.

Short answer. Read the transceiver diagnostics before swapping hardware. Then think in terms of link budget and PMD compatibility rather than absolute rules, because the interesting failures are the marginal ones.

What is actually going on. Both ends of an optical link must agree on the physical medium dependent type, wavelength and fibre type, and the received power must land inside the receiver’s usable window.

Two things go wrong in ways people describe too absolutely:

Fibre type mismatch. A single-mode optic launched into multi-mode fibre is out of specification and should not be relied on. It will sometimes link over very short distances, which is worse than failing cleanly because it produces an intermittent link that works during commissioning and fails later. Mode-conditioning patch cords exist for specific documented combinations and are not a general workaround.

Reach class mismatch. Whether two different reach classes interoperate depends on launch power, receiver sensitivity, receiver overload, wavelength and the actual link loss. A long-reach optic driving a short fibre into a short-reach receiver can *overload* it, which fails even though there is plenty of light. Attenuators exist precisely for that case.

So the useful mental model is: compare PMD types, then compare Tx power against the far end's sensitivity and overload limits, then account for the loss in between.

How to check

```
[read-only]
show interface transceiver detail
  -> type, wavelength, Tx power, Rx power
show interface <if> transceiver properties
show logging | include GBIC|SFP|TRANSCEIVER|UNSUPPORTED
show inventory

Reading Rx power
  Very low or no light      reversed Tx-Rx, a break, or far-end Tx failed
  Above the overload limit  too much power; add an attenuator
  Within spec, link still down PMD mismatch, encoding, speed, or vendor lock

Confirm both ends agree on
  PMD type, wavelength (850 / 1310 / 1550 nm), fibre type, reach class
```

Watch out for. Swap the two fibre strands as a first test. Reversed Tx and Rx costs nothing to check, accounts for a meaningful share of these tickets, and is especially common right after patching work.

References: IEEE 802.3 · SFF specifications, diagnostic monitoring

Q84. Multiple ports on the same switch card show errors after a power event

Tier: L2. Safety: read-only.

Short answer. Probable hardware damage from the surge or brownout. Check hardware and diagnostic log messages, run online diagnostics, review power and environmental status, and plan to replace the affected line card rather than chase individual ports.

What is actually going on. A surge or brownout stresses components without necessarily destroying them, so the failure shows up as intermittent errors that get worse over time rather than as a clean failure.

Multiple ports on the same card failing together points at shared components, meaning the card's PHY, ASIC or power rail, rather than at cabling. Cable faults do not correlate by card position.

The operationally important point is that partial hardware failure gets worse. A card showing post-surge errors should be scheduled for replacement, not monitored indefinitely and forgotten.

How to check

```
[read-only]
show logging | include HARDWARE|DIAG|POWER|ASIC|PARITY
show diagnostic result module <n> detail
show module
show environment all
show power status
show interface counters errors          do failures cluster by port group?
```

Evidence for hardware rather than cabling

- errors cluster on one card or one port ASIC group
- errors persist after moving the cable to a different card
- diagnostics report failures
- the error rate is trending upward over days

Watch out for. Move a suspect connection to a different line card. If the errors follow the port rather than the cable, the hardware is confirmed and the vendor RMA conversation becomes straightforward. Also check whether the UPS actually held during the event, because an untested UPS is a common contributing cause that nobody thinks to verify.

References: IEEE 802.3 · IEEE 1100, Powering Electronic Equipment

Part V. Escalation and Professional Practice

These scenarios test judgement, communication and honesty under pressure. They are answered best with your own experience, framed using STAR, rather than with memorised process.

Chapter 10. Escalation, ITIL and Communication

Scenarios Q85 to Q91. Tier mix: 1 L1, 6 L2.

Orientation

Do not memorise these answers. Interviewers hear textbook process answers constantly and discount them within a sentence.

Take the principle from each scenario, then rebuild the answer around an incident you actually worked, using **STAR**:

- **Situation.** What was happening, briefly.
- **Task.** What you specifically were responsible for.
- **Action.** What you did, in the first person. Not “we,” which hides your contribution.
- **Result.** What happened, and what changed afterwards.

Prepare three or four real stories in advance and you can adapt them to almost any behavioural question in this category. A specific account of a real Friday night outage beats a correct description of change management every single time.

If you are early in your career and genuinely have no incidents to draw on, say so honestly and describe how you would approach it. Interviewers respect that far more than an invented story, and invented stories fall apart under one follow-up question.

Each scenario below ends with a **STAR prompt** to help you build your own version.

Scenarios in this chapter

ID	Scenario	Tier
Q85	Vendor escalation	L2
Q86	Cross-team conflict	L2
Q87	Triage under load	L2
Q88	Problem management	L2
Q89	Stakeholder communication	L2
Q90	Ownership and restoration	L2
Q91	Situational awareness	L1

Q85. A vendor’s ticket for a circuit outage isn’t progressing

Tier: L2. Safety: read-only.

Short answer. Escalate inside the vendor using their published path and your contract’s SLA terms, quoting the circuit ID and case number. In parallel, open an internal bridge and progress any workaround independently, rather than waiting.

What is actually going on. Vendor tickets stall for predictable reasons: the case sat in a queue, it went to the wrong team, or it was closed as “no fault found” without telling anyone.

Two levers actually move them: the contractual SLA and a named escalation contact. Knowing both *before* the incident matters far more than anything you say during it.

Running the workaround in parallel is the part junior engineers skip. Waiting for a vendor while a redundant path sits unused is a choice, and usually the wrong one.

How to work it

Escalating effectively

- reference the circuit ID, case number, and elapsed time
- state the SLA clause and time remaining or already breached
- ask for a named engineer and a callback time, not a status
- request the duty manager if the callback is missed
- escalate through your account manager in parallel for major outages

In parallel, internally

- is there a redundant path? Fail over now, not after they answer
- keep stakeholders updated on your cadence, not the vendor's
- log every vendor interaction with a timestamp for the review

STAR prompt. Describe a time you escalated to a carrier or vendor. What evidence did you present, what did you do while waiting, and what changed as a result?

Reference: ITIL 4 Supplier Management

Q86. You suspect another team’s change caused an issue but they deny it

Tier: L2. Safety: read-only.

Short answer. Present evidence rather than an accusation. Logs and timestamps correlating the change window with symptom onset, plus the mechanism you believe links them. Ask for a joint session, and be explicit that you might be wrong.

What is actually going on. Denial is usually genuine belief, not obstruction. The other team is looking at their own domain, where everything appears healthy.

Correlation is also not causation, and a confident accusation that turns out to be wrong damages a working relationship you will need again next month. Framing it as a shared investigation keeps the other team engaged and, in practice, gets to the answer faster.

If a rollback is low risk, proposing it as a hypothesis test rather than as a demand often ends the standoff immediately.

How to work it

Present

- exact symptom onset time from monitoring
- exact change window from the change record
- the correlation, described as correlation, not proof
- a plausible mechanism linking the two
- what you have already ruled out on your own side

Propose

- a joint session, both teams looking at their own telemetry
- a controlled test: roll back in a window and observe
- blameless framing: "help me rule this in or out"

Avoid

- accusation in a group channel
- escalating to management before a technical conversation

STAR prompt. Describe a disagreement with another team during an incident. How did you establish the facts, and what did you do differently afterwards?

Reference: Google SRE, Postmortem Culture

Q87. How do you triage several simultaneous incidents with limited staff?

Tier: L2. Safety: read-only.

Short answer. Rank by business impact and affected user count rather than arrival order, work the highest-impact one first, communicate realistic timelines for the rest, and ask for extra resource early rather than after the SLAs are already at risk.

What is actually going on. The failure mode under load is context switching. An engineer bouncing between four incidents finishes none of them and resolves everything more slowly than if they had worked them in sequence.

Deliberate sequencing is faster even though it feels worse for whoever is placed later, which is why communicating the sequencing matters as much as choosing it.

Quick wins are the legitimate exception. If one incident closes in five minutes, closing it reduces the queue and the noise, and that is worth doing out of order.

How to work it

Rank by

1. Business impact: revenue, safety, regulatory, customer-facing
2. Number of users affected
3. Whether impact is expanding or contained
4. Time already elapsed against SLA
5. Effort to resolve: a genuine five-minute fix earns queue jumping

Then

- one owner per incident; avoid shared ownership
- communicate the sequencing to stakeholders explicitly
- request additional resource EARLY, not at the breach point
- re-rank if impact grows; do not defend the original order out of pride

STAR prompt. Describe a shift where you handled several incidents at once. How did you sequence them, and what did you tell the people whose incident you deprioritised?

Reference: ITIL 4 Incident Management

Q88. A recurring issue keeps getting temporary fixes

Tier: L2. Safety: read-only.

Short answer. Raise a formal problem record, separate from the incident queue, with an owner and time allocated to investigate. Bring evidence of the cumulative cost, because that is what justifies the investment.

What is actually going on. Incident management restores service. Problem management removes the cause. They compete for the same people, and incidents always win because they are urgent. Without a separate record and separate time, the root cause is never investigated.

The argument that changes minds is quantitative. An incident recurring weekly and consuming two hours each time costs more over a year than the permanent fix. Presenting it that way converts a technical preference into a business case, which is what gets it approved.

How to work it

Build the case

- frequency and trend over the last three to six months
- total downtime and total engineer hours consumed
- business impact per occurrence
- estimated cost of permanent remediation
- an explicit comparison of the two numbers

Then

- open a problem record with a named owner and a due date
- allocate investigation time outside the incident queue
- document the known error and workaround while investigating
- track to closure, then report the reduction afterwards

STAR prompt. Describe a recurring problem you drove to permanent resolution. How did you justify the time, and what was the measurable result?

Reference: ITIL 4 Problem Management

Q89. How do you communicate with stakeholders during a major outage?

Tier: L2. Safety: read-only.

Short answer. Separate the communication role from the engineering role, publish updates on a fixed cadence through one channel, and lead with business impact rather than technical detail. State uncertainty honestly rather than inventing an ETA.

What is actually going on. This question often arrives in a specific form: an executive is demanding constant updates during a major outage. The demand is legitimate. The interruption pattern is the problem.

Every ad hoc status request costs an engineer context and time, and during a serious incident those interruptions extend the outage the executive is asking about. A stated cadence satisfies the underlying need, which is confidence that the situation is under control, not raw technical detail. Updates delivered on schedule even when nothing has changed are what build that confidence.

The content matters as much as the rhythm. Non-technical stakeholders need four things: what is broken from their perspective, how long it will last, what is being done, and when they will hear next. Technical detail offered unprompted signals you are more interested in the problem than in the audience.

Analogies help when they are accurate and hurt when they are strained. Use one only if it survives a follow-up question.

How to work it

Structure	
Incident Manager	owns the incident, makes decisions
Incident Communicator	owns all stakeholder updates
Technical lead	owns the fix, fields no status requests
Update format, every 15 to 20 minutes	
1. Impact:	"Users at the Birmingham office cannot access email or files."
2. Scope:	"About 200 people. Other sites are unaffected."
3. Cause:	"The main internet circuit to that office has failed."
4. Action:	"The carrier is engaged; we are moving to the backup link."
5. Timeline:	"Backup should restore service within 30 minutes."
6. Next:	"I will update you at 14:20 either way."
Avoid	
jargon, invented ETAs, unrequested technical detail	

Watch out for. Never invent an ETA under pressure. "We do not have a reliable estimate yet; the next update is at 14:20" preserves credibility. A missed invented ETA destroys it, and you will need that credibility for the rest of the incident.

STAR prompt. Describe explaining a serious technical failure to a senior non-technical stakeholder. What did you include, what did you leave out, and how did they respond?

Reference: ITIL 4 Incident Management

Q90. A change you made caused unexpected downtime. What's your next step?

Tier: L2. Safety: change approval.

Short answer. Stop the change, invoke incident and change authority, and restore service by the safest available means according to the approved decision criteria. In most cases that is the pre-approved rollback, but not always. Then notify stakeholders promptly and honestly.

What is actually going on. Two things are being tested here, and only one of them is technical.

The technical judgement is about restoration. “Roll back immediately” is the right instinct most of the time, and diagnosing before restoring extends the outage while you satisfy your curiosity. But rollback is not universally safe or even available:

- A firmware upgrade or a schema migration may not be cleanly reversible
- Rolling back may itself require a reload and extend the outage
- Partial rollback can leave the system in a state neither version expects
- If the change has already been partly consumed by other systems, reverting can cause a second, larger failure

So the correct framing is: restore service by the safest available means, using the rollback decision criteria that were agreed *before* the window opened. Sometimes a controlled forward fix genuinely is lower risk, and a senior engineer is expected to make that call with the change authority rather than alone.

The second thing being tested is disclosure. Attempting to fix quietly and hoping nobody noticed is the answer that ends interviews. Prompt, plain disclosure is what a mature organisation expects and what a blameless culture exists to make safe.

How to work it

Order of operations

1. Stop the change. Make no further modifications.
2. Declare the incident and engage change authority.
3. Restore service by the safest available means:
 - pre-approved rollback, if it is available and safe
 - controlled forward fix, if rollback is riskier
 - failover to a redundant path, if one exists
4. Verify restoration end to end, as a user would.
5. Notify stakeholders: what happened, current state, next steps.
6. Preserve evidence: logs, config diffs, timestamps.
7. Update the change record with the actual outcome.
8. Blameless review: why was this risk not identified at approval?

Do not

- attempt an unplanned forward fix alone, under pressure
- conceal or minimise the cause
- close the change as successful

STAR prompt. Describe a change you made that caused an outage. What did you do in the first five minutes, how did you decide between rollback and forward fix, and what control did you add afterwards?

References: ITIL 4 Change Enablement • Google SRE, Postmortem Culture

Q91. How do you stay aware of network changes across departments?

Tier: L1. Safety: read-only.

Short answer. Review the change calendar and CAB notifications as routine, keep a shift handover log that carries context forward, and build change visibility into the incident workflow so checking is a required step rather than a good habit.

What is actually going on. Change awareness fails when it depends on individual diligence. Making it a process step is what makes it reliable.

The strongest control is automated configuration-diff alerting, because it catches changes nobody declared. The changes that cause outages are disproportionately the ones that bypassed the process, and those are exactly the ones a calendar will never show you.

How to work it

Routine

- change calendar reviewed at the start of every shift
- NOC on the distribution list for all CAB notifications
- shift handover log carrying open items and recent changes forward
- a short daily standup covering the next 24 hours of planned work

Systematic

- automated configuration-diff alerting on all network devices
- "changes in the last 24h checked" as a required incident field
- change timestamps correlated automatically against alert timestamps
- relationships with the teams whose changes affect you, built before you need them

STAR prompt. Describe a time an undeclared change caused an incident. What did you change about how your team maintained awareness afterwards?

Reference: ITIL 4 Change Enablement

Part VI. Hosts, Supporting Protocols and IPv6

Part VI covers the faults that sit just outside the network but get reported as network problems: host routing and firewalls, time and monitoring protocols, and the IPv6 behaviour that many dual-stack networks enable without planning for.

Chapter 11. Server and OS Networking

Scenarios Q92 to Q96. Tier mix: 5 L2.

Orientation

A large share of “network” tickets turn out to be host problems, and a NOC engineer who can prove that quickly saves everyone a lot of arguing.

Three host-side concepts are worth understanding because they cause faults that look exactly like network faults:

The host has its own routing table. A server can be perfectly connected and still fail to reach anything because it has no default route or the wrong one.

The host has its own firewall. iptables, nftables, firewalld and Windows Defender Firewall all drop traffic that arrived over a healthy network.

Multi-homed hosts behave differently by operating system. Linux and Windows make different assumptions about which interface owns an address and which one replies. Those differences cause genuinely confusing, intermittent faults.

One command is worth memorising: `ip route get <destination>` on Linux tells you exactly which route and which source address the kernel will use. It answers in one line what reading the routing table and reasoning about it takes several minutes to work out.

Scenarios in this chapter

ID	Scenario	Tier
Q92	Host routing and filtering	L2
Q93	Driver and firewall profile	L2
Q94	Multi-homing behaviour	L2
Q95	Host throughput	L2
Q96	Certificates behind load balancing	L2

Q92. A Linux server can ping its gateway but not any other host

Tier: L2. Safety: read-only.

Short answer. Reaching the gateway shows the local interface, ARP and the local segment are working for ICMP. The fault is beyond the local subnet: routing, host filtering, or reverse path filtering.

What is actually going on. Local subnet traffic needs only ARP and a correct interface address. Anything beyond it needs a route.

A missing or badly weighted default route is the most common cause, particularly on a multi-homed host or after a NetworkManager change. Host firewalls are next: a default DROP on OUTPUT, or a rule set permitting established connections while denying new outbound ones, produces exactly this.

Reverse path filtering is the subtler third cause on multi-homed hosts. The kernel drops replies that would leave by a different interface than the one they arrived on, which looks like the network dropping traffic and is not.

How to check

```

# [read-only]
ip route show                # default route present? right interface?
ip route get 8.8.8.8         # which route will actually be used?
ip addr show
ip neigh show                # ARP: is the gateway resolved?

iptables -L -v -n            # legacy
nft list ruleset              # nftables
firewall-cmd --list-all     # firewalld

sysctl net.ipv4.conf.all.rp_filter
sysctl net.ipv4.ip_forward   # only relevant if it should route

# [active test]
traceroute -n 8.8.8.8

```

Watch out for. `ip route get` also shows the source address the kernel will select. On a multi-homed host that is frequently the actual problem, because the reply comes from an address the far end did not expect.

References: RFC 1122, Requirements for Internet Hosts · Linux `ip-route(8)`

Q93. A Windows server loses connectivity right after an update

Tier: L2. Safety: read-only.

Short answer. Two leading candidates. A NIC driver replaced by the update, losing offload, VLAN or teaming settings. Or the network profile no longer being recognised as the domain network, so a more restrictive firewall profile applies.

What is actually going on. Windows Update can replace a working vendor NIC driver with a generic one, which sometimes drops advanced settings including teaming configuration.

The profile issue needs care, because the popular “fix” for it does not work. Windows assigns the domain profile automatically when Network Location Awareness successfully authenticates against a domain controller. You cannot simply force that assignment with a cmdlet, and attempting to set the domain category manually fails.

So when a server drops into the Public profile after an update, the profile is a *symptom*. The real question is why the machine could not contact a domain controller at boot, which usually means the NIC initialised after Network Location Awareness ran, or DNS was not answering yet, or the driver problem above delayed the interface.

Fixing DNS and domain controller reachability restores the correct profile on its own.

How to check

```
# [read-only]
Get-NetAdapter | Select Name, Status, LinkSpeed, DriverVersion, DriverDate
Get-NetConnectionProfile           # Domain, Private, or Public?
Get-NetFirewallProfile | Select Name, Enabled
Get-NetAdapterAdvancedProperty -Name "Ethernet"
Get-NetLbfoTeam                    # did teaming survive the update?

# Why is it not identifying the domain network?
Test-NetConnection <domain-controller> -Port 389
Resolve-DnsName <domain-fqdn>
nltest /dsgetdc:<domain>

# Recent updates
Get-HotFix | Sort InstalledOn -Descending | Select -First 10

Remediation [change approval]
- roll the NIC driver back (Device Manager > Driver > Roll Back Driver)
- fix DNS and domain controller reachability, then restart
  the Network Location Awareness service so the profile re-evaluates
- re-apply teaming and advanced adapter settings if they were lost
```

Watch out for. On a server with teamed NICs, check whether the team survived at all. Driver replacement can silently break teaming and leave the server on a single adapter. Everything works fine until the day that adapter's switch is rebooted, and then it does not.

Reference: Microsoft, domain profile detection troubleshooting

Q94. Two NICs on a server in the same subnet cause intermittent connectivity

Tier: L2. Safety: read-only.

Short answer. Two interfaces in one broadcast domain create ambiguity in which interface sends, which one replies, and which address the far end sees. Team or bond them, or move one to a different subnet. Two independent NICs in one subnet is a misconfiguration, not a redundancy design.

What is actually going on. The common explanation of the mechanism is wrong. Two independent NICs have two different MAC addresses, so you will not normally see one MAC flapping between switch ports. That is what happens with a Layer 2 loop, not with this.

What actually goes wrong is operating system dependent:

On Linux (weak host model by default). Any interface will answer an ARP request for any local address, so the far end may learn either MAC for the address it wants. This is ARP flux. The reply may then leave by the other interface entirely, and source address selection may pick an address the far end did not send to. `arp_ignore` and `arp_announce` control this behaviour, and reverse path filtering may then drop the return traffic.

On Windows (strong host model since Vista). An interface will not accept traffic for an address bound to a different interface, which prevents some of the Linux behaviour but introduces its own:

outbound traffic follows interface metric, and a reply can leave via the interface with the better metric rather than the one the request arrived on, producing asymmetry that stateful devices drop.

On both. Two default gateways in the same subnet gives you two equally valid exit paths and no deterministic choice, which is where the intermittency comes from.

How to check

Linux	[read-only]
ip route show	two routes to the same subnet?
ip route get <remote-host>	which interface and source address?
sysctl net.ipv4.conf.all.arp_ignore	
sysctl net.ipv4.conf.all.arp_announce	
sysctl net.ipv4.conf.all.rp_filter	
cat /proc/net/bonding/bond0	if bonded
Windows	[read-only]
Get-NetIPInterface Sort InterfaceMetric	
Get-NetRoute -DestinationPrefix 0.0.0.0/0	two default gateways?
Get-NetLbfoTeam	is teaming configured?
From the far end	[read-only]
arp -a <server-ip>	which MAC did it learn?
Does the answer change over time?	
Correct designs	[change approval]
1. NIC teaming or bonding: one logical interface, one MAC, one address	
2. Two interfaces in genuinely different subnets, with deliberate routing and a single default gateway	

Watch out for. Say explicitly that dual-homing into the same subnet without teaming is a misconfiguration rather than a redundancy strategy. That framing is what the interviewer is checking for, and it is more useful than reciting sysctl names.

References: RFC 826, ARP · IEEE 802.1AX, Link Aggregation

Q95. A server's throughput is capped well below the NIC's rated speed

Tier: L2. Safety: active test.

Short answer. Work down the stack: negotiated speed and duplex, offload settings, virtual switch or hypervisor limits, CPU saturation on a single queue, then the TCP stack itself. Measure with iperf3 before and after each change so you know what actually helped.

What is actually going on. Several unrelated bottlenecks produce the same symptom.

A half-duplex negotiation caps throughput and generates late collisions. Disabled offloads push per-packet work onto the CPU, and a single saturated core limits the whole interface. In virtualised environments the vSwitch, port group or a hypervisor shaping policy can cap a VM regardless of the guest configuration.

On high-latency paths the limit may not be the host at all. If the TCP window is too small for the bandwidth-delay product, the pipe never fills and no amount of NIC tuning changes that.

How to check

```

# [read-only]
ethtool eth0          # negotiated speed and duplex
ethtool -k eth0       # offload settings
ethtool -S eth0       # per-queue drops and errors
ethtool -g eth0       # ring buffer sizes
mpstat -P ALL 1       # is one core pinned at 100 percent?
ss -ti               # cwnd, rtt, retransmits per socket
sysctl net.core.rmem_max net.ipv4.tcp_rmem

# [active test]
iperf3 -c <server> -t 30 # single stream
iperf3 -c <server> -t 30 -P 8 # parallel streams

Bandwidth-delay product
required window (bytes) = bandwidth (bits/s) x RTT (seconds) / 8
Example: 1 Gbps at 80 ms RTT needs roughly a 10 MB window

```

Watch out for. Compare single-stream against parallel-stream results. If eight streams reach line rate and one does not, the limit is per-flow, meaning window size, a single CPU queue, or a shaper, and not the interface itself.

References: RFC 6349, TCP Throughput Testing · RFC 7323, TCP Extensions for High Performance

Q96. A hostname resolves fine but TLS certificate errors happen intermittently

Tier: L2. Safety: active test.

Short answer. The name resolves to more than one server and they are not all presenting the same certificate, or one node holds a stale or expired one. Test each backend individually rather than through the load-balanced name.

What is actually going on. DNS round robin or a load balancer spreads connections across several nodes. A client reaching a node with a correct certificate succeeds; a client reaching a node with an expired, renamed or not-yet-renewed certificate fails. The intermittency is proportional to how many nodes are wrong.

Related causes look similar: a Subject Alternative Name list that omits one of the names clients use, so failure depends on which name was requested; SNI misconfiguration on a shared listener returning the default certificate; and a missing intermediate on one node only, which fails for clients that have not cached that intermediate.

How to check

```
# Find every backend, then test each one directly      # [active test]
dig +short server.example.com

for ip in 10.1.1.10 10.1.1.11 10.1.1.12; do
    echo "== $ip"
    openssl s_client -connect $ip:443 -servername server.example.com \
        </dev/null 2>/dev/null \
        | openssl x509 -noout -serial -subject -dates -ext subjectAltName
done

Check on every node
[ ] Same certificate, same serial number
[ ] Not expired, and not renewed on only some nodes
[ ] SAN list covers every hostname clients use
[ ] Full chain served, including intermediates
[ ] SNI configured so the right certificate is selected
```

Watch out for. Comparing serial numbers across backends finds the odd node in seconds. Testing through the load-balanced name is unreliable because you cannot control which backend you reach, and you may test the healthy ones five times in a row and conclude nothing is wrong.

References: RFC 5280, X.509 Certificate Profile · RFC 6125, Service Identity · RFC 6066, TLS Extensions

Chapter 12. Miscellaneous Protocols

Scenarios Q97 to Q101. Tier mix: 4 L2, 1 L3.

Orientation

These are the protocols that quietly hold everything else together, and they tend to fail in ways that look like something else entirely.

Time breaks authentication. Kerberos and TLS both compare timestamps, so a clock that drifts far enough produces “access denied” and “certificate invalid,” not “your clock is wrong.”

SNMP is how most monitoring reads device state. Change a community string or an ACL and your monitoring goes dark while the network is perfectly healthy.

Email and FTP are old protocols with unusual connection patterns that firewalls handle awkwardly.

The habit that helps most here: when a symptom seems disproportionate to the change that preceded it, look for a dependency you have not considered. Time is the classic one.

Scenarios in this chapter

ID	Scenario	Tier
Q97	SMTP retry behaviour	L2
Q98	FTP data channels	L2
Q99	Time as a dependency	L2
Q100	Monitoring credentials and ACLs	L2
Q101	Intermittent connection failure	L3

Q97. Users complain email is delayed by up to an hour but eventually delivers

Tier: L2. Safety: read-only.

Short answer. Delivery that is delayed but eventually succeeds points at retry behaviour: greylisting, a queue backlog, or DNS failures forcing retries. Read the message headers, because each hop stamps its own time and the gap tells you exactly where the delay was.

What is actually going on. Greylisting deliberately rejects the first delivery attempt from an unknown sender with a temporary failure, on the assumption that legitimate mail servers retry and spam sources do not. Retry intervals of 15 to 60 minutes match the reported symptom precisely.

Queue backlogs from a rate limit or reputation throttle look similar. So do DNS failures: if the sender cannot resolve the recipient’s MX record, or the receiver cannot resolve SPF for the sender, delivery is deferred and retried.

The headers separate these cleanly, which is why they are the first thing to read rather than the last.

How to check

```

Read the message headers                                [read-only]
  Each "Received:" line carries a timestamp.
  The gap between two consecutive hops is where the delay lives.

Sending side                                            [read-only]
  Exchange:  Get-Queue ; Get-MessageTrackingLog -MessageId <id>
  Postfix:   postfix -p ; grep <message-id> /var/log/mail.log

DNS                                                    [active test]
  dig MX example.com
  dig TXT example.com                                SPF record present and valid?
  dig -x <sending-ip>                                does reverse DNS match the HELO name?

Look for 4xx temporary rejections in the logs.
That is greylisting or throttling, stated explicitly.

```

Watch out for. Missing or mismatched reverse DNS on the sending IP triggers greylisting and throttling at many receivers. It is a frequently overlooked cause of exactly this symptom, so check PTR alongside SPF rather than only SPF.

References: RFC 5321, SMTP · RFC 6647, Email Greylisting · RFC 7208, SPF

Q98. FTP fails in passive mode through the firewall but works in active mode

Tier: L2. Safety: read-only.

Short answer. The firewall is not handling the data connection for that mode. Active and passive open the data channel in opposite directions on dynamic ports, so each needs different handling. Either enable the FTP inspection engine, or define a fixed passive port range and permit it explicitly.

What is actually going on. FTP separates control and data. The control channel uses TCP 21 in both modes.

Mode	Data connection	Blocked by
Active	Server port 20 connects back to the client's high port	Client-side firewalls, as an unsolicited inbound connection
Passive	Client connects to a high port on the server	Server-side firewalls, if the dynamic range is not permitted

An application layer gateway reads the control channel and opens the pinhole dynamically. That works right up until someone encrypts the control channel with FTPS, at which point the gateway can no longer read it and the whole mechanism fails.

How to check

Server side, passive	[change approval]
vsftpd: pasv_min_port=50000 ; pasv_max_port=50100	
IIS: set the passive port range, then permit it on the firewall	
Firewall	[read-only]
show service-policy inspect ftp	Cisco ASA
Permit the fixed passive range explicitly if inspection is disabled	
Diagnose	[active test]
ftp -d <server>	verbose: watch PORT vs PASV
tcpdump -i <if> -n port 21	read the PASV response

Watch out for. Inspection engines cannot see inside FTPS, so an encrypted FTP deployment must use a fixed passive port range with explicit rules. It also illustrates why application layer gateways are falling out of favour generally.

References: RFC 959, FTP · RFC 2428, FTP Extensions for NAT · RFC 4217, Securing FTP with TLS

Q99. NTP sync failures are causing certificate and authentication errors

Tier: L2. Safety: read-only.

Short answer. Clock drift beyond tolerance. Kerberos rejects tickets outside a default five-minute skew, and TLS validation compares the certificate validity window against the local clock. Verify NTP reachability, stratum and offset, then fix the time hierarchy.

What is actually going on. Time is a dependency of authentication and encryption, so a time failure presents as a security failure. That is why it gets diagnosed slowly.

Kerberos includes timestamps specifically to prevent replay and enforces a tolerance, commonly five minutes. Exceed it and authentication fails across the domain. TLS validation fails if the local clock sits outside the certificate's validity period, which is why a device with a dead battery and a clock reset to 1970 rejects every certificate it sees.

Common causes: UDP 123 blocked, a stratum loop where devices synchronise to each other rather than to an authoritative source, and virtual machines where host time sync and in-guest NTP fight each other.

How to check

```

Cisco [read-only]
  show ntp status          synchronised? stratum? offset?
  show ntp associations detail
  show clock detail

Linux [read-only]
  timedatectl status
  chronyc tracking ; chronyc sources -v
  ntpq -p

Windows [read-only]
  w32tm /query /status
  w32tm /query /source
  w32tm /stripchart /computer:<dc> /samples:5

Confirm
[ ] UDP 123 permitted to the time source
[ ] Stratum is sane; 16 means unsynchronised
[ ] Offset well within tolerance
[ ] One authoritative hierarchy, not devices syncing to each other
[ ] VMs: host time sync and guest NTP not both enabled

```

Watch out for. In a Windows domain, time flows from the PDC emulator down through the hierarchy, and that server should be the only one synchronising externally. Domain members pointed straight at an internet time source is a common misconfiguration that causes quiet, intermittent authentication failures.

References: RFC 5905, NTPv4 · RFC 4120, Kerberos V5

Q100. SNMP polling fails for a subset of devices after a hardening project

Tier: L2. Safety: read-only.

Short answer. The hardening changed SNMP on those devices. Community strings rotated, the version moved to v3, or a new ACL restricts which sources may poll. Work out what the failing subset has in common, because that names the change.

What is actually going on. SNMP hardening typically does three things, and any of them breaks polling if the monitoring platform is not updated in step:

- **Community strings rotated.** v1 and v2c strings travel in clear text, so rotating them is genuinely worthwhile, and forgetting to tell monitoring is genuinely common.
- **Version moved to v3,** which needs authentication and privacy credentials the monitoring system must be given.
- **An ACL applied** restricting SNMP to specific sources, which fails if the monitoring server polls from a different address than the one documented.

The “subset” detail is the clue. Whatever those devices share, a template, a site, a platform, is where the change landed.

How to check

Device	[read-only]
show snmp	
show snmp community	
show snmp user	
show snmp group	
show run include snmp-server	
From the monitoring server	[active test]
snmpwalk -v2c -c <community> <device> system	
snmpwalk -v3 -l authPriv -u <user> -a SHA -A <authpass> \	
-x AES -X <privpass> <device> system	
Reconcile	
[] Version matches on both sides	
[] Credentials match exactly, including case	
[] The ACL permits the monitoring server's ACTUAL source address	
[] The correct VRF is specified if management sits in a VRF	

Watch out for. A monitoring server with several interfaces may poll from a different source than the one in the ACL. Confirm the actual source with a capture on the device rather than trusting documentation, because that mismatch is invisible from both ends until you look at the packets.

Also: do not paste live community strings or v3 credentials into tickets or chat. Reference the credential store instead.

References: RFC 3411, SNMP Management Frameworks · RFC 3414, SNMPv3 USM

Q101. One particular TCP port randomly fails while others work

Tier: L3. Safety: active test.

Short answer. Capture at both ends simultaneously and compare. The packet evidence narrows the field considerably, but treat each observation as an interpretation that needs corroboration rather than as proof on its own.

What is actually going on. Randomness is itself diagnostic. Consistent failure is policy. Random failure is state, capacity, or path variation.

The evidence maps to hypotheses like this, and the caveats matter:

Observation	Leading interpretation	But also consider
SYN never arrives at the server	Path drop: firewall, ACL, routing	ECMP with one bad path; capture missing offloaded traffic
SYN arrives, server sends RST	Nothing listening at that instant	A host firewall configured to REJECT rather than DROP; a middlebox forging RST
SYN arrives, no response at all	Listen backlog full, or host firewall DROP	SYN cookies changing behaviour under load; capture placed after the drop point
Handshake completes, session dies	Stateful timeout or application close	Idle timers on an intermediate device; application crash

Observation	Leading interpretation	But also consider
Fails roughly 1 in N attempts	N backends, one unhealthy	N ECMP paths, one broken

Captures themselves can mislead. Hardware offload can hide traffic from a host capture, and a capture taken at the wrong point in the path shows you a clean stream while the drop happens elsewhere. Corroborate with counters and logs from the devices in between.

How to check

```
# Simultaneous capture, both ends                                # [active test]
tcpdump -i any -n "host <client> and port <port>" -w server.pcap
tcpdump -i any -n "host <server> and port <port>" -w client.pcap

# Server-side state                                              # [read-only]
ss -ltnp | grep <port>
ss -s                                                           # socket summary, TIME_WAIT volume
netstat -s | grep -i -E "listen|overflow|drop"
dmesg | grep -i "possible SYN flooding"
cat /proc/sys/net/core/somaxconn
ulimit -n                                                       # file descriptor limit
```

Watch out for. Listen backlog overflow is under-diagnosed. It appears in `netstat -s` as listen queue overflows, produces exactly this intermittent pattern under load, and is invisible to every network-side test. Naming it demonstrates real depth, because most candidates never get past the firewall hypothesis.

References: RFC 9293, TCP · RFC 5382, NAT Requirements for TCP

Chapter 13. IPv6

Scenarios Q102 to Q108. Tier mix: 2 L1, 3 L2, 2 L3.

Orientation

IPv6 is not “IPv4 with longer addresses.” Several things work differently in ways that produce faults you will not recognise from IPv4 experience, and it comes up increasingly often in interviews because most candidates have not studied it.

The four differences that matter most operationally:

There is no ARP. Neighbour Discovery replaces it, and it runs over ICMPv6. Which leads directly to the next point.

You cannot block ICMPv6. In IPv4, filtering ICMP is merely unwise. In IPv6 it breaks the protocol: address resolution, router discovery, duplicate address detection and path MTU discovery all depend on it. Blanket-blocking ICMPv6 is one of the most common self-inflicted IPv6 outages.

Hosts usually have several addresses at once. A link-local address starting fe80::, one or more global addresses, and often a temporary privacy address that changes. “The server’s IPv6 address” is rarely one thing.

Routers advertise themselves. Instead of DHCP handing out a gateway, routers send Router Advertisements and hosts listen. That is convenient and it means any device on the segment can claim to be a router.

One habit worth building now: whenever you troubleshoot a dual-stack service, establish which protocol the client actually used before anything else. Half of all “dual-stack” tickets are IPv6 problems being reported as general slowness, because the client tried IPv6 first and fell back.

Scenarios in this chapter

ID	Scenario	Tier
Q102	Dual-stack fallback delay	L2
Q103	RA and default route	L1
Q104	Rogue Router Advertisement	L2
Q105	SLAAC, DHCPv6 and the M/O flags	L2
Q106	ICMPv6 filtering and PMTUD	L3
Q107	Dual-stack policy asymmetry	L3
Q108	Premature AAAA publication	L1

Q102. A website is slow to load for some users but fine for others on the same network

Tier: L2. Safety: active test.

Short answer. Suspect a broken IPv6 path with the client falling back to IPv4 after a delay. Test both protocols separately, because a dual-stack client hides the failure by eventually succeeding.

What is actually going on. Modern clients implement Happy Eyeballs. They start the IPv6 connection first and, if it has not completed after a short delay (RFC 8305 recommends 250 ms), start IPv4 in parallel and use whichever connects first. On such a client a broken IPv6 path costs a fraction of a second per connection, which adds up on a page that opens many connections. Multi-second stalls point elsewhere: an application or older client that does not implement Happy Eyeballs and waits for a full TCP timeout, or an IPv6 path where the handshake succeeds and full-size packets then disappear (see Q106). Happy Eyeballs cannot help in that second case, because the connection already looked healthy.

That produces a very specific signature: the page eventually loads, so nobody reports it as an outage, but it feels sluggish. Users on IPv4-only clients on the same network are unaffected, which makes it look random.

The underlying IPv6 fault could be almost anything: a route missing at the ISP, a firewall rule that exists for IPv4 and not IPv6, an MTU problem, or a AAAA record published for a service that is not actually listening on IPv6.

How to check

```
# [active test]
curl -6 -v -o /dev/null -w '%{time_total}\n' https://example.com
curl -4 -v -o /dev/null -w '%{time_total}\n' https://example.com
# compare the two. A large gap is your answer.

dig AAAA example.com
dig A example.com
ping6 example.com
traceroute6 example.com
mtr -6 -rwc 100 example.com

On the client                                [read-only]
ip -6 addr show
ip -6 route show
Get-NetIPConfiguration                        Windows
```

Watch out for. Always test with -6 and -4 explicitly. A plain curl or browser test succeeds and tells you nothing, because the fallback is doing its job. Isolating the protocol is the first and most important step.

References: RFC 8305, Happy Eyeballs Version 2 · RFC 6724, Address Selection for IPv6

Q103. Clients have an IPv6 address but no IPv6 connectivity

Tier: L1. Safety: read-only.

Short answer. Check which kind of address they have. If it only starts with fe80::, they have a link-local address and nothing more, which means no Router Advertisement is reaching them. If they have a global address but no default route, the RA is arriving without the router flag set properly.

What is actually going on. Every IPv6 interface generates a link-local address automatically, whether or not anything else is configured. Seeing an IPv6 address is therefore not evidence of IPv6 working, which trips people up constantly.

A global address requires a Router Advertisement carrying a prefix. The default route comes from the RA's router lifetime field, and if that is zero the host installs the prefix but no gateway.

Common causes: IPv6 not enabled on the router interface, RAs suppressed on that interface, RAs blocked somewhere on the Layer 2 path, or the router advertising a prefix while not offering itself as a default router.

How to check

Client	[read-only]
ip -6 addr show	
fe80:: only	-> no RA is being received
global address	-> RA received, check the route next
ip -6 route show	
no "default via"	-> RA has a zero router lifetime, or none received
Get-NetIPAddress -AddressFamily IPv6	Windows
Get-NetRoute -AddressFamily IPv6	
Router	[read-only]
show ipv6 interface <if>	
is IPv6 enabled? is RA suppressed? what lifetime is advertised?	
show ipv6 routers	what is being seen?
	[active test]
tcpdump -i <if> -n 'icmp6 && ip6[40] == 134'	capture RAs

Watch out for. fe80:: addresses only ever work on the local link. A host with nothing else has no IPv6 connectivity at all, regardless of how healthy the interface looks. Knowing that distinction is the L1 answer here.

References: RFC 4861, Neighbor Discovery for IPv6 · RFC 4862, IPv6 Stateless Address Autoconfiguration

Q104. Users on one VLAN suddenly lose connectivity after someone plugs in a home router

Tier: L2. Safety: read-only.

Short answer. A rogue Router Advertisement. The device is announcing itself as an IPv6 router, hosts on the segment accept it, and traffic is black-holed to a device that cannot forward it. Find the source MAC from a capture, trace it to a switchport, and shut the port.

What is actually going on. Hosts accept Router Advertisements from any device on the link by default. There is no authentication in the base protocol.

A consumer router plugged in the wrong way round happily advertises a prefix and offers itself as a default gateway. Hosts configure a global address from the bogus prefix and install a default route pointing at a device with no upstream connectivity.

The symptom is confusing on a dual-stack network, because IPv4 keeps working while anything preferring IPv6 fails. Users describe it as “some things work,” which sends people looking in the wrong place.

How to check

Identify	[active test]
<pre>tcpdump -i <if> -n 'icmp6 && ip6[40] == 134'</pre>	
-> read the source link-local address and MAC of each RA	
rafixd or ndpmon, if you run them	
Client	[read-only]
ip -6 route show	
is the default route via an unexpected address?	
ip -6 neigh show	
show ipv6 routers	
on a Cisco host or device	
Trace it	[read-only]
show mac address-table address <rogue-mac>	
-> which switchport? shut it.	
Prevent	[change approval]
RA Guard on access ports (IPv6 first-hop security)	
ipv6 nd raguard policy HOST	
ipv6 nd raguard attach-policy HOST	
on user-facing ports	
Combine with DHCPv6 Guard and IPv6 Source Guard	

Watch out for. RA Guard is the IPv6 equivalent of BPDU Guard, and most enterprise networks that enabled the IPv4 protections never enabled the IPv6 ones. Pointing that out unprompted is a strong signal, because it shows you think about dual-stack parity rather than treating IPv6 as an afterthought.

References: RFC 6104, Rogue IPv6 RA Problem Statement · RFC 6105, IPv6 Router Advertisement Guard

Q105. Clients get an IPv6 address but no DNS server

Tier: L2. Safety: read-only.

Short answer. The Router Advertisement’s M and O flags decide whether clients also ask DHCPv6, and for what. If neither flag is set and the RA carries no RDNSS option, hosts get an address and no resolver.

What is actually going on. IPv6 address configuration has more combinations than IPv4, and this is where people get lost. Two flags in the Router Advertisement control it:

M flag	O flag	Client behaviour
0	0	SLAAC only. Address from the RA prefix. No DHCPv6 at all.

M flag	O flag	Client behaviour
0	1	SLAAC for the address, DHCPv6 for other information such as DNS.
1	any	Stateful DHCPv6 for the address and other information.

So a common misconfiguration is enabling SLAAC, leaving both flags clear, and expecting DNS to arrive from somewhere. It does not.

There are two ways to supply DNS, and support varies by operating system:

- **RDNSS and DNSSL options inside the RA itself.** Widely supported now, and simpler because it avoids DHCPv6 entirely.
- **Stateless DHCPv6**, which requires the O flag set and a DHCPv6 server reachable, usually via a relay.

A further wrinkle: some platforms have historically not implemented DHCPv6 at all, so a design relying solely on DHCPv6 leaves those clients without configuration. Check what is actually on your network before choosing.

How to check

Router	[read-only]
<pre>show ipv6 interface <if> look for "Hosts use stateless autoconfig for addresses" and whether the M and O flags are advertised show run interface <if> include ipv6 nd</pre>	
Capture the RA and read the flags	[active test]
<pre>tcpdump -i <if> -vvn 'icmp6 && ip6[40] == 134' M = managed address configuration O = other configuration plus any RDNSS option present</pre>	
Client	[read-only]
resolvectl status	Linux
Get-DnsClientServerAddress -AddressFamily IPv6	Windows

Watch out for. Decide deliberately between RDNSS in the RA and stateless DHCPv6, and document which one your network uses. Half-configuring both is how you end up with clients that resolve on one VLAN and not on another, which is a miserable fault to chase.

References: RFC 4861, Neighbor Discovery · RFC 8106, IPv6 RA Options for DNS Configuration · RFC 8415, DHCPv6

Q106. IPv6 connections hang on large transfers while small requests work

Tier: L3. Safety: active test.

Short answer. Almost the same shape as the IPv4 PMTUD black hole, but worse, because IPv6 routers do not fragment in transit. If ICMPv6 Packet Too Big messages are filtered, the sender never learns and the transfer stalls permanently.

What is actually going on. This is where blocking ICMPv6 stops being merely unwise and becomes an outage.

In IPv4, a router may fragment an oversized packet if DF is clear. In IPv6, routers never fragment. Only the source may fragment, and it only knows to do so if it receives an ICMPv6 Packet Too Big message from the router that could not forward the packet.

Filter that message and path MTU discovery is dead. Small requests succeed because they fit. Anything at full size disappears silently and the sender retransmits it forever.

Tunnels make it more likely, because encapsulation reduces the effective MTU. The IPv6 minimum link MTU is 1280 bytes, which is why 1280 is the safe fallback value you will see in tunnel configurations.

Before concluding PMTUD, rule out the same alternatives as in the IPv4 case: proxy or DLP size limits, application timeouts, and server-side faults. The difference is that in IPv6 the ICMPv6 hypothesis deserves to be checked first, not last.

How to check

```
# [active test]
ping6 -M do -s 1452 <dest>      # 1452 + 48 = 1500 for IPv6
tracpath6 <dest>                # reports MTU changes along the path

# Are Packet Too Big messages arriving?
tcpdump -i <if> -n 'icmp6 && ip6[40] == 2'

Firewall policy that must exist for IPv6                                [change approval]
  permit icmpv6 type 1      destination unreachable
  permit icmpv6 type 2      packet too big          <- essential
  permit icmpv6 type 3      time exceeded
  permit icmpv6 type 4      parameter problem
  permit icmpv6 types 133-137 on-link only (ND: RS, RA, NS, NA, redirect)

Interim mitigation
  ipv6 tcp adjust-mss 1220      on the tunnel interface
```

Watch out for. If someone has written an IPv6 firewall policy by copying the IPv4 one, ICMPv6 is very likely over-filtered. Checking that is a five-minute job that resolves a class of faults people chase for days.

References: RFC 8200, IPv6 Specification · RFC 8201, Path MTU Discovery for IPv6 · RFC 4890, Filtering ICMPv6 Messages in Firewalls

Q107. A service you firewalled off is still reachable

Tier: L3. Safety: read-only.

Short answer. Check whether the IPv6 path is governed by the same policy as the IPv4 path. Dual-stack networks routinely have complete IPv4 rules and partial or absent IPv6 rules, so a service blocked on one protocol stays wide open on the other.

What is actually going on. This is one of the most consequential real-world IPv6 problems and it barely appears in training material.

IPv6 was enabled by default in every mainstream operating system for years before most organisations wrote IPv6 security policy. The result is networks where:

- Access lists were written for IPv4 and never duplicated for IPv6
- The firewall has an IPv4 rule base and a much thinner IPv6 one
- Monitoring, logging and NetFlow collection cover IPv4 more completely
- Segmentation is enforced on IPv4 and assumed on IPv6
- Hosts have IPv6 addresses nobody inventoried

An attacker or a curious user reaching the service over IPv6 bypasses the control entirely. Nothing looks wrong in the IPv4 logs, because nothing happened there.

How to check

Establish the IPv6 reality	[read-only]
show ipv6 interface brief	which interfaces are up on IPv6?
show ipv6 route	
show ipv6 access-list	does it exist at all?
Firewall: compare the IPv4 and IPv6 rule bases side by side	
On the host	[read-only]
ss -ltn	is the service bound to :: as well?
netstat -an findstr "::-"	Windows
ip -6 addr show	
	[active test]
nmap -6 -Pn -p <port> <ipv6-address>	from outside the boundary
Remediation	[change approval]
- inventory every IPv6-enabled interface and address	
- mirror every IPv4 policy into IPv6, or explicitly document why not	
- if IPv6 is not in use, disable it deliberately at the interface, rather than leaving it enabled and unmanaged	
- extend logging, NetFlow and monitoring to IPv6	

Watch out for. “We don’t use IPv6” is almost never true. It is enabled on the hosts by default and it is doing something. Say it plainly, then describe how you would verify it rather than assume.

References: RFC 7123, Security Implications of IPv6 on IPv4 Networks · RFC 9099, Operational Security Considerations for IPv6 · NIST SP 800-119, Secure IPv6 Deployment

Q108. A hostname returns an AAAA record but the service is unreachable

Tier: L1. Safety: active test.

Short answer. Something published a AAAA record for a service that is not actually working over IPv6. Confirm whether the server is listening on IPv6 and whether the path works, then either fix IPv6 properly or remove the AAAA record until you have.

What is actually going on. Publishing a AAAA record is a commitment. Clients that prefer IPv6 will try that address first, and if it does not work they wait for a timeout before falling back.

Several things produce a AAAA record without working service:

- The application binds only to an IPv4 socket, so nothing is listening on IPv6
- The load balancer has an IPv6 virtual address but IPv4-only backends with no translation
- Firewall policy permits the IPv4 address and not the IPv6 one, which is Q107 again
- The AAAA record was created during a migration and points at an address that no longer exists
- The address is correct but there is no IPv6 route to it from the internet

How to check

```
# [active test]
dig AAAA service.example.com
dig A service.example.com

ping6 <ipv6-address>
curl -6 -v https://service.example.com/
nc -6 -vz <ipv6-address> 443
traceroute6 <ipv6-address>

On the server                                     [read-only]
ss -ltn | grep -E '>:::|\[:::\]'          is it bound to IPv6 at all?
Get-NetTCPConnection -State Listen      Windows
```

Watch out for. The correct short-term action is often to remove the AAAA record. It feels like a retreat, but publishing a broken address degrades every dual-stack client that touches it, and removing it restores service immediately while you fix IPv6 properly. Say that clearly, because it separates “the right long-term design” from “the right thing to do at 2 a.m.”

References: RFC 3596, DNS Extensions to Support IPv6 · RFC 8305, Happy Eyeballs Version 2

Appendix A. Study Paths

The three study paths below sequence the same 108 scenarios by tier. Use the path for the role you are applying for, then use Appendix C to decide which chapters deserve extra time. The paths contain 25, 73 and 10 scenarios respectively.

A.1 L1 Study Path: NOC Analyst

25 scenarios · For your first NOC role, Tier 1 support, or CCNA-level knowledge

Who this path is for

You are applying for a first NOC or service desk role, or you have been in one for a few months and want to stop feeling out of your depth. You know what a VLAN is and roughly what a router does. You have probably not troubleshooted a live outage on your own yet.

That is the right starting point. Nobody expects an L1 candidate to explain BGP best-path selection. What they expect is that you gather the correct information, do not make things worse, and hand over cleanly when you reach your limit.

What interviewers are testing at L1

They ask	They are really checking
"What would you check first?"	Do you have a method, or do you guess?
"What command would you run?"	Have you actually used a CLI?
"When would you escalate?"	Will you sit on a P1 for two hours out of pride?
"How would you explain this to the user?"	Can you talk to people who are not engineers?

Notice that only half of those are technical. At L1 the process questions carry as much weight as the protocol questions, which is why this path includes all of them.

How to work through this

Week 1. Scenarios 1 to 8 below. These are the reachability and physical layer basics that come up in almost every L1 interview.

Week 2. Scenarios 9 to 16. Name resolution, addressing, and the wireless fundamentals.

Week 3. Scenarios 17 to 25. Process, escalation and communication. Do not skip these. They are the ones candidates underprepare and interviewers weight heavily.

Throughout. For each scenario, cover the answer and say yours out loud first. Reading and nodding is not studying.

Before you interview

Build at least three of these in a lab and break them deliberately. Cisco Packet Tracer is free and enough for most of this path. Being able to say “I built this and watched it fail” changes how you answer, and interviewers hear the difference.

The scenarios

#	Scenario	Chapter
Q1	A user can't reach a server in another VLAN, but can reach one in the same VLAN	Routing and Switching
Q6	A newly added VLAN isn't passing traffic across a trunk link	Routing and Switching
Q7	Users on one floor lose connectivity randomly, correlating with cleaning staff hours	Routing and Switching
Q9	A static route works for one subnet but not its neighbour subnet	Routing and Switching
Q10	Interface counters show high input errors and CRC errors	Routing and Switching
Q15	Users can reach IPs directly but not by hostname	DNS and DHCP
Q17	A new laptop can't get an IP address on the network	DNS and DHCP
Q20	Devices are getting APIPA (169.254.x.x) addresses	DNS and DHCP
Q22	After a DNS record change, some users still resolve the old IP	DNS and DHCP
Q38	A user can VPN in from home but not from a hotel Wi-Fi	VPN and Remote Access
Q44	Wireless throughput is much lower than wired for the same user	Wireless
Q48	Users near the kitchen experience 2.4 GHz Wi-Fi dropouts	Wireless
Q50	Monitoring floods with hundreds of alerts during a single outage	Monitoring and NOC Process
Q53	Ticket volume for the same issue is coming in from multiple sites	Monitoring and NOC Process
Q54	Monitoring shows 100% packet loss to a device users say is working fine	Monitoring and NOC Process
Q55	How do you decide whether to escalate or keep troubleshooting?	Monitoring and NOC Process
Q56	A maintenance window causes unexpected alerts to page on-call staff	Monitoring and NOC Process
Q57	How do you document a resolved incident for future reference?	Monitoring and NOC Process
Q79	A newly terminated cable passes a link light but has a high error rate	Physical Layer
Q80	A fibre link shows light-loss readings well beyond spec	Physical Layer

#	Scenario	Chapter
Q81	A patch panel port works but the same run at the wall jack doesn't	Physical Layer
Q82	A copper run works at 100 Mbps but fails at gigabit	Physical Layer
Q91	How do you stay aware of network changes across departments?	Escalation and ITIL
Q103	Clients have an IPv6 address but no IPv6 connectivity	IPv6
Q108	A hostname returns an AAAA record but the service is unreachable	IPv6

What to say when you do not know

You will be asked something outside this path. That is deliberate; interviewers want to see how you behave at the edge of your knowledge.

The answer that works: say what you do know, say plainly what you do not, and describe how you would find out. "I have not worked with that directly. I know it sits at Layer 3 and I would start by checking the routing table and the interface state, then look at the vendor documentation or escalate to Tier 2 if I was not making progress."

The answer that fails: guessing confidently. Every experienced engineer can tell, and it makes them wonder what else you have guessed at.

A.2 L2 Study Path: NOC Engineer

73 scenarios · For Tier 2, shift engineer, or 2 to 5 years of experience

Who this path is for

You already work in a NOC. You recognise most of these symptoms and you have fixed several of them. What you may not have done is explain *why* they happen to someone who is deliberately probing for gaps.

This is the largest path by a wide margin, and that is not an accident. Most NOC hiring happens at L2, and it is where the majority of candidates get caught out.

What interviewers are testing at L2

The single biggest difference from L1 is the follow-up question. At L1 you are asked what you would check. At L2 you are asked what you would check, and then you are asked **why**. Sometimes twice.

Level	Question	Adequate answer
L1	"OSPF neighbours will not form. What do you check?"	Timers, area ID, MTU, authentication
L2	"It is stuck in ExStart. What does that tell you?"	MTU mismatch, because DD packets carry MTU and are rejected when it differs

If you can only produce the first answer, you interview as an L1 candidate regardless of your job title. The *What is actually going on* section of each scenario is where that gap gets closed, so read those properly rather than skimming to the commands.

How to work through this

Do not work through it in order. 73 scenarios read front to back will blur together and you will retain little.

Instead:

1. **Diagnose yourself first.** Use the self-assessment tracker and score each chapter honestly. Anything below 3 is where your time belongs.
2. **Work by domain, not by tier.** Read the whole DNS chapter, then the whole VPN chapter. Related scenarios reinforce each other, and that is exactly why this guide is organised by topic rather than by tier.
3. **Use the questions-only edition** for recall practice once you have read a chapter.
4. **Lab the ones you have never seen in production.** You will be asked “have you dealt with this yourself?” and a lab is an honest answer.

The mechanisms most worth memorising

If you are short on time, these are the explanations that come up most often and differentiate most sharply:

- **OSPF adjacency states map to specific causes.** Stuck in ExStart means MTU.
- **PMTUD black holes** happen when ICMP is filtered, and they are why downloads hang over tunnels.
- **Stateful versus stateless** is why cloud NACLs need explicit return rules and security groups do not.
- **Encaps rising, decaps flat** on an IPsec SA tells you which end owns the problem.
- **Precise repeating intervals** mean a timer. Random failures mean congestion or loss.
- **Health check depth** is why a load balancer keeps feeding a dead server.
- **Wireless is half duplex and shared**, which is why airtime, not signal strength, explains most performance complaints.

The scenarios

#	Scenario	Chapter
Q2	Two switches show a spanning-tree topology change every few minutes	Routing and Switching
Q3	After adding a new switch, users report intermittent broadcast storms	Routing and Switching
Q4	A route works via traceroute but pings drop 50% of the time	Routing and Switching
Q5	OSPF neighbors won't form between two routers on the same subnet	Routing and Switching

#	Scenario	Chapter
Q8	Router CPU spikes to 99% during peak hours	Routing and Switching
Q11	Users behind an EtherChannel report slow throughput despite it showing “up”	Routing and Switching
Q12	A default route was removed accidentally and internet access failed network-wide	Routing and Switching
Q16	Internal DNS resolves fine but external domains fail intermittently	DNS and DHCP
Q18	A branch office DHCP scope is exhausted despite low headcount	DNS and DHCP
Q19	A DNS server is up, but one specific domain always times out	DNS and DHCP
Q21	A dual-homed server intermittently registers the wrong IP in DNS	DNS and DHCP
Q23	Traffic is allowed by an ACL but still blocked	Firewall and Security
Q24	After a firewall rule update, VoIP calls drop after about 30 seconds	Firewall and Security
Q25	Users can’t access a site that used to work, now blocked by URL filtering	Firewall and Security
Q26	A perimeter firewall shows a spike in denied connections from one external IP	Firewall and Security
Q27	A site-to-site tunnel is up but no traffic passes	Firewall and Security
Q28	An internal server suddenly can’t be reached from outside after a change	Firewall and Security
Q29	A user’s laptop is quarantined by NAC without explanation	Firewall and Security
Q30	Repeated login failures from many source IPs against one admin account	Firewall and Security
Q32	A misconfigured ACL blocked a critical application at 2 AM	Firewall and Security
Q35	A remote worker connects to VPN but can’t access internal file shares	VPN and Remote Access
Q36	A VPN client repeatedly disconnects at about the same interval	VPN and Remote Access
Q37	A site-to-site VPN drops whenever large file transfers start	VPN and Remote Access
Q39	After a VPN concentrator upgrade, users get “certificate not trusted” errors	VPN and Remote Access
Q40	Split-tunnel VPN users report both slow internet and slow internal resources	VPN and Remote Access
Q41	A remote office has poor VPN performance despite a good local speed test	VPN and Remote Access
Q42	Users experience random Wi-Fi disconnects only in one area of the building	Wireless
Q43	A new AP was added but clients still connect to a farther, weaker AP	Wireless
Q45	Guest Wi-Fi users can access internal VLANs unexpectedly	Wireless
Q46	Certain IoT devices fail to connect to a WPA2-Enterprise SSID	Wireless
Q47	After a wireless controller firmware update, multiple APs go offline	Wireless
Q49	Wi-Fi calling quality is choppy while data works fine	Wireless

#	Scenario	Chapter
Q51	A critical alert didn't page anyone during last night's outage	Monitoring and NOC Process
Q52	A flapping "interface down/up" alert every 5 minutes on a core switch	Monitoring and NOC Process
Q58	Bandwidth graphs show a steady climb toward saturation on a WAN link	Monitoring and NOC Process
Q59	A P1 ticket sits untouched for 20 minutes because the engineer is on break	Monitoring and NOC Process
Q60	Multiple monitoring tools report the same outage at different severities	Monitoring and NOC Process
Q61	You've fixed an issue but the dashboard still shows red	Monitoring and NOC Process
Q62	Another team's change caused an outage and the NOC wasn't told	Monitoring and NOC Process
Q63	Your shift is ending but an incident isn't resolved	Monitoring and NOC Process
Q64	Configuration drift is causing repeat incidents across similar devices	Monitoring and NOC Process
Q65	A load balancer keeps sending traffic to a backend server that's down	Cloud and Load Balancing
Q66	Users experience session drops with a round-robin load balancer	Cloud and Load Balancing
Q67	Cloud instances can't reach the internet despite having a public IP	Cloud and Load Balancing
Q69	An app in the cloud is reachable internally but not from on-prem over VPN	Cloud and Load Balancing
Q70	Auto-scaling added instances but the load balancer isn't sending them traffic	Cloud and Load Balancing
Q71	Cross-region cloud traffic has unexpectedly high latency	Cloud and Load Balancing
Q73	Calls are fine most of the day but become choppy around 5 PM	VoIP and QoS
Q74	One-way audio is reported on calls between two sites	VoIP and QoS
Q75	VoIP quality complaints only come from wireless users, not wired ones	VoIP and QoS
Q76	Softphone calls drop at a precise repeating interval	VoIP and QoS
Q77	After a WAN bandwidth upgrade, voice quality got worse	VoIP and QoS
Q78	Conference calls degrade only above about 10 participants	VoIP and QoS
Q83	An SFP transceiver shows link down despite being properly seated	Physical Layer
Q84	Multiple ports on the same switch card show errors after a power event	Physical Layer
Q85	A vendor's ticket for a circuit outage isn't progressing	Escalation and ITIL
Q86	You suspect another team's change caused an issue but they deny it	Escalation and ITIL

#	Scenario	Chapter
Q87	How do you triage several simultaneous incidents with limited staff?	Escalation and ITIL
Q88	A recurring issue keeps getting temporary fixes	Escalation and ITIL
Q89	How do you communicate with stakeholders during a major outage?	Escalation and ITIL
Q90	A change you made caused unexpected downtime. What's your next step?	Escalation and ITIL
Q92	A Linux server can ping its gateway but not any other host	Server and OS Networking
Q93	A Windows server loses connectivity right after an update	Server and OS Networking
Q94	Two NICs on a server in the same subnet cause intermittent connectivity	Server and OS Networking
Q95	A server's throughput is capped well below the NIC's rated speed	Server and OS Networking
Q96	A hostname resolves fine but TLS certificate errors happen intermittently	Server and OS Networking
Q97	Users complain email is delayed by up to an hour but eventually delivers	Miscellaneous Protocols
Q98	FTP fails in passive mode through the firewall but works in active mode	Miscellaneous Protocols
Q99	NTP sync failures are causing certificate and authentication errors	Miscellaneous Protocols
Q100	SNMP polling fails for a subset of devices after a hardening project	Miscellaneous Protocols
Q102	A website is slow to load for some users but fine for others on the same network	IPv6
Q104	Users on one VLAN suddenly lose connectivity after someone plugs in a home router	IPv6
Q105	Clients get an IPv6 address but no DNS server	IPv6

The question behind the question

At L2, interviewers are also assessing whether you can be left alone on a night shift. Several scenarios in this path are really about that: escalation timing, handover quality, and whether you check the change calendar before theorising.

Answer those with the same seriousness as the protocol questions. A candidate who is technically strong and operationally careless is a risk, and experienced interviewers screen for it deliberately.

A.3 L3 Study Path: Senior and Escalation Engineer

10 scenarios · For senior NOC, escalation engineer, or network specialist roles

Who this path is for

You are the person the L2 team escalates to. You are expected to reason about protocol behaviour from first principles, work from packet-level evidence, and design failures out rather than fix them repeatedly.

Why this path is short

Ten scenarios, against seventy-three at L2. That is deliberate and worth explaining, because the imbalance is a real property of the material rather than an oversight.

Senior interviews test differently. They rarely ask “what causes X.” They ask you to reason out loud about something ambiguous, then push on your reasoning to see whether it holds. The value is in depth per scenario, not coverage.

They also test judgement more than recall: when *not* to roll back, when a datasheet number was the real problem, when to tell a stakeholder you have no reliable ETA. Several of those live in the Escalation chapter and are marked L2 there, but they are asked at senior interviews with harder follow-ups.

So treat this path as a supplement, not a substitute. Work the full L2 path as well. What changes at L3 is the depth you are expected to reach on any of it, not the topics.

What interviewers are testing at L3

They ask	They are really checking
“Walk me through how you would approach this.”	Can you reason without knowing the answer in advance?
“Why does that happen?” repeatedly	Do you understand the protocol, or the symptom?
“What would you do if rollback was not available?”	Do you have judgement, or a memorised procedure?
“How would you stop this recurring?”	Do you think in systems or in tickets?
“What is your evidence?”	Do you reason from data or from assumption?

The last one matters most. A senior candidate who says “it is probably MTU” has given a worse answer than one who says “the symptom fits PMTUD, and I would confirm with a DF-bit sweep before committing, because a proxy file size limit produces the same complaint.”

How to work through this

Read each scenario, then argue with it. Every one of these has at least one alternative explanation the answer does not fully cover. Find it. That is the exercise.

Then explain the mechanism out loud to someone who is not a network engineer. If you cannot, you do not understand it well enough to defend it under a third follow-up question.

The scenarios

#	Scenario	Chapter
Q13	A BGP peer is up, but a prefix you expect is missing	Routing and Switching
Q14	A gateway failover didn’t happen when the primary router failed	Routing and Switching
Q31	After enabling deep packet inspection, throughput drops	Firewall and Security

#	Scenario	Chapter
Q33	Users can browse sites but file downloads consistently fail or hang	Firewall and Security
Q34	Remote sites behind different firewall vendors report inconsistent VPN compatibility	Firewall and Security
Q68	A hybrid cloud connection shows intermittent BGP flaps	Cloud and Load Balancing
Q72	A security group update blocked traffic that used to work through the NAT gateway	Cloud and Load Balancing
Q101	One particular TCP port randomly fails while others work	Miscellaneous Protocols
Q106	IPv6 connections hang on large transfers while small requests work	IPv6
Q107	A service you firewalled off is still reachable	IPv6

A note on how these were written

Several scenarios in this guide were corrected after a technical audit found overconfident or incorrect explanations, and this edition adds further corrections. The corrections made for this edition are listed in Appendix I.

That is relevant to you specifically. At L3 you will be interviewing people, reviewing designs and signing off changes, and the most useful habit in this guide is not any single mechanism. It is the practice of separating “the evidence supports this” from “this is the explanation I reached for first.”

If you find something here that is still wrong, report it using the address in Publication Details. Corrections are genuinely welcome.

Appendix B. Questions-Only Recall Edition

All 108 scenarios with no answers visible. Use this for active recall.

How to use this properly

Read the question. Say your answer **out loud**, in full sentences, as if an interviewer is sitting opposite you. Only then follow the link to check.

That last part matters more than it sounds. Reading a question, thinking “yes, I know that one,” and moving on is recognition, not recall. Recognition feels like knowledge right up until someone is watching you and you have to produce the words.

Three rules that make this work:

1. **Speak, do not think.** The gap between knowing something and being able to say it under pressure is exactly what interviews expose.
2. **Give the mechanism, not just the cause.** After your answer, ask yourself “why does that happen?” If you cannot answer that, mark the scenario.
3. **Time yourself.** Aim for 60 to 90 seconds per answer. Interview answers that run past two minutes lose the room.

Mark anything you could not answer cleanly and take it to the self-assessment tracker.

Chapter 1. Routing and Switching

Q1. A user can't reach a server in another VLAN, but can reach one in the same VLAN
L1 • answer

Q2. Two switches show a spanning-tree topology change every few minutes
L2 • answer

Q3. After adding a new switch, users report intermittent broadcast storms
L2 • answer

Q4. A route works via traceroute but pings drop 50% of the time
L2 • answer

Q5. OSPF neighbors won't form between two routers on the same subnet
L2 • answer

Q6. A newly added VLAN isn't passing traffic across a trunk link
L1 • answer

Q7. Users on one floor lose connectivity randomly, correlating with cleaning staff hours
L1 • answer

Q8. Router CPU spikes to 99% during peak hours
L2 • answer

Q9. A static route works for one subnet but not its neighbour subnet
L1 • answer

Q10. Interface counters show high input errors and CRC errors
L1 • answer

Q11. Users behind an EtherChannel report slow throughput despite it showing “up”
L2 • answer

Q12. A default route was removed accidentally and internet access failed network-wide
L2 • answer

Q13. A BGP peer is up, but a prefix you expect is missing
L3 • answer

Q14. A gateway failover didn’t happen when the primary router failed
L3 • answer

Chapter 2. DNS and DHCP

Q15. Users can reach IPs directly but not by hostname
L1 • answer

Q16. Internal DNS resolves fine but external domains fail intermittently
L2 • answer

Q17. A new laptop can’t get an IP address on the network
L1 • answer

Q18. A branch office DHCP scope is exhausted despite low headcount
L2 • answer

Q19. A DNS server is up, but one specific domain always times out
L2 • answer

Q20. Devices are getting APIPA (169.254.x.x) addresses
L1 • answer

Q21. A dual-homed server intermittently registers the wrong IP in DNS
L2 • answer

Q22. After a DNS record change, some users still resolve the old IP
L1 • answer

Chapter 3. Firewall and Security

Q23. Traffic is allowed by an ACL but still blocked
L2 • answer

Q24. After a firewall rule update, VoIP calls drop after about 30 seconds
L2 • answer

Q25. Users can't access a site that used to work, now blocked by URL filtering
L2 • answer

Q26. A perimeter firewall shows a spike in denied connections from one external IP
L2 • answer

Q27. A site-to-site tunnel is up but no traffic passes
L2 • answer

Q28. An internal server suddenly can't be reached from outside after a change
L2 • answer

Q29. A user's laptop is quarantined by NAC without explanation
L2 • answer

Q30. Repeated login failures from many source IPs against one admin account
L2 • answer

Q31. After enabling deep packet inspection, throughput drops
L3 • answer

Q32. A misconfigured ACL blocked a critical application at 2 AM
L2 • answer

Q33. Users can browse sites but file downloads consistently fail or hang
L3 • answer

Q34. Remote sites behind different firewall vendors report inconsistent VPN compatibility
L3 • answer

Chapter 4. VPN and Remote Access

Q35. A remote worker connects to VPN but can't access internal file shares
L2 • answer

Q36. A VPN client repeatedly disconnects at about the same interval
L2 • answer

Q37. A site-to-site VPN drops whenever large file transfers start
L2 • answer

Q38. A user can VPN in from home but not from a hotel Wi-Fi
L1 • answer

Q39. After a VPN concentrator upgrade, users get “certificate not trusted” errors
L2 • answer

Q40. Split-tunnel VPN users report both slow internet and slow internal resources
L2 • answer

Q41. A remote office has poor VPN performance despite a good local speed test
L2 • answer

Chapter 5. Wireless

Q42. Users experience random Wi-Fi disconnects only in one area of the building
L2 • answer

Q43. A new AP was added but clients still connect to a farther, weaker AP
L2 • answer

Q44. Wireless throughput is much lower than wired for the same user
L1 • answer

Q45. Guest Wi-Fi users can access internal VLANs unexpectedly
L2 • answer

Q46. Certain IoT devices fail to connect to a WPA2-Enterprise SSID
L2 • answer

Q47. After a wireless controller firmware update, multiple APs go offline
L2 • answer

Q48. Users near the kitchen experience 2.4 GHz Wi-Fi dropouts
L1 • answer

Q49. Wi-Fi calling quality is choppy while data works fine
L2 • answer

Chapter 6. Monitoring, Alerting and NOC Process

Q50. Monitoring floods with hundreds of alerts during a single outage
L1 • answer

Q51. A critical alert didn't page anyone during last night's outage
L2 • answer

Q52. A flapping “interface down/up” alert every 5 minutes on a core switch
L2 • answer

Q53. Ticket volume for the same issue is coming in from multiple sites
L1 • answer

Q54. Monitoring shows 100% packet loss to a device users say is working fine
L1 · answer

Q55. How do you decide whether to escalate or keep troubleshooting?
L1 · answer

Q56. A maintenance window causes unexpected alerts to page on-call staff
L1 · answer

Q57. How do you document a resolved incident for future reference?
L1 · answer

Q58. Bandwidth graphs show a steady climb toward saturation on a WAN link
L2 · answer

Q59. A P1 ticket sits untouched for 20 minutes because the engineer is on break
L2 · answer

Q60. Multiple monitoring tools report the same outage at different severities
L2 · answer

Q61. You've fixed an issue but the dashboard still shows red
L2 · answer

Q62. Another team's change caused an outage and the NOC wasn't told
L2 · answer

Q63. Your shift is ending but an incident isn't resolved
L2 · answer

Q64. Configuration drift is causing repeat incidents across similar devices
L2 · answer

Chapter 7. Cloud and Load Balancing

Q65. A load balancer keeps sending traffic to a backend server that's down
L2 · answer

Q66. Users experience session drops with a round-robin load balancer
L2 · answer

Q67. Cloud instances can't reach the internet despite having a public IP
L2 · answer

Q68. A hybrid cloud connection shows intermittent BGP flaps
L3 · answer

Q69. An app in the cloud is reachable internally but not from on-prem over VPN
L2 · answer

Q70. Auto-scaling added instances but the load balancer isn't sending them traffic
L2 • answer

Q71. Cross-region cloud traffic has unexpectedly high latency
L2 • answer

Q72. A security group update blocked traffic that used to work through the NAT gateway
L3 • answer

Chapter 8. VoIP and QoS

Q73. Calls are fine most of the day but become choppy around 5 PM
L2 • answer

Q74. One-way audio is reported on calls between two sites
L2 • answer

Q75. VoIP quality complaints only come from wireless users, not wired ones
L2 • answer

Q76. Softphone calls drop at a precise repeating interval
L2 • answer

Q77. After a WAN bandwidth upgrade, voice quality got worse
L2 • answer

Q78. Conference calls degrade only above about 10 participants
L2 • answer

Chapter 9. Physical Layer and Cabling

Q79. A newly terminated cable passes a link light but has a high error rate
L1 • answer

Q80. A fibre link shows light-loss readings well beyond spec
L1 • answer

Q81. A patch panel port works but the same run at the wall jack doesn't
L1 • answer

Q82. A copper run works at 100 Mbps but fails at gigabit
L1 • answer

Q83. An SFP transceiver shows link down despite being properly seated
L2 • answer

Q84. Multiple ports on the same switch card show errors after a power event
L2 • answer

Chapter 10. Escalation, ITIL and Communication

Q85. A vendor's ticket for a circuit outage isn't progressing
L2 • answer

Q86. You suspect another team's change caused an issue but they deny it
L2 • answer

Q87. How do you triage several simultaneous incidents with limited staff?
L2 • answer

Q88. A recurring issue keeps getting temporary fixes
L2 • answer

Q89. How do you communicate with stakeholders during a major outage?
L2 • answer

Q90. A change you made caused unexpected downtime. What's your next step?
L2 • answer

Q91. How do you stay aware of network changes across departments?
L1 • answer

Chapter 11. Server and OS Networking

Q92. A Linux server can ping its gateway but not any other host
L2 • answer

Q93. A Windows server loses connectivity right after an update
L2 • answer

Q94. Two NICs on a server in the same subnet cause intermittent connectivity
L2 • answer

Q95. A server's throughput is capped well below the NIC's rated speed
L2 • answer

Q96. A hostname resolves fine but TLS certificate errors happen intermittently
L2 • answer

Chapter 12. Miscellaneous Protocols

Q97. Users complain email is delayed by up to an hour but eventually delivers
L2 • answer

Q98. FTP fails in passive mode through the firewall but works in active mode
L2 • answer

Q99. NTP sync failures are causing certificate and authentication errors
L2 • answer

Q100. SNMP polling fails for a subset of devices after a hardening project
L2 • answer

Q101. One particular TCP port randomly fails while others work
L3 • answer

Chapter 13. IPv6

Q102. A website is slow to load for some users but fine for others on the same network
L2 • answer

Q103. Clients have an IPv6 address but no IPv6 connectivity
L1 • answer

Q104. Users on one VLAN suddenly lose connectivity after someone plugs in a home router
L2 • answer

Q105. Clients get an IPv6 address but no DNS server
L2 • answer

Q106. IPv6 connections hang on large transfers while small requests work
L3 • answer

Q107. A service you firewalled off is still reachable
L3 • answer

Q108. A hostname returns an AAAA record but the service is unreachable
L1 • answer

When you have finished

Score yourself in the self-assessment tracker, then go back to the chapters where you scored lowest. Do not re-read the ones you already know well, however satisfying that feels.

Appendix C. Self-Assessment Tracker

Score yourself honestly before you interview. The scale is deliberately harsh, because interview pressure removes roughly one level of performance from everybody.

The scale

Score	What it means
0	I do not recognise the scenario
1	I recognise it but cannot answer
2	I can state the likely cause
3	I can state the cause and the commands to confirm it
4	I can explain the mechanism and defend it under follow-up questioning
5	I have personally resolved this in production

The jump that matters is **2 to 3, and then 3 to 4**. Most candidates sit at 2, can name the cause, and fall apart on the first “why?” Getting to 4 on your weak chapters is worth more than getting to 5 on your strong ones.

Be honest about the difference between 4 and 5

You will be asked “have you dealt with this yourself?” Answering 5 when the truth is 4 is a bad trade, because the follow-up questions go somewhere you cannot follow.

“I have not hit this in production, but I understand the mechanism and I have built it in a lab” is a strong answer. An invented war story is not, and it collapses immediately.

Score sheet

Copy this table into a notebook, or photocopy the page, and fill it in.

Chapter	Scenarios	Mix	Target L1	Target L2	Your score	Date
01 Routing and Switching	Q1–Q14	5 L1 · 7 L2 · 2 L3	3	4		
02 DNS and DHCP	Q15–Q22	4 L1 · 4 L2	3	4		
03 Firewall and Security	Q23–Q34	9 L2 · 3 L3	2	4		
04 VPN and Remote Access	Q35–Q41	1 L1 · 6 L2	3	4		
05 Wireless	Q42–Q49	2 L1 · 6 L2	3	4		

Chapter	Scenarios	Mix	Target L1	Target L2	Your score	Date
06 Monitoring, Alerting and NOC Process	Q50–Q64	6 L1 · 9 L2	3	4		
07 Cloud and Load Balancing	Q65–Q72	6 L2 · 2 L3	2	4		
08 VoIP and QoS	Q73–Q78	6 L2	2	4		
09 Physical Layer and Cabling	Q79–Q84	4 L1 · 2 L2	3	4		
10 Escalation, ITIL and Communication	Q85–Q91	1 L1 · 6 L2	3	4		
11 Server and OS Networking	Q92–Q96	5 L2	2	4		
12 Miscellaneous Protocols	Q97–Q101	4 L2 · 1 L3	2	4		
13 IPv6	Q102–Q108	2 L1 · 3 L2 · 2 L3	3	4		

How to read your scores

Anything below its target is where your revision time belongs. Not the chapters you enjoy.

Anything at 4 that you have never actually done deserves a lab session before the interview. The mechanism knowledge is real, but the experience question is coming.

Anything at 0 or 1 in a chapter relevant to the role you are applying for is a red flag worth addressing first, even if it is uncomfortable. A single confident “I do not know that area at all” is survivable. Three of them is not.

Re-score after two weeks

Scores taken immediately after reading a chapter are inflated, because the material is still in short-term memory. Score once after reading, then again two weeks later without revisiting. The second number is the real one, and the gap between them tells you what needs spaced repetition rather than another read-through.

Appendix D. Command Reference

Every command carries a safety label. Read it before pasting anything into a production device.

Label	Meaning
read-only	Displays state. Safe on production.
active test	Generates traffic or probes. Low risk, but visible in logs and may trigger security alerts.
service-affecting	Can interrupt traffic. Needs a maintenance window.
change approval	Modifies configuration. Needs change control and a rollback plan.

On “exact” commands. Syntax and defaults vary by platform and software release. Commands below were written against the platforms listed in the review status. Anything you have not personally verified on your own release should be treated as a **pattern to adapt**, not as text to paste. That distinction is deliberate, and it is the honest position for a guide maintained by one author.

Cisco IOS and IOS XE

Interface and physical layer

show interface <if>	[read-only] counters, errors, duplex, load
show interface status	negotiated speed and duplex
show interface counters errors	CRC, alignment, symbol errors
show interface transceiver detail	optical Tx/Rx in dBm, thresholds
show interface trunk	allowed and active VLANs
show interface <if> capabilities	
show power inline	PoE draw and budget
clear counters <if>	[read-only, resets state] reset before measuring a rate

Layer 2

show mac address-table	[read-only]
show mac address-table address <mac>	is one MAC on two ports?
show spanning-tree detail	last topology change and its port
show spanning-tree inconsistentports	guard violations
show etherchannel summary	P bundled, s suspended, I individual
show etherchannel load-balance	
show cdp neighbors detail	what is on the other end?
show lldp neighbors detail	
show vlan brief	

Layer 3 and routing

	[read-only]
show ip route <destination>	the route that WILL be used
show ip route <prefix> longer-prefixes	
show ip cef exact-route <src> <dst>	which ECMP path a flow takes
show ip arp	
show ip ospf neighbor	the STATE names the fault
show ip ospf interface <if>	area, timers, MTU, auth
show bgp ipv4 unicast summary	
show bgp neighbor <ip> received-routes	needs soft-reconfig or refresh
show bgp neighbor <ip> advertised-routes	
show bgp ipv4 unicast <prefix>	why it won or lost best-path
show standby brief	HSRP state and priority
show vrrp brief	
show track	is anything actually tracked?
show bfd neighbors detail	

IPv6

	[read-only]
show ipv6 interface brief	
show ipv6 interface <if>	RA settings, M/O flags, lifetime
show ipv6 route	
show ipv6 neighbors	
show ipv6 routers	what RAs is this device seeing?
show ipv6 access-list	does an IPv6 policy exist at all?

Control plane and platform

	[read-only]
show processes cpu sorted exclude 0.00	
show processes cpu history	5-second, 1-minute, 72-hour graphs
show ip cef	CEF enabled and populated?
show controllers cpu-interface	punted packet counters
show policy-map control-plane	CoPP drops
show environment all	
show power status	
show diagnostic result module <n> detail	
show inventory	
show logging include <keyword>	

QoS

	[read-only]
show policy-map interface <wan-if>	priority queue drops
show mls qos interface <if>	trust boundary state
show run interface <if> include bandwidth service-policy	

Testing and debugging

```

[active test]
ping <dest> repeat 100 size 1500 df-bit
traceroute <dest>

[service-affecting]
debug ip ospf adj
debug crypto ikev2
debug capwap client events
! Always: terminal monitor, then undebug all when finished.
! debug output on a loaded control plane has taken routers down.

```

Change safety

```

[change approval]
archive
  path <secure-transfer-target>      SCP or SFTP, never TFTP
  write-memory
  log config
    logging enable
    notify syslog

configure replace flash:pre-change.cfg force
configure terminal revert timer 5     IOS XE auto-revert

```

Cisco ASA

```

[read-only]
packet-tracer input <if> tcp <src> <sport> <dst> <dport> detailed
show access-list <name> | include hitcnt
show conn address <ip>
show nat detail
show xlate
show asp drop                      accelerated path drop reasons
show crypto ikev2 sa                Phase 1
show crypto ipsec sa                Phase 2: compare encaps vs decaps
show vpn-sessiondb detail anyconnect
show service-policy inspect sip
show run timeout

```

packet-tracer is the first command to learn on this platform. Run it from the interface the traffic actually enters, with the real source address.

Palo Alto

```

[read-only]
test security-policy-match from <zone> to <zone> source <ip> \
    destination <ip> protocol 6 destination-port 443
show session all filter source <ip>
show session info                session count vs platform maximum
show system resources follow
show running resource-monitor hour
show vpn ike-sa ; show vpn ipsec-sa ; show vpn flow
show log url query equal "( addr.src eq <ip> )"
show log decryption

```

Fortinet

```

[read-only]
get system performance status
diagnose sys session stat
diagnose sys session list
diagnose npu npu-feature          is hardware offload engaged?

[service-affecting]
diagnose debug flow filter addr <ip>
diagnose debug flow trace start 20
diagnose debug application urlfilter -1
diagnose debug disable           always turn it off afterwards

```

Cisco Wireless LAN Controller

```

[read-only]
show ap join stats summary all
show ap join stats detailed <ap-mac>  names the join failure reason
show ap auto-rf <ap-name>             channel, power, noise, interference
show ap uptime <ap-name>             has it been rebooting?
show ap image all
show client detail <mac>              RSSI, SNR, retries, data rate, AC
show ap dot11 5ghz summary            utilisation and client count
show wlan <id>                       VLAN mapping, WMM, 11k/v/r
show mobility anchor
show time                             cert validation depends on it

[service-affecting]
debug client <mac>
debug capwap client events
debug pm pki enable

```

Linux

Addressing and routing

```
# [read-only]
ip addr show
ip route show
ip route get 8.8.8.8      # the route AND source address that will be used
ip neigh show            # ARP cache
ip -6 addr show
ip -6 route show
ip -6 neigh show
```

`ip route get` is the command to learn first here. It answers in one line what reading the routing table and reasoning about it takes several minutes to work out.

Interface and performance

```
# [read-only]
ethtool eth0              # speed, duplex, link
ethtool -S eth0           # per-queue statistics and drops
ethtool -k eth0           # offload settings
ethtool -g eth0           # ring buffer sizes
ss -tulpn                 # listening sockets
ss -ti                    # per-socket cwnd, rtt, retransmits
ss -s                     # socket summary
netstat -s | grep -i -E "listen|overflow|drop"
mpstat -P ALL 1           # is one core pinned?
```

Firewall

```
# [read-only]
iptables -L -v -n
nft list ruleset
firewall-cmd --list-all
sysctl net.ipv4.conf.all.rp_filter
sysctl net.ipv4.conf.all.arp_ignore
sysctl net.ipv4.conf.all.arp_announce
```

Diagnosis

```
# [active test]
mtr -rwzc 100 <host>      # loss and latency per hop; run from both ends
traceroute -n <host>
tracepath <host>          # reports path MTU changes
tracepath6 <host>
ping -M do -s 1472 <dest>  # DF-bit test, IPv4
ping6 -M do -s 1452 <dest> # DF-bit test, IPv6
iperf3 -c <server> -t 30   # single stream
iperf3 -c <server> -t 30 -P 8 # parallel: isolates per-flow limits
tcpdump -i any -n "host <ip> and port <p>" -w capture.pcap
```

DNS

```
# [active test]
dig <name>
dig <name> +trace           # walk the delegation from the root
dig @<server> <name> +norecurse
dig <name> +tcp             # does TCP 53 work?
dig <name> +bufsize=4096    # large EDNS0 response
dig <name> +dnssec          # validation behaviour
dig <name> +cd              # checking disabled, isolates DNSSEC
dig MX <domain> ; dig TXT <domain> ; dig -x <ip>
```

TLS

```
# [active test]
openssl s_client -connect <host>:443 -showcerts
openssl s_client -connect <ip>:443 -servername <host> </dev/null 2>/dev/null \
| openssl x509 -noout -serial -subject -dates -ext subjectAltName
```

Time

```
# [read-only]
timedatectl status
chronyc tracking ; chronyc sources -v
ntpq -p
```

Windows and PowerShell

```
# [read-only]
Get-NetAdapter | Select Name, Status, LinkSpeed, DriverVersion, DriverDate
Get-NetIPConfiguration
Get-NetIPInterface | Sort InterfaceMetric
Get-NetRoute
Get-NetRoute -AddressFamily IPv6
Get-NetConnectionProfile           # Domain, Private or Public
Get-NetFirewallProfile
Get-NetLbfoTeam                    # did teaming survive?
Get-NetTCPConnection -State Listen

# [active test]
Test-NetConnection <host> -Port 443 -InformationLevel Detailed
Resolve-DnsName <name> -Server <dns>
nslookup <name> <dns-server>

# [read-only]
Get-DnsClientServerAddress
Get-DnsClientNrptPolicy            # split DNS rules
w32tm /query /status               # NTP state
w32tm /stripchart /computer:<dc> /samples:5
nltest /dsgetdc:<domain>
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4625}
Get-HotFix | Sort InstalledOn -Descending | Select -First 10
```

A caution on Set-NetConnectionProfile. You cannot force the domain profile with this cmdlet. Windows assigns it automatically once Network Location Awareness authenticates against a domain controller. If a machine has dropped to Public, fix DNS and domain controller reachability instead of trying to override the category. See Q93.

Windows DHCP and DNS server

```
# [read-only]
Get-DhcpServerv4Scope
Get-DhcpServerv4ScopeStatistics
Get-DhcpServerv4Lease -ScopeId <x> | Group-Object ClientId
Get-DnsServerZone
Get-DnsServerForwarder
Get-DnsServerResourceRecord -ZoneName <zone> -Name <host>
Clear-DnsServerCache
```

AWS CLI

```
# [read-only]
aws ec2 describe-route-tables \
  --filters Name=association.subnet-id,Values=<subnet>
aws ec2 describe-internet-gateways \
  --filters Name=attachment.vpc-id,Values=<vpc>
aws ec2 describe-security-groups --group-ids <sg>
aws ec2 describe-network-acls \
  --filters Name=association.subnet-id,Values=<subnet>
aws ec2 describe-nat-gateways --nat-gateway-ids <nat>
aws ec2 describe-vpn-connections --vpn-connection-ids <vpn>

aws elbv2 describe-target-health --target-group-arn <arn>
aws elbv2 describe-target-group-attributes --target-group-arn <arn>
aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names <asg>
aws autoscaling describe-auto-scaling-activities --auto-scaling-group-name <asg>

# Reachability Analyzer: answers "what is blocking this?" definitively
aws ec2 create-network-insights-path ...
aws ec2 start-network-insights-analysis ...
```

Flow logs. `aws ec2 describe-flow-logs` lists flow log **configurations**. It does not return records. To read actual REJECT entries you must query the destination:

```
# [read-only]
aws logs start-query \
  --log-group-name <flow-log-group> \
  --start-time <epoch> --end-time <epoch> \
  --query-string 'fields srcAddr, dstAddr, dstPort, action
                  | filter action = "REJECT"
                  | sort @timestamp desc | limit 50'
# Or query S3-delivered logs with Athena.
```

Handling credentials safely

Several commands here take community strings, pre-shared keys or passwords. Three rules:

1. **Never paste live credentials into tickets, chat or screenshots.** Reference the credential store instead.
2. **Prefer SNMPv3 and SSH** over SNMPv2c and Telnet. The older protocols carry credentials in clear text and a packet capture reveals them.
3. **Back up configurations over SCP, SFTP or HTTPS**, never TFTP. Device configurations contain SNMP strings, pre-shared keys and password hashes, so treat the backup repository as a secret store rather than a file share.

Appendix E. Ports, Protocols and Markings

Ports worth knowing by heart

Service	Port	Transport	Why it matters in a NOC
FTP control	21	TCP	The data channel is separate. See Q98
FTP data (active)	20	TCP	Server initiates back to the client
SSH, SCP, SFTP	22	TCP	Management baseline. Use this, not Telnet
Telnet	23	TCP	Clear text. Should be disabled everywhere
SMTP	25	TCP	Frequently blocked outbound by ISPs
DNS	53	UDP and TCP	TCP is required for large responses. Policies that permit only UDP cause Q16
DHCP	67 server, 68 client	UDP	Broadcast. Needs a relay to cross a router
TFTP	69	UDP	No auth, no encryption. Do not use for config backup
HTTP	80	TCP	
Kerberos	88	TCP + UDP	Time-sensitive. Breaks when clocks drift. See Q99
NTP	123	UDP	Blocking it breaks TLS and Kerberos, not just clocks
NetBIOS	137–139	TCP + UDP	Legacy Windows name services
SNMP	161 poll, 162 trap	UDP	See Q100
BGP	179	TCP	Sessions drop on sustained keepalive loss. See Q68
LDAP / LDAPS	389 / 636	TCP	Directory authentication
HTTPS	443	TCP	Also the fallback transport for TLS-based VPN
SMB	445	TCP	File shares. See Q35
Syslog	514	UDP	6514 for syslog over TLS
IKE	500	UDP	IPsec phase 1
RADIUS	1812 auth, 1813 acct	UDP	802.1X and NAC
NAT-T (IPsec)	4500	UDP	Required when a peer is behind NAT
CAPWAP	5246 control, 5247 data	UDP	Lightweight AP to controller. See Q47
SIP	5060, 5061 TLS	TCP + UDP	Signalling only. Media is separate
RTP media	16384–32768 typical	UDP	Range varies by vendor. See Q74

Service	Port	Transport	Why it matters in a NOC
QUIC / HTTP/3	443	UDP	Increasingly common. Policies permitting only TCP 443 silently force fallback

Modern additions worth knowing

Service	Port	Note
DNS over TLS (DoT)	853	TCP. Easy to identify and block by port
DNS over HTTPS (DoH)	443	Indistinguishable from web traffic. Bypasses your DNS filtering entirely unless you manage the client
DNS over QUIC (DoQ)	853	UDP
HTTP/3	443	Over QUIC on UDP

Why this matters operationally. A browser configured for DNS over HTTPS ignores the resolver you handed out via DHCP, which means your internal names may not resolve and your DNS-based filtering and logging see nothing. If your organisation relies on DNS policy as a control, DoH is a gap to manage deliberately rather than discover during an incident.

QUIC matters for a different reason. It runs over UDP 443, so a firewall permitting only TCP 443 causes clients to fall back to TCP after a delay. The symptom is “the internet feels slow” with nothing in the logs, which is the same shape as Q102.

IP protocol numbers

These appear in ACLs and firewall rules where a port number would not apply.

Protocol	Number	Note
ICMP	1	
TCP	6	
UDP	17	
GRE	47	Tunnelling. Adds 24 bytes of overhead
ESP	50	IPsec encrypted payload
AH	51	IPsec authentication header
OSPF	89	
VRRP	112	
ICMPv6	58	Do not blanket-block this. See below

ICMPv6 message types you must permit

In IPv4, filtering ICMP is unwise. In IPv6 it breaks the protocol, because Neighbour Discovery and path MTU discovery both run over ICMPv6.

Type	Message	Consequence of blocking
1	Destination unreachable	Poor error reporting, slow failures
2	Packet too big	Path MTU discovery dies. Large transfers hang. See Q106
3	Time exceeded	Traceroute stops working
4	Parameter problem	Poor error reporting
133	Router solicitation	Hosts cannot discover routers
134	Router advertisement	No address, no gateway
135	Neighbour solicitation	No address resolution at all
136	Neighbour advertisement	No address resolution at all
137	Redirect	Suboptimal routing

Types 133 to 137 should be permitted on-link and generally not forwarded across boundaries. Type 2 must be permitted end to end.

Full guidance is in RFC 4890, which is the document to cite if anyone insists on blocking ICMPv6 wholesale.

DSCP values and QoS markings

This table is where a lot of published material goes wrong, so read the note underneath before using it.

RFC 4594 service classes

Traffic	DSCP	Decimal	RFC 4594 service class
Network control	CS6	48	Network Control
Telephony (voice media)	EF	46	Telephony
Signalling	CS5	40	Signaling
Multimedia conferencing	AF41–AF43	34, 36, 38	Multimedia Conferencing
Real-time interactive	CS4	32	Real-Time Interactive
Multimedia streaming	AF31–AF33	26, 28, 30	Multimedia Streaming
Broadcast video	CS3	24	Broadcast Video
Low-latency data	AF21–AF23	18, 20, 22	Low-Latency Data

Traffic	DSCP	Decimal	RFC 4594 service class
OAM	CS2	16	OAM
High-throughput data	AF11–AF13	10, 12, 14	High-Throughput Data
Standard	DF	0	Standard
Low-priority data	CS1	8	Low-Priority Data

About call signalling specifically

You will see **AF31 described as call signalling** in a great deal of documentation and in older Cisco design guides. Be careful with it.

- **RFC 4594** assigns **CS5** to its Signaling service class, and uses **AF31 for Multimedia Streaming**.
- **Cisco's older QoS baseline** used **AF31** for call signalling, which is where the widespread convention came from.
- **Cisco's later enterprise medianet models** moved signalling to **CS3**, aligning more closely with RFC 4594.

All three exist in production networks today. None of them is universally “correct.”

What to do about it. State which profile your network follows, document it, and apply it consistently end to end. In an interview, saying “we mark signalling CS3 following the Cisco enterprise model, though RFC 4594 specifies CS5 and older designs used AF31” is a much stronger answer than confidently asserting any single value.

The value that is genuinely consistent everywhere is **EF (46) for voice media**. That one you can rely on.

Voice quality targets

Worth memorising, because they turn a vague complaint into a measurable fact.

Metric	Target	Beyond it
One-way latency	under 150 ms	People talk over each other
Jitter	under 30 ms	Choppy, robotic audio
Packet loss	under 1 percent	Dropouts and clipped words

Source: ITU-T G.114 for latency, RFC 4594 for the service class targets.

Wireless thresholds

Metric	Data	Voice
RSSI	better than -70 dBm	better than -67 dBm

Metric	Data	Voice
SNR	20 dB or better	25 dB or better
Retries	under 15 percent	under 10 percent
Channel utilisation	under 50 percent	under 40 percent
Roam time	not critical	under 50 ms, needs 802.11r

Useful numbers

Encapsulation overhead

Encapsulation	Bytes added
GRE	24
IPsec ESP tunnel mode	roughly 50 to 60
NAT-T (UDP 4500)	8
VXLAN	50
802.1Q VLAN tag	4

MTU reference points

Value	Meaning
1500	Standard Ethernet payload
1492	PPPoE
1400 / 1360	Common tunnel MTU / TCP MSS clamp
1280	IPv6 minimum link MTU. Safe fallback
9000	Typical jumbo frame

Optical loss budget

Element	Typical loss
Mated connector pair	0.3 to 0.75 dB
Fusion splice	around 0.1 dB
Multi-mode fibre at 850 nm	around 3.0 dB per km
Single-mode fibre at 1310 nm	around 0.4 dB per km

Propagation delay

Light in fibre travels at roughly **200 km per millisecond**.

$$\text{round-trip distance (km)} / 200 = \text{minimum round-trip time (ms)}$$

A 15,000 km round trip cannot be faster than about **75 ms** of pure propagation. Real routes are longer than straight-line distance and equipment adds more, so 90 to 110 ms would be normal. Work this out before escalating a latency complaint to a carrier.

Appendix F. Standards Index

Every standard referenced in this guide, grouped by topic. Links go to the primary source.

Why primary sources matter. A great deal of published networking material repeats claims that were true once, true for one vendor, or never true. The RFC or the IEEE document is what actually governs the behaviour. When something in this guide disagrees with a standard listed here, the standard is right and this guide has an error worth reporting.

Core protocols

Topic	Reference
Internet Protocol	RFC 791
ICMP	RFC 792
TCP (current specification)	RFC 9293
ARP	RFC 826
Private address space	RFC 1918
Requirements for Internet hosts	RFC 1122
Requirements for IPv4 routers	RFC 1812
TCP extensions for high performance	RFC 7323
HTTP semantics	RFC 9110
HTTP state management (cookies)	RFC 6265

MTU and fragmentation

Topic	Reference
Path MTU Discovery	RFC 1191
TCP problems with PMTUD	RFC 2923
Packetization Layer PMTUD	RFC 4821
MTU and fragmentation issues with tunnels	RFC 4459
Path MTU Discovery for IPv6	RFC 8201

Routing

Topic	Reference
OSPFv2	RFC 2328

Topic	Reference
BGP-4	RFC 4271
BGP operations and security	RFC 7454
Bidirectional Forwarding Detection	RFC 5880
VRRPv3	RFC 5798
Protecting the router control plane	RFC 6192

DNS

Topic	Reference
Domain names: concepts	RFC 1034
Domain names: implementation	RFC 1035
DNS terminology	RFC 8499
EDNS(0)	RFC 6891
DNS transport over TCP	RFC 7766
DNSSEC protocol modifications	RFC 4035
Dynamic updates in the DNS	RFC 2136
Serving stale data	RFC 8767
DNS extensions to support IPv6	RFC 3596

DHCP and addressing

Topic	Reference
DHCP	RFC 2131
DHCP relay agent information option	RFC 3046
IPv4 link-local addresses (APIPA)	RFC 3927
DHCPv6	RFC 8415

IPv6

Topic	Reference
IPv6 specification	RFC 8200
Neighbor Discovery for IPv6	RFC 4861
IPv6 stateless address autoconfiguration	RFC 4862
Default address selection for IPv6	RFC 6724

Topic	Reference
Happy Eyeballs version 2	RFC 8305
RA options for DNS configuration	RFC 8106
Rogue IPv6 RA problem statement	RFC 6104
IPv6 Router Advertisement Guard	RFC 6105
Filtering ICMPv6 messages in firewalls	RFC 4890
Security implications of IPv6 on IPv4 networks	RFC 7123
Operational security considerations for IPv6	RFC 9099

Security, VPN and PKI

Topic	Reference
IKEv2	RFC 7296
Security architecture for IP	RFC 4301
IP Encapsulating Security Payload	RFC 4303
NAT traversal negotiation in IKE	RFC 3947
UDP encapsulation of IPsec ESP	RFC 3948
IKEv2 fragmentation	RFC 7383
Split DNS configuration for IKEv2	RFC 8598
Traditional IP NAT	RFC 3022
NAT behavioural requirements for UDP	RFC 4787
NAT behavioural requirements for TCP	RFC 5382
X.509 certificate and CRL profile	RFC 5280
Service identity verification	RFC 6125
TLS extensions including SNI	RFC 6066
TLS 1.3	RFC 8446
EAP	RFC 3748
EAP-TLS	RFC 5216
Kerberos V5	RFC 4120
Firewall behaviour and requirements	RFC 2979
Effects of pervasive encryption on operators	RFC 8404
Captive portal architecture	RFC 8952

Voice, video and QoS

Topic	Reference
SIP	RFC 3261
SDP	RFC 4566
RTP	RFC 3550
RTP topologies	RFC 7667
Session timers in SIP	RFC 4028
DiffServ field definition	RFC 2474
DiffServ service class guidelines	RFC 4594
Mapping DiffServ to IEEE 802.11	RFC 8325
One-way transmission time	ITU-T G.114

Measurement and monitoring

Topic	Reference
Framework for TCP throughput testing	RFC 6349
Benchmarking methodology for network devices	RFC 2544
RFC 2544 on production networks considered harmful	RFC 6815
IPFIX protocol specification	RFC 7011
sFlow	RFC 3176
Interfaces group MIB	RFC 2863
SNMP management frameworks	RFC 3411
SNMPv3 user-based security model	RFC 3414

Read RFC 6815 before citing RFC 2544. RFC 2544 methods are designed to overload a device under test in order to measure its capacity, and RFC 6815 states plainly that they belong in an isolated test environment. Running them on a production network harms user traffic. For live path measurement, use RFC 6349 instead.

Wireless and CAPWAP

Topic	Reference
Wireless LAN (802.11 family)	IEEE 802.11
Port-based network access control	IEEE 802.1X
CAPWAP protocol specification	RFC 5415

Topic	Reference
ITU-R Radio Regulations (ISM bands)	ITU-R

Ethernet, switching and cabling

Topic	Reference
Ethernet	IEEE 802.3
1000BASE-T over Category 5 (interpretation)	IEEE 802.3ab interpretation
VLAN bridged networks	IEEE 802.1Q
Spanning Tree	IEEE 802.1D
Link aggregation	IEEE 802.1AX
Powering electronic equipment	IEEE 1100
Generic cabling for customer premises	ISO/IEC 11801
Fibre optic connector cleanliness	IEC 61300-3-35
Structured cabling standards	TIA

Network automation and telemetry

Topic	Reference
NETCONF	RFC 6241
RESTCONF	RFC 8040
YANG 1.1	RFC 7950

Email

Topic	Reference
SMTP	RFC 5321
Email greylisting	RFC 6647
Sender Policy Framework	RFC 7208
FTP	RFC 959
FTP extensions for NAT	RFC 2428
Securing FTP with TLS	RFC 4217

Time

Topic	Reference
NTPv4	RFC 5905

Operations, security and process frameworks

Topic	Reference
ITIL 4 service management	Axelos
Site Reliability Engineering	Google SRE Book
SRE practices	Google SRE Workbook
Incident response	NIST SP 800-61 Rev. 3
Firewall policy guidelines	NIST SP 800-41 Rev. 1
Digital identity guidelines	NIST SP 800-63 Revision 4
Securing wireless LANs	NIST SP 800-153
IoT device cybersecurity	NIST SP 800-213
Secure IPv6 deployment	NIST SP 800-119
Adversary tactics and techniques	MITRE ATT&CK

Note on NIST currency. SP 800-61 Rev. 2 was withdrawn on 3 April 2025 and superseded by Rev. 3. The SP 800-63-3 family has been superseded by Revision 4. If you are citing NIST in your own work, check the publication page rather than a search result, because withdrawn documents remain online and rank well.

Vendor documentation

Vendor behaviour changes between releases far more often than standards do, so treat these as current-state references rather than as settled facts.

Topic	Reference
AWS VPC	docs.aws.amazon.com/vpc
AWS EC2 instance IP addressing	EC2 user guide
AWS target group health checks	ELB documentation
Windows domain profile detection	Microsoft Learn
Linux ip-route	man7.org
SFF specifications (optics diagnostics)	SNIA SFF

Lab platforms

Build the scenarios rather than only reading them.

Platform	Best for
Containerlab	Fast container-based topologies, modern NOS images
GNS3	Mixed vendor labs with real images
EVE-NG	Larger multi-vendor topologies
Cisco Packet Tracer	Free, enough for the whole L1 path
Wireshark sample captures	Packet analysis practice without building anything

Appendix G. Research Foundations

The primary references in this book are standards: RFCs, IEEE documents, NIST publications and vendor documentation. Standards tell you how a protocol is supposed to behave. The peer-reviewed studies below measure how networks actually behave in production, and several of them explain *why* a scenario in this book is common. They are useful background for L3 interviews, where “what is your evidence?” is a standard follow-up question.

Each entry gives the full citation, a DOI link where one could be verified, and the scenarios it supports. Two entries are given without a DOI because none could be confirmed; they are cited by venue and page range instead.

Transport performance and path MTU

[G1] V. Jacobson, “Congestion avoidance and control,” in *Proceedings of ACM SIGCOMM '88*, Stanford, CA, USA, 1988, pp. 314–329. DOI: <https://doi.org/10.1145/52324.52356>

The paper that introduced slow start, round-trip-time variance estimation and TCP congestion avoidance, written in response to the Internet congestion collapses of 1986. It is the origin of the TCP behaviour behind throughput questions such as Q41, Q58 and Q95.

[G2] M. Mathis, J. Semke, J. Mahdavi and T. Ott, “The macroscopic behavior of the TCP congestion avoidance algorithm,” *ACM SIGCOMM Computer Communication Review*, vol. 27, no. 3, pp. 67–82, 1997. DOI: <https://doi.org/10.1145/263932.264023>

Derives and validates a model that predicts the bandwidth of a sustained TCP connection under light to moderate loss. Throughput is proportional to segment size and inversely proportional to round-trip time and to the square root of the loss rate. This is the quantitative reason a fast local speed test can coexist with a slow tunnel (Q41), and why a single flow cannot fill a long-distance path (Q71, Q95).

[G3] M. Luckie and B. Stasiewicz, “Measuring path MTU discovery behaviour,” in *Proceedings of the 10th ACM Internet Measurement Conference (IMC '10)*, Melbourne, Australia, 2010, pp. 102–108. DOI: <https://doi.org/10.1145/1879141.1879155>

Measured path MTU discovery for 50,000 popular websites and found an IPv4 failure rate of between 5 and 18 percent, depending on the MTU of the constraining link. The authors also found that many failures were caused by software bugs rather than by firewalls discarding ICMP. Background for Q33, Q37 and Q106, and a useful caution against assuming every stalled transfer is a filtered ICMP message.

DNS and dual-stack behaviour

[G4] J. Jung, E. Sit, H. Balakrishnan and R. Morris, “DNS performance and the effectiveness of caching,” *IEEE/ACM Transactions on Networking*, vol. 10, no. 5, pp. 589–603, 2002. DOI: <https://doi.org/10.1109/TNET.2002.803905>

A measurement study of DNS lookup performance and of how effective caching is in practice. Background for why caches and TTLs shape what clients see after a change (Q22), and for resolver-path problems (Q16).

[G5] V. Bajpai and J. Schönwälder, “Measuring the effects of Happy Eyeballs,” in *Proceedings of the 2016 Applied Networking Research Workshop (ANRW '16)*, Berlin, Germany, 2016. DOI: <https://doi.org/10.1145/2959424.2959429>

Used a three-year dataset from 80 dual-stacked measurement probes to study how Happy Eyeballs chooses between IPv6 and IPv4. It found that only around 1 percent of IPv6 connection times ever exceeded the 300 ms timer then in use, so clients preferred IPv6 almost all of the time. Background for Q102 and Q108.

[G6] J. Czyz, M. Luckie, M. Allman and M. Bailey, “Don’t forget to lock the back door! A characterization of IPv6 network security policy,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, USA, 2016. DOI: <https://doi.org/10.14722/ndss.2016.23047>

A measurement of 520,000 dual-stack servers and 25,000 dual-stack routers. It found services such as SSH, Telnet and SNMP more than twice as open on routers over IPv6 as over IPv4. Every operator the authors contacted confirmed that the difference was unintentional. This is the research evidence behind Q107.

Wireless

[G7] G. Bianchi, “Performance analysis of the IEEE 802.11 distributed coordination function,” *IEEE Journal on Selected Areas in Communications*, vol. 18, no. 3, pp. 535–547, 2000. DOI: <https://doi.org/10.1109/49.840210>

The standard analytical model of 802.11 DCF throughput with a finite number of contending stations. It formalises why wireless is a shared, contended medium and why adding clients reduces what each one receives (Q44, Q75).

[G8] M. Heusse, F. Rousseau, G. Berger-Sabbatel and A. Duda, “Performance anomaly of 802.11b,” in *Proceedings of IEEE INFOCOM 2003*, San Francisco, CA, USA, 2003, pp. 836–843. DOI: <https://doi.org/10.1109/INFCOM.2003.1208921>

Showed that one host transmitting at 1 Mb/s can reduce the throughput of hosts transmitting at 11 Mb/s to below 1 Mb/s, because CSMA/CA gives every host equal access opportunity regardless of rate. This is the mechanism behind disabling low legacy data rates in Q43 and the airtime point in Q44.

[G9] J. Martin, T. Mayberry, C. Donahue, L. Foppe, L. Brown, C. Riggins, E. C. Rye and D. Brown, “A study of MAC address randomization in mobile devices and when it fails,” *Proceedings on Privacy Enhancing Technologies*, vol. 2017, no. 4, pp. 365–383, 2017. DOI: <https://doi.org/10.1515/popets-2017-0054>

The first wide-scale study of MAC address randomisation in the field, broken down by operating system, manufacturer and model. It found adoption and implementation varied widely between vendors. Background for why client identity is unreliable in DHCP pools and on guest networks (Q18).

Security policy and configuration

[G10] A. Wool, “A quantitative study of firewall configuration errors,” *IEEE Computer*, vol. 37, no. 6, pp. 62–67, 2004. DOI: <https://doi.org/10.1109/MC.2004.2>

Analysed real corporate firewall rule sets and found that they often enforce policies violating well-established security guidelines, with the number of errors rising with rule-set complexity. Background for rule-order and review questions such as Q23 and Q32.

Failures, operations and availability

[G11] P. Gill, N. Jain and N. Nagappan, “Understanding network failures in data centers: measurement, analysis, and implications,” in *Proceedings of ACM SIGCOMM 2011*, Toronto, Canada, 2011, pp. 350–361. DOI: <https://doi.org/10.1145/2018436.2018477>

A large-scale study of failures across tens of data centres over more than a year. Load balancers dominated failure occurrences, many of them short-lived software faults, and network redundancy was only 40 percent effective in reducing the median impact of a failure. Useful evidence for Q14 (redundancy that does not take over), Q65 (load balancer behaviour) and Q50 (correlating many alerts to one cause).

[G12] D. Turner, K. Levchenko, A. C. Snoeren and S. Savage, “California fault lines: understanding the causes and impact of network failures,” in *Proceedings of ACM SIGCOMM 2010*, New Delhi, India, 2010. No DOI verified; available from the first author’s research page at the University of California, San Diego.

Showed how a history of network failure events can be reconstructed by mining “low-quality” data that networks already collect, such as syslog and configuration records, rather than deploying new instrumentation. Background for working from logs and counters in Q10, Q52 and Q57.

[G13] R. Govindan, I. Minei, M. Kallahalla, B. Koley and A. Vahdat, “Evolve or die: high-availability design principles drawn from Google’s network infrastructure,” in *Proceedings of ACM SIGCOMM 2016*, Florianópolis, Brazil, 2016. DOI: <https://doi.org/10.1145/2934872.2934891>

An analysis of more than 100 high-impact failure events across Google’s data centres and two WANs. A large number of failures occurred while a network management operation was in progress. The authors’ design principles include defence in depth, consistency across planes and assessing root

cause quickly. This is research support for the change-control scenarios Q12, Q32, Q62 and Q64, and for Chapter 10 generally.

Time

[G14] D. L. Mills, “Internet time synchronization: the Network Time Protocol,” *IEEE Transactions on Communications*, vol. 39, no. 10, pp. 1482–1493, 1991. No DOI verified; cited by volume and page range.

The paper that brought the NTP architecture, protocol and algorithms to the wider engineering community. It describes the hierarchical, self-organising subnet of time servers that Q99 depends on.

Appendix H. Glossary

Terms are defined as they are used in this book. Where a term has a governing standard, it is listed in Appendix F.

Term	Meaning
AAAA record	DNS record that maps a name to an IPv6 address
AC_VO	The Wi-Fi Multimedia access category for voice, given the highest priority for airtime
ACK	Acknowledgement. In TCP, confirms received data; in SIP, confirms a final response to an INVITE
ACL	Access control list. An ordered set of permit and deny rules evaluated top to bottom until the first match
ALG	Application layer gateway. A firewall or NAT function that reads application signalling, such as SIP or FTP, and rewrites addresses or opens pinholes
APIPA	Automatic Private IP Addressing. A self-assigned IPv4 link-local address in 169.254.0.0/16, used when DHCP fails
ARP	Address Resolution Protocol. Maps an IPv4 address to a MAC address on the local link
BFD	Bidirectional Forwarding Detection. A lightweight protocol that detects path failure in milliseconds for routing and redundancy protocols
BGP	Border Gateway Protocol. The path-vector routing protocol used between autonomous systems
Blast radius	The set of users, sites and services affected by a fault or a change
BPDU	Bridge Protocol Data Unit. The frame spanning tree uses to elect a root and detect loops
CAB	Change Advisory Board. The group that reviews and approves changes
CAPWAP	Control And Provisioning of Wireless Access Points. The protocol between lightweight access points and a controller (UDP 5246 and 5247)
CEF	Cisco Express Forwarding. The hardware or optimised software forwarding path that keeps transit traffic off the CPU
CIDR	Classless Inter-Domain Routing. Prefix notation such as 10.1.0.0/16
Control plane	The part of a device that runs routing protocols and management, as distinct from the data plane that forwards traffic
CRC error	A frame whose checksum does not match its contents, meaning it was corrupted in transit
Dead peer detection	An IKE mechanism that detects when the far end of a tunnel has stopped responding
DF bit	The IPv4 “Don’t Fragment” flag. Routers must drop an oversized packet with DF set and should return ICMP Type 3 Code 4

Term	Meaning
DHCP relay	A router function (for example <code>ip helper-address</code>) that forwards DHCP broadcasts as unicast to a server on another subnet
DSCP	Differentiated Services Code Point. The six-bit IP header field used to mark traffic for QoS treatment. EF (46) is the usual voice marking
ECMP	Equal-cost multipath. Load sharing across several routes of equal cost, usually by hashing each flow onto one path
EDNS0	Extension mechanisms for DNS that allow UDP responses larger than 512 bytes
EtherChannel	Cisco's term for link aggregation. See LACP
FHRP	First-hop redundancy protocol, such as HSRP, VRRP or GLBP, providing a shared virtual default gateway
Greylisting	An anti-spam technique that temporarily rejects mail from unknown senders, relying on legitimate servers to retry
Happy Eyeballs	The client algorithm that races IPv6 and IPv4 connection attempts and uses whichever succeeds first
Health check	A probe a load balancer sends to decide whether a backend should receive traffic
ICMPv6	ICMP for IPv6. It carries Neighbor Discovery and Packet Too Big messages and cannot be blocked wholesale
IKE	Internet Key Exchange. Negotiates IPsec security associations (UDP 500, and UDP 4500 with NAT traversal)
IPsec	A suite that provides encryption and integrity for IP packets, using ESP in almost all modern deployments
Jitter	Variation in packet delay. Voice targets are usually under 30 ms
LACP	Link Aggregation Control Protocol (IEEE 802.1AX). Bundles several physical links into one logical link
Link-local address	An address valid only on the local link: 169.254.0.0/16 in IPv4, fe80::/10 in IPv6
Longest prefix match	The rule that a router forwards using the most specific route that covers the destination
MSS clamping	Rewriting the TCP maximum segment size during the handshake so segments fit a smaller path MTU. It affects TCP only
MTU	Maximum transmission unit. The largest packet a link carries without fragmentation
MTR	A tool that combines traceroute and continuous ping to show loss and latency per hop
NAC	Network access control. Admission based on authentication (usually 802.1X) and often device posture
NACL	Network ACL. In AWS, a stateless subnet-level filter that needs explicit rules for return traffic

Term	Meaning
NAT	Network Address Translation. Rewrites source or destination addresses, commonly private to public
Native VLAN	The VLAN whose frames cross an 802.1Q trunk untagged
Neighbor Discovery	The IPv6 protocol, carried in ICMPv6, that replaces ARP and provides router and prefix discovery
NTP	Network Time Protocol. Synchronises clocks through a hierarchy of servers identified by stratum
OTDR	Optical time-domain reflectometer. Locates breaks, bends and bad splices on a fibre by distance
P1 to P4	Incident priority levels derived from business impact and urgency. P1 is a major outage
PMTUD	Path MTU discovery. The process by which a sender learns the smallest MTU on a path from ICMP feedback
PoE	Power over Ethernet. Supplies power to devices such as access points and phones over the data cable
PortFast	A switch port setting that moves an edge port straight to forwarding without generating topology changes
Posture assessment	A NAC check that a device meets compliance rules, such as current antivirus definitions
Problem management	The ITIL practice that finds and removes the root cause of recurring incidents
RA	Router Advertisement. The ICMPv6 message in which IPv6 routers announce prefixes and themselves as gateways
RA Guard	A switch feature that blocks Router Advertisements from ports that should not send them
RCA	Root cause analysis. The investigation, and the document, explaining why an incident happened
RDNSS	Recursive DNS Server option. Carries DNS server addresses inside an IPv6 Router Advertisement
Rollback	Returning a system to its state before a change. It is not always available or safe, see Q90
RSSI	Received signal strength indicator, in dBm. Voice designs usually target better than -67 dBm
RTP	Real-time Transport Protocol. Carries voice and video media, separately from signalling
Runbook	The documented procedure for handling a known scenario
SA	Security association. A negotiated set of IPsec keys and parameters for one direction of traffic
SAN	Subject Alternative Name. The certificate field listing every hostname the certificate is valid for

Term	Meaning
SDP	Session Description Protocol. Describes media addresses, ports and codecs inside SIP messages
Security group	In AWS, a stateful instance-level filter; allowing outbound traffic implicitly allows its return
SIP	Session Initiation Protocol. Sets up, modifies and tears down voice and video calls
SLA	Service level agreement. The contractual response and resolution times for incidents
SLAAC	Stateless Address Autoconfiguration. IPv6 hosts build their own addresses from an advertised prefix
SNI	Server Name Indication. The TLS extension that tells a shared server which hostname the client wants
SNMP	Simple Network Management Protocol. Used by monitoring systems to poll device state (UDP 161, traps on UDP 162)
SNR	Signal-to-noise ratio, in dB. Voice designs usually target 25 dB or better
STAR	Situation, Task, Action, Result. A structure for answering behavioural interview questions
STP	Spanning Tree Protocol and its variants (RSTP, MSTP). Blocks redundant Layer 2 paths to prevent loops
Stateful	Describes a device that tracks connections and permits return traffic automatically
Stratum	An NTP server's distance from a reference clock. Stratum 16 means unsynchronised
SVI	Switched virtual interface. A routed Layer 3 interface for a VLAN on a multilayer switch
TCN	Topology Change Notification. A spanning tree message that shortens MAC table ageing across the domain
TTL	Time to live. In IP, a hop limit; in DNS, how long a cached answer may be used
VLAN	Virtual LAN. A logical broadcast domain on a switched network
VRF	Virtual routing and forwarding. Separate routing tables on one device
WMM	Wi-Fi Multimedia. The 802.11e-based prioritisation of traffic into access categories

Thank you for choosing this guide to prepare for your NOC Engineer interview. I continually strive to improve these resources and welcome your feedback; if you notice any errors or have suggestions, please reach out at contact@iteducationnepal.org or contact@jiwanbhattarai.com. For additional role-based IT interview guides, please visit iteducationnepal.org. Best of luck in your upcoming interviews and your continued career in IT.
